TECHNICAL AND USER DOCUMENTATION · 1 NOVEMBER 2026

Energy Data Preparation and the H2RES Model

A version-specific reference covering public data sources, preparation methods, model equations, input contracts, scenario assumptions, operational workflows, numerical quality and references.

Documented model: H2RES_1.11.2026 · Separate preparation and optimization workflows · Source hashes and machine-readable inventories included

H2RES documentation library

Edition: 1 November 2026. **Language:** English. **Model examined:** `D:\POSAO\H2RES_1.11.2026`.

This library documents two separate applications: the standalone internet data preparation tool in this repository and the supplied H2RES optimization model. Data preparation writes files. Model execution, investment optimization and dispatch belong to the separate model environment.

Start here

Document	Audience and contents
Data preparation manual	Installation, interface, internet sources, country selection, caching, sector methods, input assumptions, validation and troubleshooting
H2RES model manual	Model architecture, planning modes, mathematical formulation, sector coupling, policy constraints, cost accounting, execution, solver diagnostics and outputs
Input and output reference	File contracts, workbook families, field meanings, units, annual investment limits, joins, schema and examples
Assumptions and limitations	What is observed, derived, inherited or assumed; applicability and outstanding calibration needs
References	Scientific literature, dataset/API documentation, technology references, inherited bibliography and verification notes
Code reference	Configuration values, optimization variable/constraint call sites and source line locations extracted from this exact model copy
Machine-readable inventories	Complete CLI catalogue, workbook headers, CSV headers, configuration assignments, source URLs and hashes
BibTeX bibliography	Bibliography suitable for a publication or reference manager

Open the complete PDF manual for reading or sharing, or the searchable offline HTML for navigation between sections and references. Markdown files remain the editable source of the documentation.

Reading paths

1. **First use:** preparation manual sections 1–5, then its troubleshooting section.
2. **Prepare Italy:** preparation manual sections 4, 6 and 14; compare generated `System` / `zone` fields with the selected country.
3. **Prepare process industry:** preparation manual section 10 and assumptions register A06–A10.
4. **Run a study:** model manual sections 1–5 and 15–18, followed by the data reference.
5. **Review a paper:** formulation, assumptions register, source registry, exact source hashes and validation evidence.
6. **Maintain code:** code reference and inventories; regenerate them after changing implementation or templates.

Scope and evidence

The descriptions are based on the actual source files and live input schemas, not just the folder name or historical README files. Some notes supplied with the model describe older copies, experimental corrections or staged candidates. They are provenance, not proof that a feature or correction is active in this copy. The exact examined files are recorded in documented_version.json.

The mathematical notation is a readable representation of the implemented constraint families. It deliberately suppresses asset-specific indices and conditional branches. For the exact executable form, consult the named implementation and line-level inventories. A static call-site count is not the number of rows or columns in an instantiated optimization problem.

References explain scientific background or data provenance. They do not make inherited scenario values empirically calibrated. Unattributed or incompletely verified assumptions are identified explicitly. Preparing these documents did not run or solve H2RES and did not change the original model's inputs or code.

Rebuilding the documentation

From the preparation repository:

```
python docs/build_reference_inventory.py --model-root "D:\POSAO\H2RES_1.11.2026"
python docs/build_documentation.py
python docs/validate_documentation.py
python docs/render_pdf.py
```

The first command reads the model and writes only documentation inventories. The second creates the reference lists and combined HTML. The third checks internal links, references, inventory consistency and source hashes. The optional final command prints the HTML to PDF using a hidden Chrome or Edge session and an isolated browser profile. These commands are documentation tools, not model runs.

Internet data sourcing and input preparation manual

Version documented: source snapshot of 1 November 2026. See version inventory. References use identifiers in the bibliography.

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1. Purpose and boundaries

The tool obtains public energy and spatial data, caches downloads, transforms observations into sector input tables and hourly profiles, and packages those files in the interface required by the supplied H2RES model. It supports power, industry, transport and heating. It can prepare one country, several countries or a shared custom geography.

The graphical application is **Energy Data Sourcing and Preparation**. Its only preparation runner is `generate_all_sector_inputs.py`. It does not operate the H2RES solver. A separate adapter can accompany an output bundle so an independently run H2RES installation reads the right variants and policy settings; its existence does not mean preparation ran an optimization.

The model-compatible package combines newly prepared sector data with portable template inputs. Selecting one sector regenerates that sector; it does not automatically recalibrate every remaining workbook, network, cost, policy or demand series for a new country. The coverage report is essential when moving beyond the template geography.

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2. Package layout

Path	Role
`START_DATA_PREPARATION.bat`	Main Windows shortcut
`launch_data_preparation.ps1`	Python discovery, dependency probe, startup logging and safe path handling
`data_preparation_gui.py`	Public interface entry point
`h2res_input_gui.py`	Interface implementation; historical module name retained for compatibility
`START_H2RES_GUI.bat`	Compatibility shortcut to the same preparation interface
`data_preparation_config.json`	Saved settings, excluding the ENTSO-E token
`generate_all_sector_inputs.py`	Shared orchestration, geography and calls to sector generators
`preparation_dependencies.py`	Selected-sector dependency checks before downloads/output changes
`model_reference/data`	Portable format, technology and scenario templates
`model_input_bundle.py`	Native file export, worksheet replacement and coverage checks
`scenario_config.py`	Input-year, renewable-target and investment-limit assumptions
`industry`, `power`, `transport`, `heating`	Sector implementations and bundled support resources

Path	Role
`data_all*`	User-selected download/cache roots; they may contain multinational source data
`outputs_all*`	User-selected prepared output roots
`validation`	Developer test evidence; not a guarantee that a fresh national scenario is feasible

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3. Installation and environment

Use a consistent 64-bit Python environment. The launcher accepts Python 3.10 or later and checks the basic table packages; selected pipelines additionally need geospatial, raster and weather dependencies. Python 3.12 is the interpreter used in the recorded local verification. A solver package/license is not required to source and prepare data.

Install from the repository directory using the Python that will run the application:

```
& "C:\Users\lukah\AppData\Local\Programs\Python\Python312\python.exe" -m pip install -r requirements.txt
```

The default requirements include NumPy, pandas, openpyxl, Requests, GeoPandas, Shapely, pyproj, pyogrio, rasterio, rasterstats, pyxlsb, pycountry, xarray and netCDF4. The optional CDS downloader adds `requirements-cds.txt`. Tkinter normally comes with the Windows Python installation. GeoPandas installation guidance is maintained by its developers R26.

The launcher searches a local `.venv`, standard Python directories, Anaconda/Miniconda locations and a usable PATH interpreter. `DATA_PREPARATION_PYTHON` can identify another `python.exe`. It verifies the interpreter before starting the interface. Installing a package into a different Python does not fix the interpreter used by the application.

Check the complete default stack without opening a window:

```
.\START_DATA_PREPARATION.bat -Check
python preparation_dependencies.py
```

The preparation command and interface start action also check dependencies for the selected sectors. Complex industry alone needs fewer packages than raster-based category industry. Missing or broken imports produce an install command containing the active Python path before the tool starts source downloads or deletes old raw build files.

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4. First run and interface

Double-click `START_DATA_PREPARATION.bat`, or run `python data_preparation_gui.py`. The old shortcut remains compatible. Paths containing spaces and parentheses are handled by the PowerShell launcher. Startup information is written to `DATA_PREPARATION_STARTUP.log`.

4.1 General

Select a country code such as `IT` or `HR`, a comma-separated list such as `HR,SI`, or `ALL`. Choose the sectors, spatial resolution, output root and cache root. Enable e-fuel-compatible inputs when required by the intended model representation. Refresh forces source updates; leave it off when intentionally reusing cached observations.

4.2 Power and profiles

Choose aggregated units, individual units or both. Hydro profile resolution can follow power resolution or be selected separately. Choose weather/statistical years, Open-Meteo or Renewables.ninja, and IRENA calibration. Enter the ENTSO-E token if electricity demand is being downloaded. Skip controls deliberately omit the corresponding profile work; a successful plant-table run with skipped curves is not a complete power-data package.

4.3 Heating and advanced

Choose Open-Meteo temperatures or an existing/CDS NetCDF source, and optionally supply a population raster. Reusing raw heating files is useful only when their country, source year and geography match the desired case.

4.4 Output format

`generate` sources/prepares the selected data. `export existing outputs` repackages files already prepared. Select regular, category or complex industry and the conservative or technical-2050 electricity case. The portable templates directory supplies schemas and technology/policy assumptions. Packaging in H2RES format is optional; strict coverage can make an incomplete bundle return an error.

4.5 Industry sources

These controls apply to **complex** industry: JRC-IDEES-2023 source folder, calibration year, explicit sector selection, regional weights, missing-source downloads and waste-allocation policy. The source-folder field is not currently forwarded as the JRC-IDEES-2021 source directory for regular/category industry. That is a documented interface limitation, not evidence that category preparation used the selected local folder.

4.6 Input assumptions

Set planning years, renewable target and annual investment multiplier, or supply scenario JSON. These choices affect prepared future-year input/policy tables. Historical/statistical source years remain separate. `reference` retains supplied assumptions unless other explicit settings request changes. `res100` records the chosen renewable policy and requires e-fuel-compatible inputs for the whole-energy option.

4.7 Run controls

The command preview is the exact argument list passed to the runner. **Source and prepare data** starts it in a worker process. The run log combines output and errors. **Stop** terminates the tracked orchestration process; do not assume it can roll back downloaded or partly written files or every spawned child. Review outputs and rerun with matching settings. **Save settings** preserves selections; the ENTSO-E token is omitted. Legacy configuration is read only if the new configuration file does not exist.

5. Country selection and multinational downloads

The requested countries are passed to each sector generator. The shared map is restricted to those countries. Normal processing loops operate on that list, and `ALL` expands only when explicitly selected.

Country selection does **not** mean every upstream download is country-specific. The category industry path downloads the complete JRC-IDEES-2021 archive and SECURES archive, then finds the requested country's workbooks/profiles. The inspected archives contained 27 JRC countries and 29 SECURES demand-country codes. GISCO boundaries, EIGL rasters, the powerplants database and JRC hydro database also cover broad regions R06, R08, R09, R10, R11.

Transport unpacks national tables for many countries from bundled assets. Those folders are not separate internet downloads and do not imply that all countries were allocated. Its OSM downloader requests the chosen country extracts; the inspected Italy cache contained `IT.osm.pbf` only R13.

Complex industry uses separate public JRC-IDEES-2023 country archives and selects the Industry, EnergyBalance and EmissionBalance workbooks R07. Do not confuse this method with the category pipeline's older bulk archive.

Verify selection in the output, not by counting country names in a cache. Power rows use `System` and `Zone`; most other sector tables use `System` and `zone`. For national Italy these should be `IT` and `IT1`. Intermediate industry files can still use Italian NUTS3 codes before shared aggregation. The read-only checks performed during development found 1,073 Italian power rows and Italy-only industry/transport tables; this is evidence for those files, not certification of every future run.

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6. Spatial and temporal conventions

The shared geography is based on 2021 NUTS3 boundaries. The default CLI spatial level remains `nuts3`. Country resolution maps the selected country's NUTS3 units to a zone such as `IT1`. NUTS2/NUTS1 use the corresponding code prefixes. Custom maps have `country,nuts3,zone`; optional `system_new` permits a shared model system.

```
country,nuts3,zone,system_new
HR,HR021,Adriatic1,Adriatic
SI,SI031,Adriatic1,Adriatic
```

This is a schema example, not a complete territorial allocation. Every required source region must have an unambiguous target. Zone names must be globally unique unless the countries genuinely share the same modeled system/zone. A custom aggregate does not automatically create cross-border transmission, resource potentials or consistent policy targets.

The model calendar has 8,760 hourly positions, numbered 1–8760. Timestamped leap years drop February 29, preserving the rest of the calendar. Do not relabel an unexplained 8,759-row or duplicated-hour series as complete. For one-hour intervals MW and MWh/hour have the same numeric value, but capacity and annual energy remain different quantities. Source weather, source statistics and future decision years must be stated separately.

Country aliases matter: EU datasets commonly use `EL` where other providers use `GR`. Individual modules handle supported aliases at source boundaries; the model's System labels must remain consistent. Do not silently mix country-code conventions across tables.

7. Power inventory and electricity demand

7.1 Inventory construction

The power builder combines non-hydro rows from powerplantmatching with hydro rows from the JRC hydro database, avoiding a direct second inclusion of hydro powerplantmatching rows. It normalizes fuel and technology labels, assigns locations to NUTS polygons, and fills technical fields from supplied reference/default tables R09, R10.

It can exclude unsupported battery rows from this generation inventory and rejects missing/non-positive hydro capacities. The raw builder assigns unresolved border/outside locations using nearest spatial matches. For more detailed geographies, inspect these corrections: nearest placement is a spatial proxy. Capacity, location, decommissioning and commissioning status need a study-specific inventory audit. Downloading a fresh CSV does not prove that every row is an operating plant in the intended base year.

Plant parameters such as efficiency, remaining life and costs can come from template medians rather than individual observed plants. A Croatian donor parameter is a technology default, not a Croatian capacity observation being asserted for Italy. Preserve the distinction when reporting results.

7.2 Aggregated and individual outputs

The orchestrator builds a detailed inventory and both aggregation schemas. Aggregated units pool compatible technology/fuel rows by target zone. Individual mode retains plant rows and requires profiles keyed to those individual names. `both` provides the canonical chosen aggregation plus supplementary individual files. Never combine an individual inventory with only aggregated availability columns.

Main outputs are `power/GENCO.csv`, `power_inputs.xlsx`, `_genco_aggregation_schema.csv` and `_genco_individual_schema.csv`. The exported bundle renames the canonical inventory to `generators.csv`.

7.3 Electricity demand

The demand module requests actual total load from the ENTSO-E Transparency Platform by month using `documentType=A65` and `processType=A16`. Requests use the applicable national/control-area code and selected year. The token is obtained from the GUI/environment; it is not a data parameter to publish. API requirements and code lists are maintained by ENTSO-E R12.

Regional shares are derived from the JRC EIGL 2019 total-electricity raster. They allocate national hourly magnitude; the raster year is not the ENTSO-E load year. If A(t) is national load and w(r) are normalized spatial weights, then L(r,t)=w(r)A(t), and regional values reconcile to national load at every hour. Demand is subsequently aggregated through the shared map.

Do not subtract/add the same industrial or heating electricity twice. The prepared baseline and the model's endogenous sector electricity must obey the baseline convention documented in the model manual and the source-year assumptions of the scenario.

8. Hydro, wind and solar profiles

8.1 Hydro

Normal execution uses bundled plant runoff shapes, rather than deriving a new European hydrology model. The bundled `runoff_shapes_2020.npz` and supporting unit/quality tables therefore have their own provenance and year. The JRC database provides plant information R10; it is not by itself the source of all hourly hydrology.

Run-of-river curves are normalized to annual generation and capped by capacity. Reservoir inflows are scaled with generation/discharge assumptions and expressed using the model's capacity-normalized convention. Pumped-storage plants without natural inflow are treated as closed-loop. Several generator components matched to one hydro complex share its annual generation by MW to avoid duplicating the whole complex's inflow. Aggregate curves are capacity-weighted.

Read matching/quality outputs for missing profiles, nearest mappings, equivalent full-load hours and excluded units. Plant runoff shapes and inflow-derived generation are approximations; inflows, dispatch and reservoir stock have distinct physical roles.

8.2 Wind and solar

The default module requests Open-Meteo historical weather at plant sites and transforms radiation/wind into proxy PV/turbine capacity factors. Requests are cached by their parameters. Renewables.ninja is an alternative simulation backend and requires `RENEWABLES_NINJA_TOKEN` R15, R18.

These are weather-derived profiles, not measured plant output. Reanalysis information and temporal coverage are documented by the weather providers R16, R17. Profile uncertainty includes turbine parameters, hub height, PV orientation, irradiance conversion, weather resolution and missing operational losses.

IRENA calibration uses a local statistics extract. The target is annual generation divided by capacity and hours:

```
target_CF = generation_GWh × 1000 / (capacity_MW × hours)
```

A technology-level scaling factor is solved against the national capacity-weighted mean while clipping hourly factors to [0,1]. Scaling precedes aggregation. Record the exact statistics year and local workbook hash; do not cite the current IRENA website as proof of the values in a previously extracted workbook R19.

9. Regular and temperature-category industry

This path uses JRC-IDEES-2021 annual information, national SECURES demand shapes and spatial allocation helpers. It has its own reference years and should not be described as the JRC-IDEES-2023 complex method. The helper scripts are compressed into the standalone file and unpacked for execution; their geospatial/raster imports are real dependencies.

The allocation stage combines national sector/fuel observations with NUTS3 spatial weights and temperature bins. A complete new country-specific process-by-temperature cross-tabulation has not been rebuilt by the handoff integration. Inherited allocation rules and optional facility/EIGL evidence must be documented as such.

Regular mode aggregates delivered industrial services by sector. Categories retain temperature/service classes, including `NON_HEAT`, so electrical mechanical and other non-heat services are not lost from the total. Legacy process-heat-only views remain available for a different purpose and must not substitute for the full native demand file.

Source-energy allocation, assumed operating hours/reserves and technology efficiencies are used to derive equipment baselines. These derived MW values are not an observed industrial plant census. Equipment parameters and available future alternatives can come from portable templates.

The generator writes regular/e-fuel and category/non-category variants. The chosen representation is canonical `Industry_data`; exact `Industry` and `category` columns must match profile labels. Baseline electricity is exported separately. The bundle recalculates the pooled `simple_industry` remainder from excluded source services and demand weights; retaining old cached pooling formulas would risk double counting.

National aggregation should conserve each allocated service's demand and baseline equipment totals. Inspect the country `01_allocated`, `02_units_hourly`, `03_non_energy` stages before relying on combined results.

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10. Complex process industry

10.1 Observation extraction

The new compiler uses the supplied handoff's process-tree extraction and service-screening methods. The three country source workbooks are JRC-IDEES-2023 Industry, EnergyBalance and EmissionBalance R07. It traverses useful-energy, final-energy and emissions trees by indentation and matching paths, rather than adding every subtotal as a separate service.

It produces eleven native sectors: Iron_steel; Non_ferrous_metals; Chemical_petrochemical; Non_metallic_minerals; Paper_pulp_printing; Food_beverages_tobacco; Transport_equipment; Machinery; Textile_leather; Wood_wood_products; Not_elsewhere_specified_(industry).

Every observation retains source workbook hashes and sheet/cell addresses. Requested calibration year defaults to 2020. A branch with no positive activity in that year can use its latest positive observation; fallbacks are recorded and must be considered when comparing the aggregate with a single-year national total. Numeric formulas without cached results are rejected.

For a process/fuel stream:

```
useful_MWh = useful_ktoe × 11630
final_MWh  = final_ktoe × 11630
efficiency = useful_MWh / final_MWh
useful_share(f) = useful_MWh(f) / sum_f useful_MWh(f)
initial_capacity_MW(f) = useful_MWh(f) / 8760
combustion_factor(f) = observed_CO2_t(f) / final_MWh(f)
```

Observed aggregate useful/final ratios above one are retained; arbitrary unobserved combustion alternatives do not inherit a ratio above one. The capacity formula is explicitly an average-utilization assumption, not measured installed nameplate MW.

10.2 Product activity and process emissions

Physical output reported in kt is converted to tonnes. Output indices remain indices, using `_per_activity` intensities; the legacy `_per_t` fields are blank for index sectors. Useful energy is activity × intensity. Observed branch process emissions are assigned once to an emitter component, kept separate from fuel combustion. Capturable fractions remain assumptions.

Observed product activity is used through the final source year; afterward the last observation is held constant through 2050. This is an editable scenario extension, not a forecast produced from the internet.

The supplied material-feedstock convention maps legacy non-energy gas/oil services to H2-fed demand. Croatia's exact documented 2020 basic-chemicals ammonia-reforming term (535,320.11 tCO2) is retained in historical fields but excluded from additional process emissions under that convention. The exception is guarded by country, branch, source year and amount. Other-country process emissions remain unchanged pending their own evidence R23, R24.

10.3 Biomass and fossil waste

The source can group renewable biomass and fossil waste. Combustion CO2 remains the observed stream factor. Where matching balance evidence shows one fossil-waste class, its factor permits the exact energy split. Default `sector-blend` mode otherwise assumes that the observed sector fossil-waste blend applies to its processes and records that assumption. `strict` rejects ambiguous splits. Neither method silently labels observed fossil-waste emissions as renewable.

10.4 Alternatives and heat pumps

The inherited method screens substitutes by service and compatible incumbent fuel families. Material/feedstock services and identified reducing chemistry are protected. Biomass share growth is blocked; lignin is limited to a small local eligible solid-heat expansion. Equipment replacement options do not assert that the unmodified original burner or motor accepts every new fuel.

Conservative electrical screening preserves established electrical/mechanical/cooling/low-heat services and uses a partial additional envelope for unresolved furnaces; technical-2050 broadens eligible future equipment. The partial-coverage coefficient is a scenario rule, not a measured national adoption statistic. The supporting industrial-heat literature is listed in R20–R22 and the inherited bibliography.

Heat pumps have 60, 80, 125 and 150 °C equipment classes. Their COP, lifetime, cost proxies, cumulative sink-temperature budgets and source-heat fractions are assumptions. Shared budgets prevent adding the individual maxima as if each were independent eligible demand.

10.5 Regional allocation and cost proxies

National source observations need regional assumptions. Country resolution needs no weighting file. A regional CSV has `country,zone,Industry,weight`, with `Industry=*` as a common fallback. Every selected country/sector must cover its zones and sum to one. `--industry-regional-allocation equal` is an explicit equal-weight proxy.

Energy, emissions, physical output and finite national annual build budgets are apportioned. Activity indices remain unchanged and their intensities are apportioned instead. Blank limits remain unrestricted, and zero remains blocked. Cross-country merged zones use country-prefixed process keys to preserve distinct intensities/activity indices.

New component cost columns copy explicit generic equipment donor curves, preserving existing component-specific costs and unrelated workbook XML/cached formulas. This fills an input interface; it does not create a new equipment-price survey.

10.6 Outputs and reconciliation

Read `complex_generation.json`, `source_reconciliation.csv`, `industry_allocation.csv`, `normalized/*` and the selected complex workbook. National HR1 runs also write `reference_calibration_diff.csv`. Reconciliation covers useful energy, final energy, combustion CO2 and historical process CO2 by sector. The compiler uses inherited country hourly shapes rescaled to source totals, or records a flat fallback; it does not claim measured process-level hourly production.

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11. Transport

The transport preparer unpacks the national base tables and availability/activity curves supplied with the package, then allocates country service demand to NUTS3/target zones using spatial evidence such as energy-demand rasters and OSM transport indicators. Much of this national data is bundled; it is not freshly retrieved from a national statistics API on every run.

Road/rail/marine/aviation activity, vehicle counts, primary/secondary efficiencies, storage sizes and plug-in/V2G availability are different data types. `transport_demand.csv` contains activity shapes; `transport_profiles.csv` contains equipment/parent plug-in availability. The native model multiplies activity by annual demand/stock inputs. Do not treat a normalized activity profile as MW load.

GeoPBF downloads are country-specific. If the installed GDAL/OSM stack cannot provide rail, shipping or aviation indicators, documented allocation fallbacks may be used. An output can therefore be structurally valid while containing proxy spatial weights. Inspect warnings rather than interpreting absence of a traceback as complete infrastructure evidence.

The bundle preserves paid engine conversions and `compat_*` metadata from the technology templates, including primary-fuel recipes and parent availability profiles. Marine pilot fuel, blend ceilings, newbuild premiums and conversion lifetime/cost assumptions are scenario inputs. They are not a statement of current fuel certification or a calibrated Italian/Croatian fleet R25.

Source shipping identifiers are normalized to the current model contract: `ICE_MARINE_DOM` becomes `ICE_MARINE_SHORT` in `Domestic_shipping`, and `ICE_MARINE_INT` becomes `ICE_MARINE_LONG` in `international_shipping`. This preserves the source loads, counts and efficiencies. Normalization precedes generation of the domestic electric-vessel option and lookup of the model's paid conversion families. Cached hourly profile columns receive the same renaming when exporting older results. Unknown combustion-engine identifiers still cause an explicit compatibility error.

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12. Heating and cooling-related inputs

The heating preparer combines JRC residential/tertiary annual energy with population, weather/HDD and fuel-demand/density rasters. Hotmaps density informs district-heating allocation; EIGL layers inform fuel/spatial shares R11, R14. It then prepares hourly useful demand and equipment tables for individual and district heating, and baseline electric heating.

The default weather path uses Open-Meteo temperatures. The alternative is an existing or CDS-downloaded ERA5-Land NetCDF. CDS access requires the provider's current account/token setup and dataset terms; use the provider's instructions and placeholders rather than publishing credentials R15, R27.

Population weighting and degree-day allocation are spatial/temperature proxies. They do not identify every building, heat network or retrofit opportunity. Separate solar/DH/boiler/HP technology rows include template assumptions. The preparation application does not simulate the economic operation of those assets. Cooling input files retained in a bundle should not be presumed freshly recalibrated merely because heating was generated.

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13. Planning years, policy targets and investment defaults

Choose an increasing, equally spaced planning grid within available cost/input years. Current supplied templates cover 2020–2050; adding 2055 requires supporting data rather than extrapolation by label alone. The target year must be a modeled decision year after the start for the renewable preset.

`investment_multiplier=1` retains finite build limits; 2 doubles them; 0 sets finite limits to zero. It does not scale installed stock, all costs or total installed-capacity ceilings. Blanks and scalar fallback semantics remain intact. The model's runtime technology multipliers can additionally apply, so examine both layers.

The whole-energy `res100` bundle prepares the electricity RPS path plus integration metadata/constraints for fossil/nuclear energy, unverified imports and uncertified boundary storage at/after the target year. It is not a new historical dataset and is not the same target as net-zero CO2. Process emissions remain separately modeled. Actual enforcement belongs to the separately executed model adapter; copying CSVs alone does not insert its additional equations.

An example JSON is `scenario_defaults_res100_2050.json`. CLI fields override JSON fields. Per-row investment overrides use the native workbook's units; do not confuse MW/year, MWh/year, tCO2/hour or vehicle counts. Complex-workbook per-row overrides are not exposed as an independent GUI editor; check supported sheet names in `scenario_config.py` and edit the intended native complex inputs explicitly when necessary.

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14. Worked commands

Run from the preparation repository. These examples prepare data; they do not run H2RES.

14.1 Full Italy preparation

Set an ENTSO-E token in the current shell, or enter it in the GUI. Do not substitute an actual token into published scripts.

```
$env:ENTSOE_API_TOKEN = "YOUR_TOKEN"
python generate_all_sector_inputs.py --countries IT --sectors power,industry,transport,heating --spatial-level country --
efuels --industry-mode categories --out-root outputs_IT --data-root data_IT --require-model-ready
```

A new country still requires complete matching template/network/policy coverage. If strict validation fails, read the reported gaps; do not relabel the bundle as complete.

14.2 Plant-table-only check using cached sources

```
python power/generators/generate_genco_power_sector.py --mode raw --countries IT --out
outputs_IT/power/_genco_nuts3_detailed.csv --output-format detailed --nuts3-geojson
power/data/NUTS_RG_20M_2021_4326_LEVL_3.geojson --no-download-sources --no-clean-built-dir
```

This deliberately does not create all weather, hydro and electricity-demand profiles. `--no-clean-built-dir` preserves other previously built country files; the combine step still selects IT.

14.3 Croatian process-industry inputs and renewable assumptions

```
python generate_all_sector_inputs.py --countries HR --sectors industry --spatial-level country --industry-mode complex --
efuels --scenario-config scenario_defaults_res100_2050.json --out-root outputs_complex_HR --data-root data_all --require-
model-ready
```

Unselected sectors come from the portable reference. This demonstrates an industry replacement within that baseline, not a fresh all-sector statistical build.

14.4 Industry regional weights

```
country,zone,Industry,weight
HR,coast,*,0.4
HR,inland,*,0.6
```

Use a matching custom map and `--industry-zone-weights weights.csv`. The example fractions are illustrative assumptions.

14.5 Package already prepared files

```
python generate_all_sector_inputs.py --export-only --countries HR --sectors industry --industry-mode complex --efuels --
out-root outputs_complex_HR --require-model-ready
python validate_model_inputs.py --bundle outputs_complex_HR/model_inputs --industry-mode complex --efuels
```

The export variant must match the generated e-fuel/electricity case. Sector selection metadata comes from generation. Changing these assumptions should normally be followed by generation with matching flags.

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15. Output packaging and quality gates

Sector outputs remain under their own subdirectories. When enabled, `model_inputs/data` contains the native names and the selected technology/policy workbooks. `bundle_manifest.json` records selected sectors, country/zone coverage, template/source information, repairs, scenario settings and missing files. `validation.json` records structural/coverage errors and warnings.

An export starts from the template set and replaces selected sector inputs. Unchanged files/sheets can still contain other countries or old scenario assumptions. The `active_zones` sheet and each generated sector's coverage determine the selected case. A template file containing a country label is not proof that all its parameters are

calibrated for that country.

Coverage checks include 8,760 unique hours per active zone, matching unit/profile names, required columns, active-zone coverage, category/process keys and finite nonnegative investment limits. Exact repeated transport-demand rows in the supplied template are removed and recorded; conflicting duplicates must not be silently accepted.

`ready=true` means the implemented input checks passed. It does not establish optimality, future demand realism, complete scientific provenance, lifecycle sustainability, security of supply or feasibility of a renewable policy. Actual-reader and solver tests are separate evidence levels.

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16. Caching, repeatability and disk use

Keep the same cache root to reuse downloads. A new root is a new cache, even for the same country. Refresh can redownload large complete archives, and extraction can consume substantially more disk space than the ZIP. Some power raw data/build caches are module-global under `power/data` and `power/generators`, rather than fully isolated by `--data-root`; avoid concurrent runs that clean/write those same locations.

Record source release, retrieval time, exact request parameters, workbook/file hashes, statistical/weather years, country list, map and template version. Master-branch CSV endpoints change over time. Store the actual source snapshot used in a published study. Cache reuse improves repeatability but does not prove the cached data are the latest release.

Excel cached values matter. XML-level replacement preserves unrelated members/caches. An uncached numeric formula cannot safely be read as zero. Recalculate a source workbook in a compatible spreadsheet application where necessary, preserving the original as provenance.

---

17. Troubleshooting

Symptom	Likely cause and response
Window closes before Python starts	Use the repaired PowerShell-backed launcher; inspect startup log, especially paths with parentheses
`No module named geopandas` or another package	Install `requirements.txt` with the exact interpreter printed in the log; restart the interface to load new checks
Many country files appear in cache	Bulk archives or unpacked bundled tables; inspect generated System/zone fields
Same large archive downloads again	Refresh enabled or a different cache root
Local JRC-2021 folder seems ignored	Industry-source-folder GUI control applies to complex JRC-2023, not the current category runner
Complex regional run refuses to proceed	Supply valid country/sector weights or explicitly select equal allocation
No SECURES profile for a country	The source coverage differs by country; `ALL` is not proof that all four pipelines can run for every code
Missing ENTSO-E token/invalid API response	Verify provider access, code/year and token without printing credentials

Symptom	Likely cause and response
OSM fallback warnings	Installed reader cannot extract the intended infrastructure; inspect allocation weights and install/validate a suitable reader if required
`No current primary-fuel compatibility mapping for ICE_MARINE_DOM` (or `ICE_MARINE_INT`)	Older shipping names reached the export step. The current preparer normalizes both fleet rows and profile columns. Re-export existing results with matching settings and `--export-only`; use the transport script's `--skip-allocation` first to recreate its full option set from cached NUTS3 tables. Heating success messages before this error indicate that heating finished; the export subsequently failed.
`ready=false` after sector generation	Native input coverage is incomplete; read errors, generate missing profiles/sectors and provide missing country policies/networks
Complex workbook/export mismatch	Regenerate with the selected e-fuel/electricity variant
Source formula has no cached value	Recalculate the source workbook; do not fill zero automatically
A renewable target later proves infeasible	Review resources, build rates, fixed processes, trade/origin assumptions and turnover limits in the separate model

---

18. Validation and reproducible reporting

The dependency/preparation regression suite passed 45 tests after the shipping-identifier correction. The Italy recovery check is recorded in `validation/transport_shipping_recovery.json`: source shipping loads/counts/efficiencies, paid conversions, primary-fuel coverage and hourly keys were checked using the actual cached files. That export still reports a missing electricity-demand profile when its download was skipped. Process-industry verification additionally reconciled Croatian calibration and checked Croatian/Hungarian inputs through the actual model adapter; the full Croatian reader was exercised. Italy's plant-table pipeline generated 1,073 individual rows. These are bounded tests documented in local `validation/*.json`; they do not certify a complete national optimization.

Run relevant checks after a change rather than treating this count as permanent:

```
python -m unittest discover -s tests -v
python preparation_dependencies.py
python validate_model_inputs.py --bundle outputs_IT/model_inputs --industry-mode categories --efuels
```

A publication should report the archived input bundle and hashes, command/settings, assumptions, source years, coverage/fallback warnings and a separate model-solution quality report. Cite the datasets used and the model formulation, not just the preparation interface.

H2RES model manual

Documented model: `D:\POSAO\H2RES_1.11.2026`. **Documentation edition:** 1 November 2026. The source/version inventory identifies the exact files reviewed. The model is separate from the data-preparation interface.

---

1. Model purpose, lineage and scientific use

H2RES is used to study integrated energy systems and transition pathways across electricity, heat, transport, hydrogen and other energy carriers. Its historical island-energy modelling lineage and the later optimization formulation are distinct developments. The 2004 island study and the 2022 planning formulation are appropriate background references; the newer local code contains additional processes and workflows that those articles do not document R01, R02.

The formulation examined here is a linear investment-and-operation model implemented in Python with Gurobi. The static inventory of active runtime source files found continuous variable declarations, including the `_add_vars` wrapper. Vehicle numbers may therefore be fractional optimization quantities; they are not automatically integer purchase decisions. A fuel called a binary choice in ordinary language does not imply a binary solver variable. Confirm the constructed model if additional user modules are added.

The model minimizes a configured economic objective subject to physical, stock, investment, service, network, resource and policy constraints. It can help compare pathways under alternative assumptions. It does not discover future policy targets, determine a unique forecast, prove causation in a real market, or by itself establish full lifecycle sustainability. Published H2RES applications are examples of model use, not a validation certificate for this source snapshot R03, R04.

---

2. Architecture and execution flow

2.1 Main runtime files

File	Responsibility
`main.py`	Working directory, scenario name, feature flags, file selection, caching, planning/dispatch execution and exports
`policy/model_policy.py`	Entered economic, transition, resource and policy assumptions
`policy/runtime_policy.py`	Numerical tolerances, sampling, solver and recovery settings
`policy/transport_pathways.py`	Transport conversion/compatibility assumptions and default migration values
`scripts/read_process_data.py`	Reads/scopes input tables, constructs sets/dictionaries, annual limits and processed demand
`scripts/reading_prev_and_vars_def_dual_industry_categories1.py`	Loads historical decisions/stocks and declares optimization variables
`scripts/Build_model_dual_industry_categories_named_debug.py`	Core physical/policy constraint construction
`scripts/CAPEX_OPEX_Gurobi_settings_dual_industry_categories1.py`	Economic expressions, objective, solver configuration, optimization and recovery

File	Responsibility
`scripts/complex_industry_support.py`	Validates/adapts process activity and replaces selected ordinary industry services
`scripts/industry_annual_s_curve.py`	Demand-weighted adoption envelopes and shared electric-service transition policies
`scripts/transport_pathways.py`	Fuel compatibility, paid conversion pathways and stock/energy links
`scripts/energy_accounting.py`	Shared CHP fuel-equivalent and refinery physical accounting
`scripts/physical_solution_audit.py`	Physical-unit checks for solutions and history
`scripts/run_stage2_rolling_dispatch.py`	Rolling operational checks with fixed stage-1 capacities
`scripts/export_plot*.py` and `combine_plot.py`	Investment/history exports, balances, summaries and visualizations

Scripts are executed in a shared namespace through `exec_script`. A variable can therefore be created by a reader/helper and consumed later without a conventional function argument. Configuration provenance must include the imported policy modules as well as `main.py`. Static code review must account for this execution pattern.

2.2 Runtime sequence

```
main + three policy modules
    -> select technology/policy/profile files
    -> read and normalize inputs
    -> choose planning/dispatch mode and time grid
    -> load previous accepted investment/stock history
    -> declare variables
    -> build balances and constraints
    -> define costs/objective and solve
    -> check solution/status/physical quality
    -> export accepted results and historical handoffs
```

A separate data-preparation run stops before the model sequence above. Its optional bundle launcher configures the separate model in memory, redirects data paths and inserts the whole-energy renewable constraints before solving. The original model's `main.py` does not automatically read the preparation GUI settings.

---

3. Exact configuration and precedence

`main.py` changes its working directory to its own model directory, then executes the three policy modules in its namespace. It subsequently sets feature flags and derives file names. Reader/workbook settings supply additional numerical inputs. A preparation-bundle adapter can override selected flags and data paths at its supported execution boundary. The resulting precedence is contextual rather than a single universal JSON configuration layer.

The configuration reference lists source assignments and line positions. Values described below are the examined local snapshot, not universal H2RES defaults. Historical README files include other scenario names and mode selections.

3.1 Observed local selection

The reviewed `main.py` uses scenario name `industry_testx`, `Myopic=True`, `partial_perfect_foresight=True`, heating and transport/industry enabled, e-fuels enabled, category industry enabled and complex industry disabled. Category electricity is `conservative`, industrial category storage is enabled, and industry fuel-share/S-curve controls are enabled. Solar individual/DH heating and cooling switches are disabled in this snapshot; model availability and a currently enabled option are different matters.

The snapshot also has transport pathway corrections/electrification/share ceilings enabled, standalone H2 transmission and standalone terminals disabled, and `disable_transport_infrastructure=True`. The latter is labelled a debugging setting in the source: do not describe EV/FCEV infrastructure investment as active in a study using that setting.

3.2 Mutually exclusive or dependent modes

Category and complex industry are alternatives. Complex mode requires transport/industry and cannot use the legacy single-sector industry option. Category industrial storage requires category mode. Standalone hydrogen pipelines are disabled when e-fuels handle carrier transmission. Standalone H2 terminals use a separate non-e-fuel implementation. Oil/gas product supply depends on e-fuels. These guards are executed, not merely suggestions.

The model policy defaults complex sector selection to `Non_metallic_minerals`; preparation may generate all eleven sectors. An exported selection must be applied explicitly in the model configuration or supported adapter. Otherwise a workbook containing all sectors can still run only the selected subset.

3.3 Input workbook selection

Technology file selection follows regular/category/complex and e-fuel flags. Category e-fuel conservative mode selects `flex_tech_efuel_categories_electricity_review.xlsx`; its technical alternative selects the corresponding `_electricity_technical_2050.xlsx`. Complex electricity variants change the complex workbook name. The retained legacy combined category/complex workbooks are not a supported simultaneous mode. See data reference for the exact mapping.

---

4. Spatial and temporal representation

Let s denote a model System, z a zone belonging to it, y a modeled decision year, t an operational period, g a generation asset, k another technology, f a carrier, i an industrial service/process and a an investment vintage. Assets and flows are created for active mappings; country names elsewhere in a workbook do not imply active modeled systems.

The full input calendar is normally 8,760 hourly positions. Operational subsets and representative weights depend on mode. A constraint using `_periods_for_year(y)` need not cover the same periods in every optimization year. Temporal metadata must be exported with results.

4.1 Perfect foresight

With `Myopic=False` and the applicable mode switches disabled, the model jointly chooses investments and operations across the modeled horizon. It knows the deterministic future input trajectories provided to it. Perfect foresight is an information assumption, not a claim that actual planners possess those forecasts.

4.2 Myopic planning

With `Myopic=True`, overlapping windows are built from the planning grid and setup controls. The model solves a window, accepts/exports its decision-year results, and carries historical decisions into later windows. Already accepted years are not reoptimized as freely as future years. New scenario names and coherent histories are necessary after material changes to costs, units or policies.

4.3 Partial perfect foresight

This mode combines detailed current-year operations with sampled future-year operations in each myopic solve. Future samples guide current investment, but they are not an independently validated full-year dispatch for those future years. The current source uses explicit per-year period sets and sparse variable creation. Read the configuration snapshot for representative-day settings; do not assume all years in a chunk contain 8,760 variables per asset.

4.4 Representative capacity planning and two-stage operation

Representative-capacity mode can sample configured weeks, with weights representing the full year. Lightweight mode exports only the capacity stage. Two-stage mode can subsequently export a fixed-capacity dispatch bundle and perform rolling operation. These approximations have different abilities to represent storage chronology, seasonal scarcity and ship arrivals. A representative-week optimum does not prove that a full chronological year is feasible.

4.5 Rolling dispatch

Rolling mode advances through operational windows with capacity fixed from a stage-1 bundle. It carries stocks, relevant fuel/terminal states and guidance into subsequent windows. Resume and stitching options can reuse accepted window files. Review terminal conditions, guidance penalties and automatic slack settings; they can relax a pure hard-feasibility interpretation.

---

5. Mathematical notation and decision variables

The following equations summarize implemented families. Asset-specific dimensions, history maps and flag branches are suppressed. The code reference and machine-readable call inventories give executable locations and conditional context.

Symbol	Meaning	Typical physical basis
X(k,z,s,y)	New capacity decision	MW, MWh, vehicles, tCO2/h or another explicitly declared asset unit
K(k,z,s,y)	Available installed capacity	Same unit as the asset's capacity
P(g,z,s,t,y)	Electricity generation	MW for one-hour periods
Q(k,z,s,t,y)	Delivered heating/industrial service	MW useful heat/service
U(f,k,z,s,t,y)	Carrier/fuel input	MWh of carrier per hour
E(k,z,s,t,y)	Storage stock	MWh, tCO2 or other carrier stock unit
C,D	Storage charging/discharging flows	MW or corresponding hourly carrier flow

Symbol	Meaning	Typical physical basis
N(v,z,s,y)	Transport stock	Continuous vehicle-equivalent count in the LP
F(f,z,z',t,y)	Carrier transfer	MW-equivalent flow
H(i,z,s,t,y)	Industrial delivered-service demand	Useful MWh per hour
A(b,z,s,t,y)	Product activity	Tonnes or index-activity units per hour
w(t,y)	Representative period weight	Represented hours

Variable bounds and scales differ by family. Scaled transport variables have physical-value wrappers; numerical solver column values must be converted back before comparing with raw vehicles, stock or energy data. State and investment symbols in this manual denote physical quantities.

---

6. Economic objective and investment accounting

6.1 Objective implemented in this source

The production objective has the following structure, in million currency units before an explicit solver scaling factor:

```
min  sum_(y,z,s) DF(y) × [CAPEX_year(y,z,s)
                        + multip × fixed_OPEX_year(y,z,s)
                        + variable_OPEX_MEUR(y,z,s)]
     + rolling_carbon_slack_penalty
     + rolling_terminal_fuel_SOC_slack_penalty
     + stage2_quantity_guidance_slack_penalty
```

`DF(y)=(1+discount_rate)^(-(y-discount_base_year))`. The reader defines `multip=(max(years)-min(years)+1)/len(years)` for its solve grid. Variable-OPEX expressions also apply the configured annualization/multiplier and EUR-to-MEUR conversion. Those exact conventions matter on sparse decision-year grids. They must not be replaced in a report by an unexplained sum over every calendar year or by a different textbook objective.

The examined policy has discount rate 0.05, discount base year 2025, capital finance rate 0.05 and numerical objective scale 1000. Numerical scaling is intended to change conditioning, not physical economic units. Read exported objective/cost variables consistently with that scale.

6.2 Capital service and vintages

The cost script treats capital service as an annuity-due payment for each investment vintage. The first payment is in the build year, and payment continues while that vintage is within technical life. Its unit capital cost is the build-year cost, not each later year's learning-curve price. Payments after the modeled horizon are omitted under this accounting convention.

```
annuity_due_factor(L,r) = 1 / sum_(n=0 to L-1) (1+r)^(-n)
annuity_due_factor(L,0) = 1/L
payment(vintage a, year y) is due when a <= y < a + lifetime
```

Costs may be MEUR per MW, per MWh of stock, per vehicle, per tCO2/hour or per terminal TWh/year. Applying one unit cost to another basis can dominate the result while leaving the LP numerically feasible. The live source includes literature-linked annual fixed-OPEX data and generic/fallback technology mappings. Matched rows and fallback paths should be audited separately.

6.3 Operating costs, trade and penalties

Variable OPEX includes the appropriate fuel/commodity purchases, non-fuel operation, ramping, external trade and carbon charges, with special accounting for e-fuel recipes. Do not add a fuel bill a second time merely because its energy passes through several conversion assets. Revenue signs depend on import/export terms and market configuration. The code contains separate national/external-carbon options and fuel-specific prices; study documentation must identify the active option.

Unserved electricity, rolling heat/service shortages and selected guidance or carbon shortfalls can be penalized. These are scarcity/slack representations, not real zero-cost supplies. The configured electricity unserved-load cost is a scenario penalty, not a nationally estimated VOLL. A low objective obtained with nonzero slack must be reported as such.

---

7. Capacity, lifetime, decommissioning and build constraints

For an asset class, the general available-capacity relation is:

```
K(y) = surviving_existing_capacity(y)
       + sum_a survival(a,y) × X(a)
```

Survival/retirement is asset-specific. Existing capacity uses remaining-life/decommissioning inputs; new vintages use technical life and the implemented retirement shape. Some families retain a final-life fraction. The exact helpers and history conventions should be used when interpreting the asset's investment CSVs.

Capacity may have a total ceiling and an annual build limit. They are different restrictions:

```
X(y) <= sum_(c in investment_window(y)) annual_build_limit(c)
K(y) <= total_installed_ceiling(y)
```

Investment windows use the global decision-year grid. For endpoints 2025 and 2030, the first endpoint represents 2025 only and the next represents 2026–2030, including on resume. Blank annual limits use the legacy scalar if present; absent/blank scalar means unrestricted. Zero disallows additions but does not delete existing capacity. A period with an unrestricted year remains unrestricted under the relevant industry-limit reader. Negative/nonfinite supplied limits are rejected.

Generation-specific runtime build multipliers can further scale workbook limits. In the reviewed policy four Croatian wind/PV names have multiplier 2; that is a scenario configuration, not a universal country potential or model constant. Transport, terminal, capture and storage additions use their native physical units and may have separate active flags.

8. Electricity generation, networks and storage

8.1 Dispatch and renewable availability

Dispatchable output is bounded by available capacity and applicable technical constraints. Renewable output uses matching availability factors. Hydro run-of-river has flag-dependent equality/inequality behavior in the source. Curtailment or excess electricity is represented where enabled; do not assume every renewable MWh available must be delivered.

The principal electricity balance includes generators, fuel cells, relevant storage discharge, vehicle-grid services, net imports and an explicit unserved-load variable. Demand includes baseline electricity plus endogenous heating, industry, transport, electrolysis, synthesis/CC and storage charging, adjusted under the code's baseline conventions.

```
generation + fuel_cells + discharge + delivered_imports + unserved_load
 = baseline_load + endogenous_sector_electricity + charging + exports + excess
```

The exact terms and signs are in `el_balance`. This is a zonal energy balance, not a full nonlinear AC network. Do not infer voltage, frequency, transient stability or N-1 security from hourly energy feasibility alone.

8.2 Network transfers

Primary/secondary links, outside links and carrier networks use input mappings, transfer capacities, losses and costs. One delivered-flow conversion appearing in the electricity balance uses `1/(1+distance_multiplier*loss)`; substituting a different loss formula in a methodology description would misstate this implementation. Outside-system availability can restrict imports/exports using hourly supply-share/curve tables.

Transmission duplicates can be resolved by the configured `first` policy. The log reports removed rows and conflicts. A deterministic rule does not prove that conflicting network data are correct. New country/system aggregation requires deliberate network and outside-trade calibration.

8.3 Electricity and generic storage

An illustrative physical stock relation is:

```
E(t) = retained_stock(t-1) + eta_charge × C(t) × dt
                           - withdrawal_for_discharge(t) × dt + natural_inflow(t)
0 <= E(t) <= storage_capacity
0 <= C(t) <= charge_power_limit
0 <= D(t) <= discharge_power_limit
```

Efficiency may be applied to the stock transition or to the delivered balance depending on family; the source is the authority. Representative weights used for annual accounting are not automatically multi-hour storage transition durations. Initial/final/cyclic conditions can materially change apparent flexibility. The reviewed policy has `storage_year_continuity_mode="cyclic"`, with additional rolling/fuel boundary controls.

8.4 Hydro reservoirs and pumped storage

Hydro includes plant-specific turbine capacity, storage, inflow and pumping where applicable. Water-energy stock balances and available generation are constrained separately. Pumped storage consumes electricity when charging; it is not a new primary renewable source merely because its fuel label is Hydro. The source/profile year and reservoir assumptions need to accompany any annual generation interpretation.

---

9. Buildings, district heating and CHP

The heating block meets individual and district-heating service demand using enabled fuel boilers, electric heaters/heat pumps, solar heat, CHP and storage. Each class has capacity, efficiency/COP and investment/retirement constraints. Cooling is a separate optional representation and is disabled in the examined selection.

For a boiler or heat pump, the conceptual conversion is `useful_heat = efficiency_or_COP × input_energy`. COP above one represents heat transfer from an external source. It is not electricity generation. Heat-source constraints, DH integration and storage options must be accounted for when comparing technology outputs.

CHP is coupled production. The shared physical helper uses condensing-equivalent electricity `P + beta*Q`; its fuel input divides that quantity by electrical efficiency. That expression is used consistently for the relevant costs and emissions. CHP feasible operating regions additionally use input ratios/parameters. Treating CHP heat as cost/emission-free while retaining the full electric output would double-allocate the fuel benefit R30.

Current heating policies include base-year plant heat calibration, annual limits on CHP heat change, electric/gas heating transitions, fossil-heat non-increase and an explicit boiler phaseout. The reviewed entered phaseout years are coal 2030 and gas/oil 2035; the source notes that CHP is excluded from that boiler rule. These are local scenario settings, not automatic national legislation. Solar and storage options remain subject to their switches.

---

10. Industry

10.1 Ordinary and category formulations

Industrial equipment supplies useful service while carrier demand depends on efficiency. Regular mode uses sector/service keys; category mode adds temperature/service categories. `NON_HEAT` is retained where applicable. Excluded detailed services can be represented in a pooled remainder with its own input row. The remainder must be consistent with selected detailed demand to avoid omission or duplication.

Initial useful-service shares, maximum shares, capacity and annual build limits interact. A maximum share is applicability, not a mandatory target. An investment ceiling does not change a fuel's compatibility with the process. An asset can be technically eligible but unable to expand quickly because turnover or adoption constraints bind.

10.2 Process/activity formulation

Complex industry replaces ordinary demand/technology rows only for its selected services. It uses branch product activity and component intensities:

```
useful_service(i,t,y) = product_activity(branch(i),t,y)
                        × activity_factor(i) × energy_intensity(i)
process_CO2(branch,t,y) = sum_i product_activity(branch,t,y)
                                × activity_factor(i) × process_intensity(i)
```

Output can be tonnes or an index. The generic per-activity fields are authoritative. Process emissions are not added again as combustion of a non-energy feedstock. The adapter validates selected technology/process/product coverage, activity units, annual product demand, positive-service shares and heat-pump eligibility fields.

The supplied material convention treats non-energy gas/oil labels as H2-fed demand. That is a model convention rather than a universal chemical process representation. The Croatian ammonia exception and non-HR unresolved reclassification are documented in the preparation manual and assumptions register.

10.3 Fuel substitution budgets

The method distinguishes common low-enthalpy/steam/drying heat, process-specific heat, electrical/mechanical/cooling services, material feedstocks and reducing chemistry. Compatible substitute families include H2 and enabled synthetic fuels, with local/shared limits so the same replaceable pool is not repeatedly allocated without restriction. Biomass has a baseline-share ceiling, and lignin expansion is constrained to at most two additional percentage points of eligible local service in the supplied screening method.

These are replacement-equipment assumptions. They are not certification of blending in an unchanged incumbent burner. The authoritative latest fuel-compatibility handoff supersedes older zero-ammonia/narrow-hydrogen/unlimited-lignin comments R20, R21, R22.

10.4 Adoption trajectories and S-curves

Demand-weighted annual fuel/electric-service adoption constraints are separate from hourly dispatch. Technology switching can be optimized hour by hour within the accepted annual transition and applicability envelope. The electric service share combines direct electric equipment and industrial HP service; it is not simply their electrical input fraction.

The reviewed S-curve policy uses an endogenous envelope with initial fraction 0.02, growth parameter 0.50 and reference milestones every five years. Shorter model steps interpolate toward those milestones so seeds/tolerances are not restarted every year. Directions can be explicitly increasing/decreasing by fuel, unit or asset tuple; gas/coal/oil/biomass directions are entered policies. These parameters are scenario choices, not nationally measured adoption rates.

10.5 Industrial heat pumps and storage

Industrial HP units share cumulative temperature and source-heat budgets. If eligible shares below 60, 80, 125 and 150 °C are cumulative, their sum is not independent addressable demand. The builder also respects source and service restrictions. Category industrial heat storage is a separate optional asset family; enabling it outside category mode is rejected.

11. Transport and fleet transition

Annual transport service is met by stocks of ICE, hybrid, electric and fuel-cell equipment in road, rail, shipping and aviation classes represented in the selected workbook. Service/activity profiles, fuel input, stock, battery availability and infrastructure constraints have different units and purposes.

The stock identity is conceptually:

```
stock(y) = surviving_legacy_stock(y)
           + surviving_new_stock(y) + accepted_conversions_in(y)
           - retired_or_converted_out_stock(y)
```

Calendar aging and historical stock are explicit. The reviewed transport settings include turnover periods/ramp policy and maximum technology service shares. Some investment/rail-link/infrastructure switches are disabled in the local snapshot. Document those switches before interpreting an optimized transport pathway as an infrastructure plan.

ICE/hybrid primary-fuel recipes and paid upgrades use `compat_*` metadata. Drop-in hydrocarbons, FAME, alcohol blends, marine methanol/ammonia and pilot fuel have distinct configured paths. The source's default volume-share ceilings and retrofit premiums are screening assumptions; this documentation makes no claim that they constitute current legal approvals. Marine conversion parameters drawn from container-vessel literature are transferred to model proxy classes and require case-specific calibration R25.

EV plug-in availability constrains charging/V2G opportunities. Minimum battery state and activity-derived travel energy restrict storage use. The model can couple EV batteries to the power system, but a country-level normalized profile is not a location-specific charging-network observation.

Transport solver scaling is handled by physical wrappers. Very small fractional legacy stocks and saved residuals can cause degeneracy or infeasible histories. The code cleans tiny numerical residue using configured tolerances and checks physical history; changing tolerances can affect acceptance and should be reported.

12. Hydrogen, electrofuels, feedstocks and terminals

12.1 Hydrogen

Electrolyzers consume electricity according to conversion efficiencies and available capacity. Steam reforming can use gas with associated costs/emissions. Fuel cells convert hydrogen back to electricity. Zonal H2 service includes applicable transport/industry demand, conversion/feedstocks, storage and network exchanges. The active balance and supply switches must be enabled together for a meaningful carrier accounting.

12.2 E-fuel conversion recipes

The e-fuel representation includes technology rows and recipes for enabled carriers such as H2, ammonia, methanol, synthetic methane, diesel, kerosene and other workbook fuels. Electricity, heat, CO2, nitrogen and upstream carrier inputs are coupled through the recipe fields. Generic form:

```
input_j(k,t,y) = recipe_coefficient_j(k) × production(k,t,y)
production(k,t,y) <= available_conversion_capacity(k,y)
```

Energy-carrier inputs and material inputs have different units. Nitrogen and CO2 must not be counted as energy simply because they appear in a feedstock table. Recipe coefficients and output energy bases require explicit checking; do not infer a balanced reaction solely from labels.

Fuel inventory states, conversion capacities, commodity purchases and trade can interact. The current code has separate initial stock and boundary/continuity options. An initial storage fraction is an assumption about available inventory, not a free replenishable primary-energy resource.

12.3 Oil and gas product supply

An optional e-fuel-associated block represents extraction/imports, crude refining and product supply. The shared helper accounts for product energy and a configured refinery process-fuel share. The reviewed share is 0.08. Whether fossil products may be exported is separately configurable and can materially change economic incentives. Assess exports and process fuel in both energy and carbon balances.

12.4 Standalone H2 terminals

The separate terminal module is enabled in non-e-fuel mode by its terminal/H2-balance switches. Inputs come from `h2_terminals.xlsx`, with asset and shipping-route tables. Capacity is annual TWh/year, converted to an hourly limit using 8,760 hours. Loading, unloading, imports and exports share capacity; losses and handling electricity are charged on the appropriate flows.

This is aggregate delivered hydrogen. It does not separately model every port tank, vessel, liquefaction or ammonia-cracking process. Local H2 storage supplies buffering. Carrier-specific project calibration therefore belongs in the aggregate assumptions or an explicitly extended model.

Detailed internal shipping has departure schedules, delay, cargo size and fleet occupancy. Representative planning can use an instantaneous/average-capacity approximation. The source prevents departures that would arrive beyond the annual boundary; such cargo does not automatically cross calendar years. The terminal's standalone representation and e-fuel terminal representation should not be conflated.

---

13. Carbon accounting, capture and renewable policies

13.1 Emissions scope

The model separately accounts for power/CHP, heating, industry, transport, reforming and refining emissions where active. Combustion coefficients, process CO2 and capture/storage flows are not interchangeable. The physical electricity/CHP helpers and stream-specific industrial emission factors avoid particular double-counting errors, but a complete study still needs a stated system boundary.

The ordinary budget has the conceptual form:

```
sum(enabled sector emissions) - credited geological_storage
 <= policy_CO2_limit + permitted_LULUCF_removal
```

Exact coefficients/scaling and alternate branches are in the carbon-limit rows. LULUCF can be disabled or parameterized. The code's CO2 energy-system accounting is not a comprehensive product lifecycle, upstream methane, land-use, N2O, ammonia-slip or biodiversity assessment. `non_CO2_polutants` is explicitly marked incomplete.

13.2 Capture and geological storage

Capture capacity, rate, heat/electricity demand, reboiler capacity and optional intermediate storage impose technical and economic constraints. `model_CC`, post-generation CO2 storage and e-fuel CO2 feedstock are different switches. One can retain parts of the e-fuel CO2 handling even when a broader CC setting is disabled.

An optional separated industry-process/other-carbon representation restricts which captured streams can enter geological storage. It is disabled in the reviewed snapshot. Untagged imported CO2 does not automatically become a certified process stream. Reuse of CO2 in e-fuels should not be described as permanent sequestration.

13.3 Renewable electricity share

The native RPS policy constrains qualifying electricity generation/share according to the source's numerator/denominator definitions. It is not a universal all-energy renewable fraction. Reservoir/pumped-storage treatment, imports, curtailment and generation calibration must be interpreted consistently with the actual equations.

13.4 Whole-energy renewable preset from preparation

The preparation adapter adds stricter endpoint restrictions for the selected `res100` / `all-energy` scenario. At and after the target, it blocks fossil/nuclear energy use in covered generation, heating, industry and transport, relevant reforming/refining flows, unverified outside energy imports and imports through routes whose endpoints are outside the active model geography. It also prevents unserved electricity from serving as a zero-origin substitute.

Because generic stores lack a renewable-origin tracer, affected initial stocks are cleared at the target boundary; subsequent internally renewable stocks may carry forward. Material CO2/nitrogen remain separate. Mixed fossil-waste industrial streams are blocked under the conservative origin treatment. Earlier non-electric transition is chosen subject to the other policies rather than a claimed linear whole-energy renewable trajectory.

This policy is an energy-origin definition, not net-zero CO2. Biogenic sustainability, upstream lifecycle burdens and independent process emissions require separate decisions. Its current launcher supports the standard myopic/perfect-foresight workflows and rejects the unsupported whole-energy two-stage combination. Feasibility may be prevented by fixed processes, feedstock conventions, build rates or resource limits.

---

14. Policy calibration and scenario design

Scoped policy workbooks can contain generation targets, hydro caps, transport targets, biomass caps and outside trade calibration. Rows are selected by active System/zone and modeled years. Some calibration applies only to the global base year; resource caps can apply in every year. Calibration targets are not all equivalent to future investment opportunities.

Before creating a new national study, review demand forecasts, technology prices, currency/base year, lifetimes, reserves, resource potential, fuel/commodity availability, network/outside trade, heat-share policies, transport turnover, industrial compatibility and process routes. Automatically sourced historical data do not determine those future scenarios.

Use one coherent baseline and document each changed assumption. A suggested sensitivity matrix varies demand growth, resource potentials, grid expansion, storage chronology, external trade, capital/operating costs, industrial transition rates, biomass eligibility and renewable-origin definitions. Optimized technology shares should always be interpreted conditional on this matrix.

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15. Preparing and running a study

15.1 Environment and input validation

The separate model needs a suitable Python/Gurobi environment. Its supplied local reference lists Python-tested numerical packages and `gurobipy==12.0.3`; these are recorded local versions rather than a requirement that another compatible release is scientifically identical. Large national models need appropriate solver licensing and sufficient memory. No license credentials belong in published inputs.

For a native model run, work in the model's own folder, set a fresh scenario name and review the active flags, policies and file names. `python main.py` executes that model, including optimization; it is not a data-preparation command. Run an input-reader-only check before a costly solve, using a supported validation harness.

For a preparation-generated bundle, read `bundle_manifest.json`, `validation.json`, `scenario.json` and process-generation metadata. The generated integration launcher points at an existing model installation. Its default is input-reader-only; `--solve` explicitly requests construction/optimization. The preparation GUI does not invoke it.

15.2 Separate integration example

These commands belong to the model integration workflow, after preparation:

```
python outputs_complex_HR/model_inputs/run_model_scenario.py --model-root "D:\POSAO\H2RES_1.11.2026"
python outputs_complex_HR/model_inputs/run_model_scenario.py --model-root "D:\POSAO\H2RES_1.11.2026" --solve
```

The first command reads inputs. The second builds and solves, using the configured model workflow and a fresh scenario name. Do not substitute a stale bundle merely because filenames exist. Adapter compatibility is version-specific; inspect model/source hashes when moving to another release.

15.3 Resume and historical decisions

Set resume options only for coherent, accepted results. Investment handoffs, fuel/vehicle/heat histories and terminal/stock files must belong to the same input convention and scenario. Changing costs, unit names, decision grid, physical scaling or technology representation while reusing histories can produce mathematically valid but inconsistent comparisons or an infeasible model.

A partial foresight run exports the accepted decision year, not every sampled future-year solution as a committed history. Failed/provisional outcomes need their quality gates before they can be used as later stock constraints. Preserve the exact logs and history files when investigating a transition failure.

16. Solver behavior and numerical quality

The cost script selects the primary solver method and applies numerical scaling/tolerances from runtime policy. The examined settings favor barrier for planning, with specialized annual-S-curve behavior and recovery options. A configured time limit of zero can mean unlimited in the implemented wrappers. Recovery calls may differ from the primary solve and can run for a long time.

Solver status and engineering quality must be reported separately. `OPTIMAL` means the solver's optimality criteria were met for the constructed problem and tolerances; it does not validate input realism. `TIME_LIMIT` or `SUBOPTIMAL` may have a usable incumbent but are not equivalent to a certified optimum. `INFEASIBLE`, `INF_OR_UNBD` and numerical failures require different diagnoses R28, R29.

Inspect primal constraint/bound violations, dual/residual information where meaningful, objective/bound gap and the model's physical balance audit. Numerical row scaling can make a small solver residual correspond to a different MW/MWh/tCO2 violation. Read the physical-unit gates before accepting exports. Do not use a higher tolerance merely to remove an inconvenient policy or stock inconsistency.

A barrier solution without crossover can lack a conventional simplex basis; shadow prices or recovery behavior need careful interpretation. Memory exhaustion can occur during building, presolve, factorization or export, and should not be labelled proof of physical infeasibility. Gurobi's numerical guidelines explain the general distinctions; the local configured thresholds are in the configuration inventory.

17. Results and interpretation

Result family	Interpretation and checks
Investment/capacity CSVs	New versus surviving capacity, native units, vintage/retirement and decision windows
Hourly/yearly electricity balances	Generation, demand components, charging/discharge, losses, imports/exports and unmet load
Heat and industry outputs	Useful service versus final fuel input; direct-electric/HP shares; process replacement accounting
Transport stocks/flows	Vehicle-equivalent stock, demand service, primary/secondary fuel recipes and conversions
Hydrogen/e-fuel outputs	Production, upstream inputs, storage, trade and terminal throughput
Carbon outputs	Sector emissions, independent process emissions, capture/reuse/storage and budget/slack
Cost outputs	Unit bases, annualization, discounting, source price year and solver scale
Solver/quality logs	Status, method, runtime, recovery, objective/bound and physical residuals
Combined dashboards/plots	Derived presentation of accepted exports; not a substitute for balance tables

Myopic results are commonly organized under `results/<scenario>/<decision-year>` with annual investment/history files; detailed paths depend on exporter and mode. Stage-2 bundles and rolling window/stitching outputs have their own metadata. Terminal outputs can include `h2_terminal_flows.csv`, annual

capacity/summary tables and `transfer_terminal_new_YEAR.csv`.

Check that representative-period additive quantities are annualized exactly once. Installed capacities and shares are not summed like hourly energy. Compare runs on matching economic/energy boundaries and time grids. A shadow price is conditional on the active constraints and an accepted economic solution, not a forecast of a regulated tariff.

---

18. Validation evidence, inherited warnings and known limits

This documentation was produced by static source/input review and reference checking. It did not execute a fresh national solve. Local preparation reports establish bounded reader/schema/source-reconciliation results; they do not independently establish all model modes or years.

The supplied folder also contains a critical-correction/rebuild harness and verification files describing unpromoted candidates. In particular its `verification.json` distinguishes component tests, synthetic workflow fixtures and user-reported remote transition evidence, with no real local checkpoint promoted in that report. Historical run instructions and comments must not be cited as proof that those candidate fixes are active or that a full transition passed locally.

Validate the exact production objective and physical constraints in the documented source, then compare any proposed correction with its checkpoint/diff and acceptance certificate. A source hash, accepted transition log and scenario/input identity are stronger evidence than a filename such as `corrected` or a comment claiming a previous optimal case.

Principal study limitations are deterministic future inputs, spatial aggregation/proxies, sampled chronology in applicable modes, continuous transport stock, inherited technology/fuel compatibility assumptions, incomplete lifecycle/non-CO2 accounting, terminal aggregation, possible data-duplicate reconciliation and country-specific calibration gaps. The assumptions register makes these limitations explicit.

For a publishable run, archive the exact model/inputs, environment, flags/policies, source references/hashes, solve logs, quality certificates, decision/history files, representative weights and scripts producing reported tables/figures. Cite the H2RES scientific formulation and the actual data/technology references, while stating that this local version includes later extensions.

Input, output and units reference

Applies to: the preparation repository and the documented H2RES local model, 1 November 2026. This chapter explains the core contracts; machine-readable inventories include every examined workbook sheet/header and native CSV header.

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1. Complete schema inventories

Inventory	Contents
workbook_schema.csv	All examined workbook/sheet names, stored dimensions and exact first-row headers; blank/duplicate headers preserved
model_workbooks.json	Sheet-name lists for all live model workbooks
csv_schema.json	Exact headers for each native CSV
model_data_files.csv	All examined data files, sizes and SHA-256 digests
preparation_cli_options.csv	Argument names, source defaults/choices/help and line positions for runner and sector CLIs
model_configuration.csv	Entered source configuration assignments
workbook_reference_urls.csv	Reference URL occurrences with workbook/sheet/cell context

Some workbooks omit stored worksheet dimensions; their row count is marked unknown in the inventory rather than invented. Header inventories describe the files examined, not an assurance that all retained historical variants are active. Rows/columns can differ in another release.

---

2. Common keys and calendar

Field	Meaning and rule
`System`	Model system label; often ISO2 country, but can be a custom aggregate
`Zone` / `zone`	Model zone; capitalization is file-specific and significant
`country`	Source-country label in spatial mappings/provenance; it can differ from a merged System
`nuts3` / `NUTS3` / `nuts3_id`	Source NUTS3 geography; sector maps use documented aliases
`period`	One-hour model position, normally integer 1–8760
`year` / year-named column	Planning/observation year according to table; not interchangeable with a weather year
`unit_name`	Technology/asset identifier; must match profile, cost and compatibility mappings
`Industry`	Exact native industry identifier, not arbitrary display text
`category`	Service/temperature class, separate from Industry

Field	Meaning and rule
`branch` , `component`	Complex-industry product branch and useful-service component
`include_in_the_model` , `active` , `CC_active`	Table-specific inclusion flags; inclusion does not itself create missing demand or source data

Hourly records with long geography need unique `(System,zone,period)` keys. Wide power profiles need unique `period` and matching unit-name columns. Different files use different index orders internally; never infer an operational dictionary key order from a superficially similar variable. The native transport primary-fuel dictionary, for example, starts with physical fuel.

---

3. Units and conversion discipline

Quantity	Normal basis	Frequent mistake
Electrical/thermal power	MW	Treating MW as annual MWh
One-hour flow	MWh per one-hour interval, often numeric MW	Multiplying by 8760 twice
Annual energy	MWh/year; sometimes source GWh/ktoe	Confusing 1000 MWh/GWh with 11630 MWh/ktoe
Storage power	MW	Using an energy-capacity cost
Storage energy	MWh, or native GWh converted explicitly	Confusing stock with throughput
Fuel input	Carrier MWh, usually lower-heating-value energy conventions	Applying useful-service efficiency a second time
CO2 flow/capture capacity	tCO2/hour	Treating as tonnes of stored inventory
CO2 stock	tCO2	Applying MW-based CAPEX
Product activity	Tonnes or explicit index units	Labelling an index as tonnes
Fleet	Continuous vehicle-equivalent counts	Assuming discrete integer purchases
Capital cost	MEUR per native capacity unit	Applying MEUR/MWh to MW or €/vehicle without conversion
Variable cost	EUR/MWh or declared physical input basis	Counting purchased feedstock at multiple conversion stages
Fixed OPEX	MEUR per capacity-year or configured fraction	Treating a source annual rate as upfront CAPEX
Terminal capacity	TWh/year in standalone terminal workbook	Treating as tank energy capacity

For JRC industry extraction, `1 ktoe = 11630 MWh` is the implemented conversion. `1 GWh = 1000 MWh` ; `1 TWh = 1000000 MWh` ; `1 MEUR = 1000000 EUR` . Currency labels do not by themselves identify a price year or inflation basis. Record those separately where source metadata provide them.

4. Native technology and policy workbook families

Mode	No e-fuels	E-fuels
Regular industry	`flex_tech.xlsx`	`flex_tech_efuel.xlsx`
Category industry	`flex_tech_categories.xlsx`	Conservative: `flex_tech_efuel_categories_electricity_review.xlsx` ; technical: `flex_tech_efuel_categories_electricity_technical_2050.xlsx`
Complex industry	`flex_tech_complex_industry.xlsx`	`flex_tech_efuel_complex_industry.xlsx`

Category/complex flags are mutually exclusive. Other retained filenames, including combined category/complex variants and copies, are archived alternatives rather than an instruction to activate them together. Regular and complex technology files can be paired with case-specific process workbooks as described below.

Policy uses `policy.xlsx` or `policy_efuel.xlsx` ; e-fuel carrier prices use `policy_fuels.xlsx` where enabled. Read the active path printed/selected by the model, not the workbook whose filename happens to be open in Excel.

5. Electricity and hydro files

5.1 `generators.csv` / prepared `power/GENCO.csv`

Power assets use the source's native GENCO columns, including `unit_name,cap_mw,max_cap,fuel_type,decom_start_existing_cap,life_time,decom_start_new,final_life_cap,efficiency,cost_no_fuel,RampingCost,CO2Intensity,technology,RampUpRate,RampDownRate,CHPType,STOCapacityGWh,STOSelfDischarge,STOMaxChargingPower,STOChargingEfficiency,CHPPowerToHeat,CHPPowerLossFactor,CHPMaxHeat,System,Zone` in the examined native CSV.

`cap_mw` is installed power, `max_cap` a capacity ceiling where used, and storage energy is explicitly GWh in this native representation. A detailed intermediate can instead contain `STOCapacity` in MWh and extra coordinates/NUTS fields; use the exporter conversion rather than copying it directly into the native file. `CO2Intensity` /non-fuel costs must be interpreted on the reader's generation/CHP-equivalent basis, not assumed to be raw fuel factors.

5.2 Profiles

File	Shape	Meaning
`electricity_demand.csv`	`period,Power,System,Zone`	Baseline electric load in MW
`renewable_availability.csv`	`period` plus exact generator names	Fractions for VRE and run-of-river
`hydro_inflows.csv`	`period` plus exact HDAM/HPHS names	Normalized inflow convention expected by reader
`electricity_prices.csv`	Hourly/zone price columns in examined input	Selected trade/electricity price assumptions
`import_prices.csv`	Native import-price long format	Price input according to selected cost path
`outside_system_normalized_curves.csv`	Outside-source/period curve fields	External source availability patterns

File	Shape	Meaning
`outside_system_supply_shares.csv`	System/source/share fields	Mix for external availability calibration

Profiles must match the canonical inventory's unit naming and aggregation. Plant/source mapping and curve-quality CSVs are part of the preparation evidence. Do not rename plant rows without updating their curves and cost/investment keys.

---

6. Shared technology workbook sheets

6.1 Geography and networks

`active_zones` identifies active `(zone,System)` combinations. `keys_links`, `keys_borders`, transmission and outside-link sheets establish which modeled and external endpoints exist. Input files may retain inactive countries and templates. Only valid active mappings should contribute to a run, but inherited network/policy assumptions still require review.

6.2 Generic flexible technologies

`flex_tech` contains assets with fields such as installed/maximum capacity, annual build limits, efficiency, electric auxiliary demand, lifetime/retirement, reserve contributions, input/output carriers, storage linkage, System/zone and activity flags. `Tech_type` and carrier fields determine how a row is interpreted; identical-looking numeric `cap_mw` fields in different asset classes should not be assumed to mean identical delivered services.

6.3 `tech_cost_years`

The first column identifies year; technology/component keys are columns. Values follow each asset's documented capital-capacity basis. Complex component costs use `component_unit_name` keys, with generic donors only when explicitly recorded. Interpolated/mapped data are not all direct quotes for that component.

For the active workbook, the optional `literature_fixed_opex.csv` links annual maintenance rates to workbook, technology, year, native capacity unit, conversion and source references. Its `source_cells`, `source_sheets` and `source_price_years` should accompany a cost audit. A match is specific to the chosen workbook/version; source references alone do not prove every row has been applied.

---

7. Ordinary/category industry inputs

7.1 Equipment tables

`Industry_data` includes System/zone, Industry, optional category, `unit_name`, physical `fuel_type`, group, efficiency, inclusion, capacity, initial service share, maximum service share, lifetime/decommissioning, CO2 information and scalar/annual build limits. Category profiles must use exactly the same native sector/category identifiers.

`simple_industry` represents a pooled remainder rather than an additional copy of selected detailed sectors. It must be recalculated when the detailed inclusion selection changes. `industry_CC` is the dedicated capture authority where used. `industry_heat_stor` exists only in variants supporting category industrial thermal storage.

7.2 Demand profiles

`industry_heat.csv` contains `period,System,zone` plus sector service demands. Despite the historical filename, full native service demand includes relevant non-heat components. `industry_category_profiles.csv` uses names such as `Iron_steel[100_400C]` and includes `NON_HEAT` where generated. `industry_electricity.csv` is wide by model zone and records the baseline electricity associated with the chosen source convention.

The legacy `heat_demand_industry.csv` can be process-heat-only; do not use it as the full native file merely because its name sounds appropriate.

---

8. Complex industry workbooks

Possible native names are `industry_complex.xlsx`, `industry_complex_efuel.xlsx` and their `_electricity_technical_2050.xlsx` variants.

Sheet	Key fields	Interpretation
`complex_industry`	System, zone, Industry, branch, component, unit_name	Process-equipment alternatives and observed/assumed fuel parameters
`Process_data`	System, zone, Industry, branch, component	Useful-service/product intensity and independent process emissions
`Product_demand`	System, zone, Industry, branch, year columns	Annual product activity in declared unit
`CC`	System, zone, Industry, branch	Branch-level capture/reboiler/storage assumptions
`JRC_source_cells`	Sector/branch/component/fuel/year	Source observations and exact useful/final/emission addresses
`industry_heat_pumps`	Unit/class	COP, sink temperature, lifetime and cost assumptions
`Methodology` / screening/provenance tables	Descriptive keys	Source/version/assumption explanations; runtime tables remain authoritative

8.1 Critical process fields

`activity_unit` and `product_unit` must agree within a branch. `energy_intensity_MWh_per_activity` and `process_CO2_t_per_activity` are authoritative generic intensities. Legacy `_per_t` fields are suitable only for physical product tonnes. `reference_activity` is the observation basis. `activity_factor` modifies activity-to-service conversion and must not be applied twice.

`historical_process_CO2_*` preserves observed history; `excluded_H2_reforming_CO2_*` records the guarded Croatian accounting exception; modeled `process_CO2_*` is the remaining additional process term. `combustion_CO2_t_per_MWh` is observed final-fuel combustion, distinct from the legacy pure-fuel `CO2Intensity` and from product process intensity.

`fossil_waste_final_energy_share` applies to the mixed biomass/waste final-energy stream. `source_waste_split` must distinguish exact single-class evidence from assumed sector blends. A positive observed fossil combustion factor cannot be erased because a row is named Biomass.

`hp_below60_share`, `hp_below80_share`, `hp_below125_share`, `hp_below150_share` and `hp_source_heat_share` are shared eligibility/source budgets. They are fractions of useful service, not individual independent technology potentials that can simply be added.

8.2 Industry identifiers

The source maps JRC ISI, NFM, CHI, NMM, PPA, FBT, TRE, MAE, TEL, WWP and OIS into the eleven exact model labels listed in the preparation manual. The spelling `Not_elsewhere_specified_(industry)` and legacy branch key `cemment` are intentional compatibility identifiers. Correcting display spelling inside keys without migrating dependent inputs breaks joins.

---

9. Transport tables and profiles

`transport_all` identifies sector, equipment, type, demand/load coefficients, primary efficiencies, stock counts, maximum stock/share, storage/charging, retirement/lifetime, grid efficiency and inclusion, plus year-based investment/share fields according to variant. The extended e-fuel workbook contains persisted `compat_*` and parent/conversion metadata.

Names such as `total_load`, `number`, `tot_num`, `efficiency_primary`, `max_share` and year columns must be read together with the selected native transport method. `total_load` is a service/demand coefficient, not a universal electrical-MW field. Annual activity normalization and stock-derived coefficients determine its physical interpretation. Source table and reader equations are the authority for road versus marine/rail proxy classes; a column name alone is insufficient for a cross-sector conversion.

`transport_ICE_prim` maps primary/secondary fuels and efficiencies/shares to the applicable engine/service rows. `compat_parent` permits inherited availability for converted equipment. Paid upgrades start with no independently observed additional stock; their conversion/cost recipes are template assumptions.

Shipping source aliases use the native route keys consistently across equipment, compatibility lookup and hourly profile headers: `ICE_MARINE_DOM` → `ICE_MARINE_SHORT` (`Domestic_shipping`) and `ICE_MARINE_INT` → `ICE_MARINE_LONG` (`international_shipping`). The source service quantities and initial stocks are retained. Both newly prepared files and cached legacy files passed through export use this normalization; simultaneous legacy/current profile columns are rejected as ambiguous.

File	Values
`transport_demand.csv`	`period,zone,System` plus service activity shares, normally normalized to annual total 1
`transport_profiles.csv`	`period,zone,System` plus equipment/parent plug-in availability fractions

Fleet, service shares, fuel volume shares and fuel energy shares are different. Use the documented density/LHV/efficiency conversion in the pathway method rather than interpreting a volume percentage as an energy percentage.

10. Heating, DH, solar and cooling

Prepared technology tables are `flex_tech_heat`, `flex_tech_DH`, `thermal_tech`, `thermal_tech_DH`, `solar_indv` and `solar_DH`; storage sheets exist according to variant. Electricity/capacity and delivered-heat units must follow the class-specific reader. Heat-pump COP is not a boiler fuel efficiency.

`heating_individual.csv` and `heating_district.csv` are wide by model zone with `period`. `heating_electricity.csv` is the associated baseline electric demand. `solar_thermal_availability.csv` has zone/technology series according to the native reader. Cooling has separate individual/district files; their presence in the template does not imply they were freshly generated or enabled in a run.

11. Hydrogen, e-fuels, primary supply and capture

The selected flex workbook contains enabled e-fuel conversion, feedstock, primary-energy, trade, storage and network sheets. The exact sheet set is in the inventory because retained variants differ. Technology input/output recipes use carrier-specific energy or material units. Source notes and cost aliases are part of the contract.

Standalone terminals use `h2_terminals.xlsx`:

Terminal asset field	Unit/meaning
`cap_TWh`, `max_cap_TWh`	Annual terminal throughput capacity, TWh/year
`max_new_TWh_per_year`	New annual-throughput capacity allowed per calendar year
`loss`	Fractional handling loss
`el_MWh_per_MWh`	Handling electricity per H2 throughput energy
`var_EUR_per_MWh`	Handling cost per throughput energy
`life_years`, `existing_remaining_life_years`	New/existing lifetime inputs
`fixed_opex_pct`	Asset-specific annual maintenance fraction where present

Shipping-route fields include origin/destination, `delay_hours`, `cargo_MWh`, departure interval/offset and ships en route. A terminal row alone does not define an internal corridor. The workbook is an aggregate H2 representation and does not declare a separate ammonia vessel/cracker model.

Capture fields include `CC_cap`, capture electricity, capture rate, reboiler capacity/stripper heat and optional tank capacity. Their annual build limits have different units and cannot all be scaled as MW. Carbon-source tagging and allowed geological storage follow the chosen model switches, not merely the existence of a CC row.

12. Annual investment limits

Asset / field prefix	New capacity allowed per calendar year
Industry `max_inv_period_YYYY`	MW useful-service equipment capacity
Industry heat-storage `max_inv_period_YYYY`	MWh storage energy
`CC_max_inv_period_YYYY`	tCO2/hour capture capacity

Asset / field prefix	New capacity allowed per calendar year
`Reboiler_max_inv_period_YYYY`	MW reboiler capacity
`Tank_stor_max_inv_period_YYYY`	tCO2 intermediate inventory capacity
Generic asset or generation tables	Their declared native capacity basis
Transport/year fields	Native vehicle/service investment convention; inspect sheet and active flag

The year-specific cell takes precedence. Blank/missing annual cells fall back to the matching scalar. If that is also blank/missing, the limit is unrestricted. Zero is a binding prohibition on additions. Limits are not interpolated merely because future years are sparse.

The model sums calendar-year allowances over the global decision interval; the initial decision uses its own year. Installed stock and an annual build rate are not the same number. A total capacity limit and an annual build limit can both bind. Capture table precedence and sharing across categories prevent multiplying one physical branch capture budget by category count.

---

13. Policy tables and scenario JSON

`setup` holds the modeled years and first-row controls (`chunk_size,step_size,first_hour,last_hour`). Other policy sheets can hold RPS, CO2 limits, fuel/import/carbon prices, transport technology floors, biomass supply limits, generation calibration, hydro caps and outside trade targets. Exact names/headers are in the schema inventory. Many policy rows are System-scoped, not country-independent constants.

Scenario JSON supports `preset,scope,years,start_year,end_year,year_step,target_year,start_share,investment_multiplier,investment_overrides` . CLI choices override corresponding JSON fields. Years are increasing/equally spaced and must be supported by costs. A whole-energy RES preset requires the e-fuel representation and a compatible model adapter.

Investment overrides select supported sheets, a native column, value, optional row conditions and years. Unsupported sheet names are errors. A reference to `Industry_data` is not automatically an override of the separate `complex_industry` workbook. The supported complex multiplier can scale its build limits; use documented direct editing for unsupported per-row complex changes.

---

14. Manifests and evidence files

File	What it establishes
`model_reference/manifest.json`	Template origin, captured file/source-reader hashes
`model_inputs/bundle_manifest.json`	Prepared sectors, countries, chosen variants, replaced sheets, template-retained inputs and reported repairs
`model_inputs/validation.json`	Implemented key/schema/hour/profile/zone coverage checks
`model_inputs/scenario.json`	Requested future policy and integration assumptions
`complex_generation.json`	Process sources, allocation, case, cost donors, profile basis and limitations

File	What it establishes
`source_reconciliation.csv`	Source-to-generated energy/emission closure for selected observation bases
`reference_calibration_diff.csv`	Matching HR1 process-equipment differences for the defined comparison fields
`scenario_reader_validation.json`	Separate actual-model input-reader check, if explicitly performed
`validation/*.json`	Bounded development verification; read its scope and timestamp

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15. Minimum review before handing inputs to H2RES

Check requested geography against active/generated keys, unique hourly periods, matching unit/profile names, full sector/year coverage, useful/final-energy distinctions, scenario investment units, non-energy feedstocks, process/combustion carbon separation, finite efficiencies and stock/cost bases. Inspect warnings about fallback allocations and retained reference rows. Preserve the raw data and exact hashes so the handoff can be reproduced.

The model will also require consistent flag selection and any separately implemented whole-energy policy equations. Structural input compatibility is an interface property; historical accuracy and scenario feasibility require additional review.

Assumptions, limitations and evidence register

This register distinguishes observations from processing rules and scenario choices. It is part of both manuals. An internet source can contain analytically reconstructed data, and a referenced technology report does not validate a specific national parameter transferred from a template.

---

1. Classification

Class	Definition	Example
Source observation	Quantity taken from the identified source version/cell/response	JRC annual useful/final energy; ENTSO-E hourly load
Derived quantity	Reproducible transformation with a stated equation and units	Useful/final efficiency; capacity-weighted profile
Spatial/temporal proxy	Allocation using related evidence rather than direct local measurement	Raster shares, inherited hourly shapes, flat fallback
Inherited template assumption	Value retained from the supplied model/cost workbook	Lifetime, annual build limit, some conversion/cost rows
Explicit scenario assumption	Deliberately selected future or technical policy	2050 renewable target, investment multiplier, compatible replacement envelope
Unverified inherited provenance	A value/comment/reference whose original calibration cannot be established here	Unattributed technology defaults or missing source price-year details

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2. Parameter and method register

ID	Item	Classification / implication	Required study disclosure
A01	Country selection	Processing/output filter; bulk source geography can be wider	Selected codes, observed output System/zone, cache/source scope
A02	Geographic allocation	Source NUTS3 data plus proxy/custom mapping	Boundary vintage, raster/facility evidence, equal/custom weights, cross-border assumptions
A03	Power technical defaults	Mixed observed asset attributes and inherited technology defaults	Source snapshot, operating-status audit, donor parameters and remaining lives
A04	Weather/renewable profiles	Reanalysis/proxy conversion, optional national scaling	Weather year/backend, technology assumptions, IRENA extract and scaling
A05	Hydro profiles	Bundled runoff shapes plus scaling/matching assumptions	Runoff source/year, mapping distances, exclusions, natural/closed-loop distinction
A06	Category industry	Older annual/source profile vintages plus inherited temperature allocation	Reference years and what has/has not been recalibrated for the selected country

ID	Item	Classification / implication	Required study disclosure
A07	Complex industry initial MW	Annual useful energy / 8760	Average-utilization assumption; not observed nameplate plant census
A08	Process product activity	Observed activity to source end; constant thereafter	Future growth/structural-change scenario, fallback observation years
A09	Process electrical/HP applicability	Service screening and inherited equipment assumptions	Conservative/technical case, partial-furnace coefficient, COP/lifetime, cumulative/source budgets
A10	Biomass/waste split	Exact single-class evidence or assumed common sector blend	Strict/blend choice, exact source evidence, future mixed-stream composition assumption
A11	Material H2 convention	Non-energy gas/oil services represented as H2-fed demand	Chemical-route justification and possible process-emissions overlap
A12	Croatian ammonia exception	Guarded inherited accounting interpretation	Exact country/branch/year/amount; independent evidence required elsewhere
A13	Biomass/lignin substitution	Baseline biomass ceiling and bounded local lignin option	Share versus absolute annual MWh distinction; sustainability/resource scenario
A14	Transport stock/profiles	Bundled national data, service coefficients and allocation proxies	Source vintage, fleet/service units, availability, fractional stock and OSM fallback
A15	Paid conversions/blends	Technology-screening assumptions, including marine transfers	Engine approvals, pilot fuel, conversion cost/remaining life, actual fleet calibration
A16	Heating allocation	Annual observations plus population/HDD/density proxies	DH boundary, fuels, weather/population vintage and baseline electricity convention
A17	Costs/lifetimes/build limits	Inherited/reference or explicit scenario	Source workbook/cells, capacity basis, price year, finance/lifetime, total versus annual ceilings
A18	External energy supply	Network/capacity/price/availability assumptions	Import origin, hourly limits, fuel-specific trade prices and active endpoints
A19	Non-energy materials/CO2	Separate recipe and carbon classifications	Energy/material units, capture/reuse/storage distinction and imported stream tagging
A20	Future chronology	Full hourly or sampled/rolling mode	Period sets/weights, boundary conditions, what has actually been operationally validated
A21	Renewable preset	Explicit energy-origin constraints, not a forecast or net-zero definition	Target scope, adapter version, imports and cleared stocks, process/lifecycle exclusions
A22	Numerical acceptance	Solver tolerances and physical-unit quality gates	Status, primary/recovery, scaled/unscaled violations, slack and accepted history

3. Current boundary conditions worth checking

The live model's category/e-fuel selection does not automatically activate the process compiler's all-sector selection. The process policy defaults to non-metallic minerals until changed. Conversely, a workbook retaining data for other countries does not activate them if the active-zone scope is correct.

Several local entered values are Croatian plant/technology calibrations. Changing `--countries` can replace historical demand/inventory without supplying a new transmission network, biomass potential, outside-trade target, CAPEX forecast or heating/transport policy for that country. Complete native coverage and country-specific scientific calibration are separate review tasks.

Ordinary/category and complex industry have distinct vintages and definitions. Their totals may differ even when both are internally correct. Do not force equality by adding a remainder that duplicates replaced services. The model's process-replacement audit preserves ordinary services separately.

The documented conservative whole-energy preset blocks unverified imported energy and clears generic uncertified boundary stocks. That is stricter than permitting certified renewable imports. It can make a target infeasible even when another origin-tracing formulation would allow it. Stating "100% renewable" without the trade/storage boundary definition is insufficient.

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4. Evidence levels

Level	Typical evidence	Does not establish
Dependencies/startup	Import checks, hidden interface construction	Source availability or correct national calibration
File structure/coverage	Validation JSON, unique keys, 8760 periods	A feasible optimization scenario
Source calibration	Cell reconciliation and baseline comparisons	Future technology realism or complete lifecycle emissions
Model reader	Actual native input reader completes	All constraints/objective built or a solve accepted
Small adapter/model test	Sample periods, bounded equation/LP tests	Full national chronology and all modes
Full solve	Exact input/version, solver logs and quality certificates	Empirical truth outside the modeled assumptions
Independent study validation	Historical reproduction and external plausibility/sensitivity checks	Universal validity for other countries/scenarios

The preparation verification reports record passing bounded tests. The supplied critical-rebuild folder also contains synthetic fixtures, unpromoted candidate checks and user-reported remote results. These evidence classes must not be merged into a claim of an independently reproduced full transition in this documentation session.

5. Source and reference completeness

The bibliography contains verified public data/method references and retained identifiable references from the industry handoff and cost-source records. Exact endpoint/cell inventories support further audit. Some publisher pages block automated retrieval or large PDFs exceed the browser tool's limit; those cases are labelled rather than represented as fully re-read.

Inherited parameters without a verified primary source remain marked as inherited/unattributed. A catalogue URL, generic literature citation or software repository is not enough to reconstruct an individual numeric price, efficiency or lifetime. A publication-ready parameter table needs the exact edition, page/table/sheet/cell, physical units, price year and all conversion steps.

6. Recommended sensitivity and handoff record

Archive at least: command/configuration; selected countries/sectors/mapping; raw data/version/hashes; future years/demand; technology costs/lifetimes; annual/total investment limits; resource/biomass assumptions; industrial route and HP eligibility; fuel/engine compatibility; external prices/origins; storage initial/final policies; chronology/sampling; solver settings; accepted residuals/slack; resume history and exported results.

Useful sensitivity dimensions include high/low demand growth, renewable potential, grid/terminal expansion, weather years, profile calibration, industrial electrification scope, HP source limits, biomass ceilings, transport turnover/retrofit cost, renewable imports, financing and different chronological representations. Report which constraints bind and which inputs dominate the result.

References and source register

Documentation/source check date: **2026-11-01**.

R entries are the curated scientific/data/software references. H entries preserve additional identifiable sources from the supplied industry handoff. T entries preserve the live fixed-OPEX workbook sources. Multiple URLs can describe the same work; entries are not a count of independent validation studies.

The verification note records whether a primary page/PDF was actually inspected or metadata were retained from supplied files. No missing source, price year or parameter calibration has been invented. A reference to a technology catalogue is not proof that every local default was transferred correctly.

Citation practice

For a study, cite the H2RES optimization formulation (R02), the actual dataset releases, and the technology sources used. Also identify the local model hash, preparation version and scenario assumptions. Historical H2RES simulation and later optimization versions must not be treated as the same code.

Record download/request dates and local hashes for mutable URLs, mirrored archives and manually prepared statistics extracts. The source's licence and the web-service usage terms are separate matters; check both before redistribution. OSM attribution follows its provider's licence terms. This register documents source conditions, not a legal opinion.

Curated references

R01 — Increasing renewable energy sources in island energy supply: case study Porto Santo

Duić, Neven and da Graça Carvalho, Maria. (2004). *Increasing renewable energy sources in island energy supply: case study Porto Santo*. article.

Primary/preserved source. DOI: `10.1016/j.rser.2003.11.004`.

Use: Historical H2RES island-energy lineage; not a specification of the current Python optimization implementation.

Verification: Author-hosted paper metadata/content checked.

Additional checked page.

R02 — A long-term capacity investment and operational energy planning model with power-to-X and flexibility technologies

Feijoo, Felipe and Pfeifer, Antun and Herc, Luka and Groppi, Daniele and Duić, Neven. (2022). *A long-term capacity investment and operational energy planning model with power-to-X and flexibility technologies*. article.

Primary/preserved source. DOI: `10.1016/j.rser.2022.112781`.

Use: Published optimization formulation and model lineage; later local additions are documented from code.

Verification: Publisher search record/article text and H2RES project publication listing checked; DOI redirect sometimes blocked.

R03 — The management of an energy system in the realm of rapid energy transition and degasification as a consequence of energy crisis, examination in H2RES energy model

Herc, Luka and Feijoo, Felipe and Kodba, Ana and Dorotić, Hrvoje and Stunjek, Goran and Beljan, Doris and Pukšec, Tomislav and Krajačić, Goran and Pfeifer, Antun and Duić, Neven. (2024). *The management of an energy system in the realm of rapid energy transition and degasification as a consequence of energy crisis, examination in H2RES energy model*. article.

Primary/preserved source. DOI: `10.1016/j.enconman.2024.118782`.

Use: H2RES application background; not an independent validation of this source snapshot.

Verification: Croatian research information system publication record and project bibliography checked.

R04 — Optimal decarbonisation pathways for the Italian energy system: Modelling a long-term energy transition to achieve zero emission by 2050

Pastore, Lorenzo Mario and Groppi, Daniele and Feijoo, Felipe and Lo Basso, Gianluigi and Astiaso Garcia, Davide and de Santoli, Livio. (2024). *Optimal decarbonisation pathways for the Italian energy system: Modelling a long-term energy transition to achieve zero emission by 2050*. article.

Primary/preserved source. DOI: `10.1016/j.apenergy.2024.123358`.

Use: Italian H2RES application background. This preparation package does not automatically reproduce its input scenario.

Verification: Institutional repository PDF opened; first page confirms full author list, journal/article number, publication date and DOI. Direct DOI fetch was blocked.

Additional checked page.

R05 — H2RES Integrated Energy System Model: project website

H2RES project. (n.d.). *H2RES Integrated Energy System Model: project website*. online.

Primary/preserved source.

Use: Model lineage, publication resources and version guidance; the local model remains version-specific.

Verification: Project website opened.

R06 — JRC-IDEES-2021

Rózsai, Máté and Jaxa-Rozen, Marc and Salvucci, Raffaele and Sikora, Przemyslaw and Tattini, Jacopo and Neuwahl, Frederik. (2024). *JRC-IDEES-2021*. dataset.

Primary/preserved source. DOI: `10.5281/zenodo.15895830`.

Use: Annual source database used by category industry and buildings; local endpoint is a mirrored bulk archive.

Verification: Zenodo release metadata and its primary JRC catalogue pointer checked. Publication date 20 May 2024; mirror record created later.

Source conditions: Check the JRC source licence and the exact mirror/version; do not infer all upstream dataset rights from Zenodo hosting.

R07 — JRC-IDEES-2023

Rózsai, Máté and Jaxa-Rozen, Marc and Salvucci, Raffaele and Sikora, Przemyslaw and Gea Bermudez, Juan and Neuwahl, Frederik. (2026). *JRC-IDEES-2023*. dataset.

Primary/preserved source. DOI: `10.2905/JRC.JPXYRT8`.

Use: Country Industry, EnergyBalance and EmissionBalance source workbooks for complex industry.

Verification: Official JRC overview, technical-report metadata, country download endpoint and inherited dataset DOI checked.

Additional checked page.

Source conditions: JRC overview identifies CC BY 4.0; archive the actual release.

R08 — SECURES-Energy: Hourly electricity demand and supply profiles for historical climate and climate change projections in Europe until 2100

Schöniger, Franziska and Resch, Gustav and Suna, Demet and Widhalm, Peter and Totschnig, Gerhard and Pardo-Garcia, Nicolas and Hasengst, Florian and Formayer, Herbert and Maier, Philipp and Leidinger, David. (2024). *SECURES-Energy: Hourly electricity demand and supply profiles for historical climate and climate change projections in Europe until 2100*. dataset.

Primary/preserved source. DOI: `10.5281/zenodo.14615500`.

Use: National hourly profiles; the category pipeline downloads the historical ERA5-Land 1981–2010 archive, not every future climate scenario.

Verification: Zenodo v2 metadata, files, coverage notes and project report link checked.

R09 — powerplantmatching: tools and combined power plant data

PyPSA powerplantmatching contributors. (n.d.). *powerplantmatching: tools and combined power plant data*. software.

Primary/preserved source.

Use: Non-hydro inventory source; actual code downloads the repository master powerplants.csv.

Verification: Maintainer repository and code endpoint identified; actual source hash/retrieval date is needed because master is mutable.

R10 — JRC Hydro-power database

European Commission, Joint Research Centre. (2026). *JRC Hydro-power database*. dataset.

Primary/preserved source. DOI: `10.2905/JRC.JBF7S58`.

Use: Hydro inventory and annual plant metadata. Bundled hourly runoff profiles are an additional local resource.

Verification: Official catalogue citation and release link checked. Catalogue citation year differs from earlier literature references to the original database.

Source conditions: Official catalogue identifies CC BY 4.0.

R11 — Energy and Industry Geography Lab and demand raster distribution

European Commission, Joint Research Centre. (n.d.). *Energy and Industry Geography Lab and demand raster distribution*. dataset.

Primary/preserved source.

Use: 2019 electricity/fuel spatial-allocation rasters; spatial reference year is independent of demand/statistical year.

Verification: Official overview and directory listing checked; layer copyright file is separately listed by provider.

Additional checked page.

R12 — Transparency Platform REST API implementation guide and token management

ENTSO-E. (n.d.). *Transparency Platform REST API implementation guide and token management*. documentation.

Primary/preserved source.

Use: Hourly actual total load requests and credential/code-list requirements.

Verification: Official API guide/token document search records checked. Legacy Guide.html URL could not be opened; current account workflow should be checked with provider.

Additional checked page.

R13 — OpenStreetMap data and Geofabrik country extracts

OpenStreetMap contributors and Geofabrik GmbH. (n.d.). *OpenStreetMap data and Geofabrik country extracts*. dataset.

Primary/preserved source.

Use: Selected-country PBF transport allocation indicators; the Geofabrik index itself is multi-country.

Verification: Provider download page and OpenStreetMap copyright/licence page opened.

Additional checked page.

Source conditions: OSM data are ODbL; use required contributor attribution and consult provider terms for redistribution.

R14 — heat_tot_curr_density: current heat-density raster

Hotmaps project. (n.d.). *heat_tot_curr_density: current heat-density raster*. dataset.

Primary/preserved source.

Use: Heat-density spatial proxy used by heating/DH allocation.

Verification: Project metadata opened; dataset describes 2015 heat density. Source release and licence must be preserved.

R15 — Historical Weather API documentation

Open-Meteo. (n.d.). *Historical Weather API documentation*. documentation.

Primary/preserved source.

Use: Default site weather/temperature source for VRE and heating preparation.

Verification: Provider documentation opened; service/data usage conditions are separate from locally derived profile assumptions.

R16 — The ERA5 global reanalysis

Hersbach, Hans and others. (2020). *The ERA5 global reanalysis*. article.

Primary/preserved source. DOI: `10.1002/qj.3803`.

Use: Scientific description of ERA5 reanalysis; does not validate the local power-curve conversion.

Verification: Publisher article metadata and primary publication record checked.

R17 — ERA5-Land: a state-of-the-art global reanalysis dataset for land applications

Muñoz-Sabater, Joaquín and others. (2021). *ERA5-Land: a state-of-the-art global reanalysis dataset for land applications*. article.

Primary/preserved source. DOI: `10.5194/essd-13-4349-2021`.

Use: Scientific description of ERA5-Land underlying weather/runoff sources.

Verification: Publisher article and PDF identified.

R18 — Renewables.ninja documentation and model service

Renewables.ninja. (n.d.). *Renewables.ninja documentation and model service*. documentation.

Primary/preserved source.

Use: Optional token-authenticated PV/wind simulation backend.

Verification: Provider documentation page opened; authenticated documentation may require login. The generic /api page returned 404 and is not used as a user guide.

R19 — Renewable capacity statistics 2025 and renewable electricity statistics

International Renewable Energy Agency. (2025). *Renewable capacity statistics 2025 and renewable electricity statistics*. dataset.

Primary/preserved source.

Use: Context for annual capacity/generation calibration; actual local extract and its source year must be recorded separately.

Verification: Official statistics/PDF search records checked; /Data direct fetch was blocked. No claim that the local extract equals the latest online release.

Additional checked page.

R20 — Direct electrification of industrial process heat: An assessment of technologies, potentials and future prospects for the EU

Rehfeldt, Matthias and Bußmann, Simon and Fleiter, Tobias and Rissman, Jeffrey. (2024). *Direct electrification of industrial process heat: An assessment of technologies, potentials and future prospects for the EU*. report.

Primary/preserved source.

Use: Industrial electrical-service applicability background; national screening coefficients remain explicit local assumptions.

Verification: Publisher page lists authors, version 1.1, publication 5 June 2024 and revision 6 June 2024.

R21 — Renewables for Industry: Electrification of low-temperature heat and steam

International Energy Agency. (2025). *Renewables for Industry: Electrification of low-temperature heat and steam*. report.

Primary/preserved source.

Use: Industrial low-temperature heat/steam electrification background.

Verification: Official report page, publication date 23 December 2025 and suggested citation checked.

R22 — Technology Data for Industrial Process Heat

Danish Energy Agency. (2026). *Technology Data for Industrial Process Heat*. report.

Primary/preserved source.

Use: Technology catalogue/background and inherited industrial heat cost assumptions.

Verification: Official catalogue page checked; it lists April 2026 catalogue and January 2026 datasheet updates. Individual numeric transfers require sheet/cell provenance.

R23 — Croatian National Inventory Report 2025

Croatian national greenhouse-gas inventory authorities. (2025). *Croatian National Inventory Report 2025*. report.

Primary/preserved source.

Use: Inherited evidence for the guarded HR ammonia-reforming accounting exception, table 4.3-1 as identified in supplied method.

Verification: Original publisher PDF endpoint identified; browser retrieval exceeded its size limit. The full inventory/table was not independently reread in this documentation session.

R24 — Croatian National Inventory Report 2020

Croatian national greenhouse-gas inventory authorities. (2020). *Croatian National Inventory Report 2020*. report.

Primary/preserved source.

Use: Inherited ammonia steam-reforming process description, section 4.3.1 as identified in supplied method.

Verification: Publisher PDF endpoint identified; full file exceeded browser size limit. Country/accounting interpretation retained as inherited provenance.

R25 — Preparing Container Vessels for Conversion to Green Fuels

Mærsk Mc-Kinney Møller Center for Zero Carbon Shipping. (2022). *Preparing Container Vessels for Conversion to Green Fuels*. report.

Primary/preserved source.

Use: Context for inherited marine conversion assumptions. Transfer to national model proxy vessels is a scenario assumption.

Verification: Provider PDF identified; publication label October 2022 checked. No claim that every local conversion fraction was independently recalibrated.

R26 — GeoPandas installation documentation

GeoPandas developers. (n.d.). *GeoPandas installation documentation*. documentation.

Primary/preserved source.

Use: Python spatial dependency installation and platform guidance.

Verification: Maintainer documentation checked; locally installed versions are recorded separately.

R27 — Climate Data Store: CDSAPI setup

Copernicus Climate Change Service and ECMWF. (n.d.). *Climate Data Store: CDSAPI setup*. documentation.

Primary/preserved source.

Use: Optional CDS client/token setup and dataset terms acceptance.

Verification: Official instructions opened; no credentials copied into documentation.

R28 — Gurobi Optimizer reference manual: optimization status codes

Gurobi Optimization. (n.d.). *Gurobi Optimizer reference manual: optimization status codes*. documentation.

Primary/preserved source.

Use: Meaning of solver termination statuses; local acceptance is additionally governed by configured quality gates.

Verification: Official manual opened; current online manual can be newer than the installed solver.

R29 — Guidelines for Numerical Issues

Gurobi Optimization. (n.d.). *Guidelines for Numerical Issues*. documentation.

Primary/preserved source.

Use: General numerical conditioning and tolerance interpretation.

Verification: Official manual opened; does not certify the local model's physical residuals.

R30 — Extraction-Condensing CHP example

PyPSA developers. (n.d.). *Extraction-Condensing CHP example*. documentation.

Primary/preserved source.

Use: Background reference explicitly cited in the local CHP-equivalent energy-accounting helper.

Verification: Code-cited primary documentation URL opened and its CHP feasible-region example checked; exact local equation is documented from energy_accounting.py.

R31 — NUTS statistical regions: GISCO distribution

Eurostat GISCO. (2021). *NUTS statistical regions: GISCO distribution*. dataset.

Primary/preserved source.

Use: 2021 NUTS boundaries and hierarchy used for shared geography.

Verification: Official distribution page opened; exact scale/CRS/file/hash should be archived with each run.

R32 — Population grids and GEOSTAT

Eurostat GISCO. (2021). *Population grids and GEOSTAT*. dataset.

Primary/preserved source.

Use: Population-allocation background; package uses the supplied 2021 TIFF or its configured mirror.

Verification: Official GEOSTAT page redirects users to population grids. Exact local raster and its mirror/licence remain part of source provenance.

R33 — JRC-IDEES-2023: the Integrated Database of the European Energy System — Data update and technical documentation

Rózsai, Máté and Jaxa-Rozen, Marc and Salvucci, Raffaele and Sikora, Przemyslaw and Gea Bermudez, Juan and Neuwahl, Frederik. (2026). *JRC-IDEES-2023: the Integrated Database of the European Energy System — Data update and technical documentation*. report.

Primary/preserved source. DOI: `10.2760/6945021`.

Use: Authoritative documentation of the JRC analytical database release.

Verification: Official record lists authors, report JRC144707 and date 14 January 2026; source content describes the underlying 2025 release.

R34 — H2RES-core public source repository

H2RES-model contributors. (n.d.). *H2RES-core public source repository*. software.

Primary/preserved source.

Use: Public project code context; it is not asserted to be byte-identical to the local documented copy.

Verification: Maintainer repository opened.

Additional industry handoff references

H01 — JRC-IDEES-2023: the Integrated Database of the European Energy System - Data update and technical documentation

Rózsai, M. and Jaxa-Rozen, M. and Salvucci, R. and Sikora, P. and Gea Bermudez, J. and Neuwahl, F.. (2026). *JRC-IDEES-2023: the Integrated Database of the European Energy System - Data update and technical documentation*. inherited reference.

Primary/preserved source. DOI: `10.2760/6945021`.

Use: Identifiable reference retained from the supplied industry methodology/bibliography. Some URLs duplicate a curated work using a different endpoint.

Verification: Bibliographic metadata retained from supplied handoff; not every document or numeric transfer independently re-read in this documentation session.

Supplied bibliography: `industry\handoff\model\analysis\industry_fuel_compatibility_2026_10_31\references.csv`, key `jrc_idees_documentation`.

H02 — Renewables for Industry: Electrification of low-temperature heat and steam

{International Energy Agency}. (2025). *Renewables for Industry: Electrification of low-temperature heat and steam*. inherited reference.

Primary/preserved source.

Use: Identifiable reference retained from the supplied industry methodology/bibliography. Some URLs duplicate a curated work using a different endpoint.

Verification: Bibliographic metadata retained from supplied handoff; not every document or numeric transfer independently re-read in this documentation session.

Supplied bibliography: `industry\handoff\model\analysis\industry_fuel_compatibility_2026_10_31\references.csv`, key `iea_renewables_industry`.

H03 — Direct electrification of industrial process heat: An assessment of technologies, potentials and future prospects for the EU

{Fraunhofer ISI}. (2024). *Direct electrification of industrial process heat: An assessment of technologies, potentials and future prospects for the EU*. inherited reference.

Primary/preserved source.

Use: Identifiable reference retained from the supplied industry methodology/bibliography. Some URLs duplicate a curated work using a different endpoint.

Verification: Bibliographic metadata retained from supplied handoff; not every document or numeric transfer independently re-read in this documentation session.

Supplied bibliography: `industry\handoff\model\analysis\industry_fuel_compatibility_2026_10_31\references.csv`, key `fraunhofer_agora_electrification`.

H04 — CO2-neutral process heat using electrification and hydrogen: Technologies, barriers and required action

Fleiter, T. and Rehfeldt, M. and Neusel, L. and Hirzel, S. and Neuwirth, M. and Schwotzer, C. and Kaiser, F. and Gondorf, C.. (2024). *CO2-neutral process heat using electrification and hydrogen: Technologies, barriers and required action*. inherited reference.

Primary/preserved source.

Use: Identifiable reference retained from the supplied industry methodology/bibliography. Some URLs duplicate a curated work using a different endpoint.

Verification: Bibliographic metadata retained from supplied handoff; not every document or numeric transfer independently re-read in this documentation session.

Supplied bibliography: `industry\handoff\model\analysis\industry_fuel_compatibility_2026_10_31\references.csv`, key `fleiter_process_heat`.

H05 — High-Temperature Heat Pumps, Task 3: Applications and Transition

{IEA HPT TCP Annex 58}. (2024). *High-Temperature Heat Pumps, Task 3: Applications and Transition*. inherited reference.

Primary/preserved source.

Use: Identifiable reference retained from the supplied industry methodology/bibliography. Some URLs duplicate a curated work using a different endpoint.

Verification: Bibliographic metadata retained from supplied handoff; not every document or numeric transfer independently re-read in this documentation session.

Supplied bibliography: `industry\handoff\model\analysis\industry_fuel_compatibility_2026_10_31\references.csv`, key `hpt_annex58_task3`.

H06 — Innovation Outlook: Renewable Methanol

{IRENA} and {Methanol Institute}. (2021). *Innovation Outlook: Renewable Methanol*. inherited reference.

Primary/preserved source.

Use: Identifiable reference retained from the supplied industry methodology/bibliography. Some URLs duplicate a curated work using a different endpoint.

Verification: Bibliographic metadata retained from supplied handoff; not every document or numeric transfer independently re-read in this documentation session.

Supplied bibliography: `industry\handoff\model\analysis\industry_fuel_compatibility_2026_10_31\references.csv`, key `irena_methanol`.

H07 — Green hydrogen for industry: A guide to policy making

{IRENA}. (2022). *Green hydrogen for industry: A guide to policy making*. inherited reference.

Primary/preserved source.

Use: Identifiable reference retained from the supplied industry methodology/bibliography. Some URLs duplicate a curated work using a different endpoint.

Verification: Bibliographic metadata retained from supplied handoff; not every document or numeric transfer independently re-read in this documentation session.

Supplied bibliography: `industry\handoff\model\analysis\industry_fuel_compatibility_2026_10_31\references.csv`, key `irena_hydrogen_industry`.

H08 — Biomass boilers for industrial process heat

{Energy Efficiency and Conservation Authority}. (n.d.). *Biomass boilers for industrial process heat*. inherited reference.

Primary/preserved source.

Use: Identifiable reference retained from the supplied industry methodology/bibliography. Some URLs duplicate a curated work using a different endpoint.

Verification: Bibliographic metadata retained from supplied handoff; not every document or numeric transfer independently re-read in this documentation session.

Supplied bibliography: `industry\handoff\model\analysis\industry_fuel_compatibility_2026_10_31\references.csv`, key `eeca_biomass`.

H09 — Decarbonisation technologies

{Ceramics UK}. (n.d.). *Decarbonisation technologies*. inherited reference.

Primary/preserved source.

Use: Identifiable reference retained from the supplied industry methodology/bibliography. Some URLs duplicate a curated work using a different endpoint.

Verification: Bibliographic metadata retained from supplied handoff; not every document or numeric transfer independently re-read in this documentation session.

Supplied bibliography: `industry\handoff\model\analysis\industry_fuel_compatibility_2026_10_31\references.csv`, key `ceramics_uk`.

H10 — VOLTA: One year demonstrating hybrid melting at industrial scale

{Saint-Gobain}. (2026). *VOLTA: One year demonstrating hybrid melting at industrial scale*. inherited reference.

Primary/preserved source.

Use: Identifiable reference retained from the supplied industry methodology/bibliography. Some URLs duplicate a curated work using a different endpoint.

Verification: Bibliographic metadata retained from supplied handoff; not every document or numeric transfer independently re-read in this documentation session.

Supplied bibliography: `industry\handoff\model\analysis\industry_fuel_compatibility_2026_10_31\references.csv`, key `saint_gobain_volta`.

H11 — Arvedi ESP: Endless casting and rolling

{Primetals Technologies}. (n.d.). *Arvedi ESP: Endless casting and rolling*. inherited reference.

Primary/preserved source.

Use: Identifiable reference retained from the supplied industry methodology/bibliography. Some URLs duplicate a curated work using a different endpoint.

Verification: Bibliographic metadata retained from supplied handoff; not every document or numeric transfer independently re-read in this documentation session.

Supplied bibliography: `industry\handoff\model\analysis\industry_fuel_compatibility_2026_10_31\references.csv`, key `primetals_esp`.

H12 — Ammonia Technology Roadmap

{International Energy Agency}. (2021). *Ammonia Technology Roadmap*. inherited reference.

Primary/preserved source.

Use: Identifiable reference retained from the supplied industry methodology/bibliography. Some URLs duplicate a curated work using a different endpoint.

Verification: Bibliographic metadata retained from supplied handoff; not every document or numeric transfer independently re-read in this documentation session.

Supplied bibliography: `industry\handoff\model\analysis\industry_fuel_compatibility_2026_10_31\references.csv`, key `iea_ammonia`.

H13 — World's First 1MW Ammonia Boiler Successfully Demonstrated by UK Consortium via Amburn Project

{Amburn project consortium}. (2026). *World's First 1MW Ammonia Boiler Successfully Demonstrated by UK Consortium via Amburn Project*. inherited reference.

Primary/preserved source.

Use: Identifiable reference retained from the supplied industry methodology/bibliography. Some URLs duplicate a curated work using a different endpoint.

Verification: Bibliographic metadata retained from supplied handoff; not every document or numeric transfer independently re-read in this documentation session.

Supplied bibliography: `industry\handoff\model\analysis\industry_fuel_compatibility_2026_10_31\references.csv`, key `amburn_demo_2026`.

H14 — Technical Verifications for Practical Application Completed on Ammonia Single Firing Burner for Thermal Power Plant Boilers

{IHI Corporation}. (2026). *Technical Verifications for Practical Application Completed on Ammonia Single Firing Burner for Thermal Power Plant Boilers*. inherited reference.

Primary/preserved source.

Use: Identifiable reference retained from the supplied industry methodology/bibliography. Some URLs duplicate a curated work using a different endpoint.

Verification: Bibliographic metadata retained from supplied handoff; not every document or numeric transfer independently re-read in this documentation session.

Supplied bibliography: `industry\handoff\model\analysis\industry_fuel_compatibility_2026_10_31\references.csv`, key `ihi_ammonia_burner_2026`.

H15 — Methanol Industrial Boilers

{Methanol Institute}. (n.d.). *Methanol Industrial Boilers*. inherited reference.

Primary/preserved source.

Use: Identifiable reference retained from the supplied industry methodology/bibliography. Some URLs duplicate a curated work using a different endpoint.

Verification: Bibliographic metadata retained from supplied handoff; not every document or numeric transfer independently re-read in this documentation session.

Supplied bibliography: `industry\handoff\model\analysis\industry_fuel_compatibility_2026_10_31\references.csv`, key `methanol_boilers_2020`.

H16 — DME: Production, distribution and storage

{IEA Advanced Motor Fuels TCP}. (n.d.). *DME: Production, distribution and storage*. inherited reference.

Primary/preserved source.

Use: Identifiable reference retained from the supplied industry methodology/bibliography. Some URLs duplicate a curated work using a different endpoint.

Verification: Bibliographic metadata retained from supplied handoff; not every document or numeric transfer independently re-read in this documentation session.

Supplied bibliography: `industry\handoff\model\analysis\industry_fuel_compatibility_2026_10_31\references.csv`, key `iea_amf_dme`.

Fixed-OPEX source workbooks

T01 — Technology-data workbook identified as DEA_CCS in the supplied fixed-OPEX source table

Danish Energy Agency. (n.d.). *Technology-data workbook identified as DEA_CCS in the supplied fixed-OPEX source table*. technology workbook.

Primary/preserved source.

Use: Source endpoint retained for 416 technology/year/workbook rate records; see fixed_opex_sources.csv for an example sheet/cell trace.

Verification: Source URL and extraction provenance read from the live model CSV. Full downloaded catalogue and every numeric conversion were not independently re-audited here.

T02 — Technology-data workbook identified as DEA_FUELS in the supplied fixed-OPEX source table

Danish Energy Agency. (n.d.). *Technology-data workbook identified as DEA_FUELS in the supplied fixed-OPEX source table*. technology workbook.

Primary/preserved source.

Use: Source endpoint retained for 1456 technology/year/workbook rate records; see fixed_opex_sources.csv for an example sheet/cell trace.

Verification: Source URL and extraction provenance read from the live model CSV. Full downloaded catalogue and every numeric conversion were not independently re-audited here.

T03 — Technology-data workbook identified as DEA_HEAT in the supplied fixed-OPEX source table

Danish Energy Agency. (n.d.). *Technology-data workbook identified as DEA_HEAT in the supplied fixed-OPEX source table*. technology workbook.

Primary/preserved source.

Use: Source endpoint retained for 1144 technology/year/workbook rate records; see fixed_opex_sources.csv for an example sheet/cell trace.

Verification: Source URL and extraction provenance read from the live model CSV. Full downloaded catalogue and every numeric conversion were not independently re-audited here.

T04 — Technology-data workbook identified as DEA_NETWORK in the supplied fixed-OPEX source table

Danish Energy Agency. (n.d.). *Technology-data workbook identified as DEA_NETWORK in the supplied fixed-OPEX source table*. technology workbook.

Primary/preserved source.

Use: Source endpoint retained for 416 technology/year/workbook rate records; see fixed_opex_sources.csv for an example sheet/cell trace.

Verification: Source URL and extraction provenance read from the live model CSV. Full downloaded catalogue and every numeric conversion were not independently re-audited here.

T05 — Technology-data workbook identified as DEA_POWER in the supplied fixed-OPEX source table

Danish Energy Agency. (n.d.). *Technology-data workbook identified as DEA_POWER in the supplied fixed-OPEX source table*. technology workbook.

Primary/preserved source.

Use: Source endpoint retained for 41288 technology/year/workbook rate records; see fixed_opex_sources.csv for an example sheet/cell trace.

Verification: Source URL and extraction provenance read from the live model CSV. Full downloaded catalogue and every numeric conversion were not independently re-audited here.

T06 — Technology-data workbook identified as DEA_PROCESS in the supplied fixed-OPEX source table

Danish Energy Agency. (n.d.). *Technology-data workbook identified as DEA_PROCESS in the supplied fixed-OPEX source table*. technology workbook.

Primary/preserved source.

Use: Source endpoint retained for 2288 technology/year/workbook rate records; see fixed_opex_sources.csv for an example sheet/cell trace.

Verification: Source URL and extraction provenance read from the live model CSV. Full downloaded catalogue and every numeric conversion were not independently re-audited here.

T07 — Technology-data workbook identified as DEA_STORAGE in the supplied fixed-OPEX source table

Danish Energy Agency. (n.d.). *Technology-data workbook identified as DEA_STORAGE in the supplied fixed-OPEX source table*. technology workbook.

Primary/preserved source.

Use: Source endpoint retained for 208 technology/year/workbook rate records; see fixed_opex_sources.csv for an example sheet/cell trace.

Verification: Source URL and extraction provenance read from the live model CSV. Full downloaded catalogue and every numeric conversion were not independently re-audited here.

Provenance appendices

- URLs extracted from executable source constants: includes APIs, XML namespaces and plotting resources; not every URL is a dataset/reference.

- Workbook source/reference URL occurrences: exact workbook/sheet/cell context.
- Fixed-OPEX source groups: retained source endpoints and example extraction cells.
- Documented file hashes and input hashes.

Unresolved provenance

Generic technology defaults, some future demand/build/policy inputs, bundled profile/national-table assumptions and source price-year gaps remain inherited assumptions unless their own metadata establish a specific primary source. The manuals identify these rather than assigning a misleading scientific citation. Complete parameter-level re-auditing would require the original extraction workbooks, editions, page/table/cell references and conversion calculations.

Preparation command-line reference

Generated from the actual `add_argument` call sites. Expressions such as `SCRIPT_DIR / ...` show the source default; they are not literal shell arguments. The authoritative fully resolved help is the script's `--help` output. These commands prepare data rather than execute H2RES.

---

generate_all_sector_inputs.py

Option	Default	Choices / action	Help
`--countries`	`['HR']`	`store`	
`--sectors`	`list(SECTORS)`	`store`	'Comma-separated subset of power,industry,transport,heating.'
`--spatial-level`	`'nuts3'`	`['nuts3', 'nuts2', 'nuts1', 'country', 'custom']`	
`--zone-map`	`argparse default`	`store`	'Required with --spatial-level custom; country,nuts3,zone schema.'
`--out-root`	`SCRIPT_DIR / 'outputs_all'`	`store`	
`--data-root`	`SCRIPT_DIR / 'data_all'`	`store`	
`--nuts3`	`argparse default`	`store`	
`--refresh-sources`	`argparse default`	`'store_true'`	
`--efuels`	`False`	`argparse.BooleanOptionalAction`	
`--industry-mode`	`'categories'`	`('regular', 'categories', 'complex')`	'Categories and complex industry are mutually exclusive model modes.'
`--industry-electricity-case`	`'conservative'`	`('conservative', 'technical_2050')`	
`--industry-source-root`	`argparse default`	`store`	'Directory containing country JRC-IDEES-2023 Industry and both balance workbooks; otherwise use cache/public downloads'
`--industry-download`	`True`	`argparse.BooleanOptionalAction`	
`--industry-base-year`	`2020`	`store`	
`--industry-waste-allocation`	`'sector-blend'`	`('strict', 'sector-blend')`	'Use exact single-class waste evidence, or an explicitly recorded sector-blend assumption where needed'
`--industry-zone-weights`	`argparse default`	`store`	"Complex industry regional allocation: country,zone,Industry,weight; '*' means every sector"

Option	Default	Choices / action	Help
`--industry-regional-allocation`	`'require-weights'`	`('require-weights', 'equal')`	'National JRC data need explicit weights at regional resolution; equal is an explicit proxy'
`--complex-industry-sectors`	`'all'`	`store`	'Complex sector selection: all or comma-separated native Industry labels'
`--model-root`	`argparse default`	`store`	'Snapshot the live data from this model directory into the portable reference.'
`--model-reference`	`DEFAULT_REFERENCE`	`store`	
`--model-export`	`True`	`argparse.BooleanOptionalAction`	'Export canonical data files and the selected current model workbook.'
`--export-only`	`argparse default`	`'store_true'`	'Package existing sector outputs without downloading or regenerating sources.'
`--import-model-inputs`	`argparse default`	`'store_true'`	"Copy the supplied model's complete baseline inputs, preserving its original geography."
`--require-model-ready`	`argparse default`	`'store_true'`	'Return an error if generated curves or model zone coverage are incomplete.'
`--scenario-preset`	`argparse default`	`('reference', 'res100')`	'Keep reference assumptions or target 100%% renewable energy.'
`--scenario-scope`	`argparse default`	`('all-energy', 'electricity')`	
`--scenario-config`	`argparse default`	`store`	'JSON scenario assumptions and investment overrides.'
`--planning-years`	`argparse default`	`store`	'Comma-separated equally spaced decision years.'
`--start-year`	`argparse default`	`store`	
`--end-year`	`argparse default`	`store`	
`--year-step`	`argparse default`	`store`	
`--renewable-target-year`	`argparse default`	`store`	
`--renewable-start-share`	`argparse default`	`store`	'Starting electricity renewable-share assumption, from 0 to 1.'
`--investment-multiplier`	`argparse default`	`store`	'Scale reference annual build limits; installed-capacity ceilings stay unchanged.'
`--era5`	`argparse default`	`store`	
`--temperature-source`	`'open-meteo'`	`['open-meteo', 'cds-netcdf']`	
`--population-tif`	`argparse default`	`store`	
`--accept-population-terms`	`argparse default`	`'store_true'`	
`--skip-heating-raw`	`argparse default`	`'store_true'`	

Option	Default	Choices / action	Help
`--heating-input-root`	`SCRIPT_DIR / 'heating'`	`store`	
`--hydro-year`	`2020`	`store`	
`--hydro-profiles`	`packaged_hydro / 'runoff_shapes_2020.npz' if (packaged_hydro / 'runoff_shapes_2020.npz').exists() else bulk_hydro / 'outputs/profiles/runoff_shapes_2020.npz'`	`store`	
`--hydro-units`	`packaged_hydro / 'hydro_units.csv' if (packaged_hydro / 'hydro_units.csv').exists() else bulk_hydro / 'outputs/dispaset/hydro_units.csv'`	`store`	
`--hydro-plants`	`packaged_hydro / 'jrc-hydro-power-plant-database.csv' if (packaged_hydro / 'jrc-hydro-power-plant-database.csv').exists() else bulk_hydro / 'data/plants/jrc-hydro-power-plant-database.csv'`	`store`	
`--skip-hydro-profiles`	`argparse default`	`'store_true'`	
`--hydro-output-level`	`'match-power'`	`('match-power', 'aggregated', 'individual', 'both')`	'Hydro curve level; default follows --power-output-level.'
`--power-output-level`	`'aggregated'`	`('aggregated', 'individual', 'both')`	'Write zonally aggregated plants, individual plants, or both.'
`--skip-vre-profiles`	`argparse default`	`'store_true'`	
`--vre-year`	`2020`	`store`	
`--vre-irena-year`	`argparse default`	`store`	
`--vre-weather-source`	`'openmeteo'`	`('openmeteo', 'renewables-ninja')`	'Weather backend; Renewables.ninja reads RENEWABLES_NINJA_TOKEN.'
`--renewables-ninja-dataset`	`'merra2'`	`store`	
`--vre-calibrate`	`True`	`argparse.BooleanOptionalAction`	
`--irena-xlsx`	`SCRIPT_DIR / 'power' / 'VRES_distributions' / 'IRENA_Statistics_Extract_2025H2.xlsx'`	`store`	
`--electricity-demand-year`	`2020`	`store`	
`--skip-electricity-demand`	`argparse default`	`'store_true'`	'Skip ENTSO-E national load download and EIGL NUTS3/model-zone allocation.'

h2res_input_gui.py

Option	Default	Choices / action	Help
`--check`	`argparse default`	`'store_true'`	'Check Python and preparation dependencies without opening a window'
`--smoke-test`	`argparse default`	`'store_true'`	'Create and close the interface while keeping its window hidden'

preparation_dependencies.py

Option	Default	Choices / action	Help
`--sectors`	`'power,industry,transport,heating'`	`store`	
`--industry-mode`	`'categories'`	`('regular', 'categories', 'complex')`	

run_model_scenario.py

Option	Default	Choices / action	Help
`--model-root`	`argparse default`	`store`	
`--bundle`	`Path(__file__).resolve().parent`	`store`	
`--solve`	`argparse default`	`'store_true'`	'Build and solve; default only checks the actual input reader'
`--scenario-name`	`argparse default`	`store`	

industry/generate_industry_inputs_standalone.py

Option	Default	Choices / action	Help
`--countries`	`parse_countries('HR')`	`store`	'Comma-separated ISO2 list, e.g. HR,SI,IT, or ALL/EU to run every country found in JRC-IDEES with SECURES profiles'
`--year`	`2019`	`store`	
`--secures-year`	`2010`	`store`	
`--nuts-level`	`3`	`[0, 1, 2, 3]`	
`--reserve`	`0.15`	`store`	
`--out-root`	`RUN_DIR`	`store`	

Option	Default	Choices / action	Help
`--data-dir`	`DATA_DIR`	`store`	
`--empty-facility-dir`	`DATA_DIR / 'empty_facility_proxy'`	`store`	
`--clean`	`True`	`argparse.BooleanOptionalAction`	
`--debug`	`argparse default`	`'store_true'`	
`--adjust-hourly`	`argparse default`	`'store_true'`	
`--keep-heat-categories`	`argparse default`	`'store_true'`	'Keep temperature bands in Industry_data; default aggregates to the sector-level shape used by the legacy workbook.'
`--efuels`	`False`	`argparse.BooleanOptionalAction`	'Use the e-fuel-compatible table as canonical Industry_data. All regular/e-fuel and heat-category variants are always written.'
`--refresh-helpers`	`argparse default`	`'store_true'`	'Rewrite embedded helper scripts before running.'
`--download-sources`	`True`	`argparse.BooleanOptionalAction`	
`--refresh-sources`	`argparse default`	`'store_true'`	
`--nuts3-url`	`NUTS_URL`	`store`	
`--jrc-idees-url`	`JRC_IDEES_URL`	`store`	
`--secures-url`	`SECURES_URL`	`store`	
`--iep-zip-url`	`None`	`store`	
`--eutl-zip-url`	`None`	`store`	
`--eigl-zip-url`	`None`	`store`	
`--nuts3`	`None`	`store`	
`--jrc-root`	`None`	`store`	
`--secures-root`	`None`	`store`	
`--iep-dir`	`None`	`store`	
`--eutl-dir`	`None`	`store`	
`--eigl-dir`	`None`	`store`	
`--eigl-year`	`None`	`store`	
`--skip-eigl`	`False`	`argparse.BooleanOptionalAction`	
`--tech-template`	`None`	`store`	

industry/generate_complex_industry_inputs.py

Option	Default	Choices / action	Help
`--countries`	`'HR'`	`store`	
`--zone-map`	`argparse default`	`store`	'Shared country,nuts3,zone[,system_new] CSV; omitted means national zones'
`--reference`	`ROOT / 'model_reference'`	`store`	
`--output`	`ROOT / 'outputs_all/industry/combined'`	`store`	
`--cache`	`ROOT / 'data_all/industry/jrc_idees_2023'`	`store`	
`--source-root`	`argparse default`	`store`	
`--download`	`True`	`argparse.BooleanOptionalAction`	
`--refresh`	`argparse default`	`'store_true'`	
`--efuels`	`False`	`argparse.BooleanOptionalAction`	
`--electricity-case`	`'conservative'`	`('conservative', 'technical_2050')`	
`--base-year`	`2020`	`store`	
`--zone-weights`	`argparse default`	`store`	
`--regional-allocation`	`'require-weights'`	`('require-weights', 'equal')`	
`--selected-sectors`	`'all'`	`store`	
`--waste-allocation`	`'sector-blend'`	`('strict', 'sector-blend')`	

transport/script/generate_transport_inputs_standalone.py

Option	Default	Choices / action	Help
`--countries`	`parse_countries('HR')`	`store`	
`--nuts3`	`argparse default`	`store`	
`--national-dir`	`argparse default`	`store`	
`--pbf-dir`	`argparse default`	`store`	
`--oil-raster`	`argparse default`	`store`	
`--el-raster`	`argparse default`	`store`	
`--zone-map`	`argparse default`	`store`	
`--out-root`	`SCRIPT_DIR / 'outputs_transport'`	`store`	

Option	Default	Choices / action	Help
`--data-dir`	`DEFAULT_DATA_DIR`	`store`	
`--curves-dir`	`argparse default`	`store`	
`--refresh-sources`	`argparse default`	`'store_true'`	
`--efuels`	`False`	`argparse.BooleanOptionalAction`	'Also generate transport_ICE_prim. Without this flag only transport_all is written.'
`--metric-crs`	`'EPSG:3035'`	`store`	
`--skip-allocation`	`argparse default`	`'store_true'`	'Aggregate existing *_NUTS3.csv files without repeating geospatial allocation.'

---

heating/generate_heating_inputs_standalone.py

Option	Default	Choices / action	Help
`--countries`	`parse_countries('HR')`	`store`	
`--source-year`	`2021`	`store`	
`--efuels`	`False`	`argparse.BooleanOptionalAction`	
`--model-year`	`YEAR`	`store`	
`--zone-map`	`argparse default`	`store`	
`--out-root`	`SCRIPT_DIR / 'outputs_heating'`	`store`	
`--data-dir`	`SCRIPT_DIR / 'heating_data'`	`store`	
`--raw-output-root`	`SCRIPT_DIR / 'generated_nuts3_heating'`	`store`	
`--skip-raw-generation`	`argparse default`	`'store_true'`	'Only aggregate existing raw outputs supplied with --input-root.'
`--input-root`	`SCRIPT_DIR`	`store`	
`--refresh-sources`	`argparse default`	`'store_true'`	
`--rebuild-raw`	`argparse default`	`'store_true'`	
`--accept-population-terms`	`argparse default`	`'store_true'`	'Confirm acceptance of the Eurostat/GISCO population-grid terms.'
`--nuts3-geojson`	`argparse default`	`store`	
`--population-tif`	`argparse default`	`store`	
`--era5`	`argparse default`	`store`	'Existing ERA5/ERA5-Land NetCDF.'
`--temperature-source`	`'open-meteo'`	`['open-meteo', 'cds-netcdf']`	'Temperature source when --era5 is not supplied: cached Open-Meteo ERA5-Land centroid series (default) or a CDS NetCDF download.'

Option	Default	Choices / action	Help
`--eigl-dir`	`argparse default`	`store`	
`--hotmaps-dh-tif`	`argparse default`	`store`	
`--idees-root`	`argparse default`	`store`	
`--powerplants-csv`	`argparse default`	`store`	
`--dh-chp-lambda`	`0.9`	`store`	'DH spatial-weight blend: 0=Hotmaps only, 1=CHP only (default: 0.9).'

---

power/generators/generate_genco_power_sector.py

Option	Default	Choices / action	Help
`--mode`	`'raw'`	`['raw', 'built']`	
`--built-dir`	`None`	`store`	
`--out`	`DEFAULT_OUT`	`store`	
`--countries`	`None`	`store`	'Comma-separated ISO2 list, e.g. "HR,SI,IT".'
`--include-country-name-files`	`argparse default`	`'store_true'`	
`--keep-source-file`	`argparse default`	`'store_true'`	
`--summary`	`argparse default`	`'store_true'`	
`--output-format`	`'sdewes18'`	`['sdewes18', 'detailed']`	
`--rebuild-built`	`argparse default`	`'store_true'`	'Deprecated alias for --mode raw.'
`--powerplants`	`DEFAULT_POWERPLANTS`	`store`	
`--jrc-hydro`	`DEFAULT_HYDRO`	`store`	
`--template-genco`	`None`	`store`	
`--hr-defaults-xlsx`	`None`	`store`	
`--nuts3-geojson`	`DEFAULT_NUTS3`	`store`	
`--zone-mode`	`'country1'`	`['country1', 'country']`	
`--download-sources`	`True`	`argparse.BooleanOptionalAction`	
`--refresh-sources`	`argparse default`	`'store_true'`	
`--clean-built-dir`	`True`	`argparse.BooleanOptionalAction`	
`--powerplants-url`	`POWERPLANTS_URL`	`store`	
`--jrc-hydro-url`	`JRC_HYDRO_URL`	`store`	
`--nuts3-url`	`NUTS3_URL`	`store`	

power/generators/build_genco_union_plus_template_defaults_all.py

Option	Default	Choices / action	Help
`--powerplants`	`argparse default`	`store`	
`--jrc-hydro`	`argparse default`	`store`	
`--template-genco`	`argparse default`	`store`	
`--hr-defaults-xlsx`	`argparse default`	`store`	
`--out-dir`	`argparse default`	`store`	
`--powerplants-country-col`	`'Country'`	`store`	
`--powerplants-fuel-col`	`'Fueltype'`	`store`	
`--powerplants-tech-col`	`'Technology'`	`store`	
`--powerplants-name-col`	`'Name'`	`store`	
`--powerplants-cap-col`	`'Capacity'`	`store`	
`--hydro-country-col`	`'country_code'`	`store`	
`--hydro-name-col`	`'name'`	`store`	
`--hydro-cap-col`	`'capacity_mw'`	`store`	
`--hydro-type-col`	`'type'`	`store`	
`--zone-mode`	`'country1'`	`['country1', 'country']`	
`--countries`	`'auto'`	`store`	
`--decom-start-existing-cap`	`20.0`	`store`	
`--life-time`	`25.0`	`store`	
`--decom-start-new`	`25.0`	`store`	
`--nuts3-geojson`	`argparse default`	`store`	
`--nuts3-id-col`	`'NUTS_ID'`	`store`	
`--nuts3-level-col`	`'LEVL_CODE'`	`store`	
`--nuts3-level`	`3`	`store`	
`--nuts3-out-col`	`'NUTS3'`	`store`	

power/hydro/generate_hydro_inputs_standalone.py

Option	Default	Choices / action	Help
`--year`	`2020`	`store`	
`--countries`	`argparse default`	`store`	
`--detailed-genco`	`argparse default`	`store`	
`--aggregated-genco`	`argparse default`	`store`	
`--aggregation-schema`	`argparse default`	`store`	

Option	Default	Choices / action	Help
`--profiles`	`DEFAULT_BULK / 'outputs/profiles/runoff_shapes_2020.npz'`	`store`	
`--hydro-units`	`DEFAULT_BULK / 'outputs/dispaset/hydro_units.csv'`	`store`	
`--plants`	`DEFAULT_BULK / 'data/plants/jrc-hydro-power-plant-database.csv'`	`store`	
`--outdir`	`argparse default`	`store`	
`--max-match-distance-km`	`20.0`	`store`	
`--output-level`	`'aggregated'`	`('aggregated', 'individual', 'both')`	'Write zone-aggregated curves, individual-plant curves, or both.'

---

power/demand/generate_electricity_demand_standalone.py

Option	Default	Choices / action	Help
`--countries`	`argparse default`	`store`	
`--year`	`argparse default`	`store`	
`--out-dir`	`argparse default`	`store`	
`--cache-dir`	`argparse default`	`store`	
`--weights-dir`	`DEFAULT_WEIGHTS`	`store`	
`--zone-map`	`argparse default`	`store`	
`--token`	`argparse default`	`store`	'ENTSO-E API token; preferably set ENTSOE_API_TOKEN instead.'
`--refresh`	`argparse default`	`'store_true'`	

---

power/VRES_distributions/build_vre_capacity_weighted_cf_from_schema_openmeteo_weighted.py

Option	Default	Choices / action	Help
`--schema`	`argparse default`	`store`	'Schema CSV (genco_*_schema.csv)'
`--out-dir`	`argparse default`	`store`	
`--country`	`'HR'`	`store`	'ISO2 (matches schema System). Default HR'
`--date-from`	`argparse default`	`store`	'YYYY-MM-DD'
`--date-to`	`argparse default`	`store`	'YYYY-MM-DD'

Option	Default	Choices / action	Help
`--timezone`	`'UTC'`	`store`	'Open-Meteo timezone (default UTC)'
`--unit-column`	`None`	`store`	'Schema unit column (new_unit_name or old_unit_name)'
`--output-suffix`	`''`	`store`	'Suffix added before .csv, e.g. _individual'
`--cache-dir`	`None`	`store`	'Persistent weather-response cache'
`--weather-source`	`'openmeteo'`	`('openmeteo', 'renewables-ninja')`	
`--ninja-token`	`os.environ.get('RENEWABLES_NINJA_TOKEN')`	`store`	
`--ninja-dataset`	`'merra2'`	`store`	
`--irena-xlsx`	`None`	`store`	'IRENA_Statistics_Extract_2025H2.xlsx'
`--irena-year`	`None`	`store`	'Year to use (default from --date-from)'
`--calibrate`	`argparse default`	`'store_true'`	'Calibrate national annual CF to IRENA'
`--pv-ghi-ref`	`1000.0`	`store`	'W/m2 giving CF=1 for PV proxy'
`--wind-cut-in`	`3.0`	`store`	
`--wind-rated`	`12.0`	`store`	
`--wind-cut-out`	`25.0`	`store`	
`--sleep-s`	`0.25`	`store`	
`--max-retries`	`8`	`store`	
`--verbose`	`argparse default`	`'store_true'`	

Code and configuration reference

Generated by `build_reference_inventory.py` from the documented local copy.

These are static call sites. Loops and flags determine the instantiated optimization model; counts are not solver matrix dimensions.

Configuration snapshot

Source	Setting	Value
`main.py:20`	`scen`	`'industry_testx'`
`main.py:25`	`model_policy_dat`	`'./policy/model_policy.py'`
`main.py:26`	`runtime_policy_dat`	`'./policy/runtime_policy.py'`
`main.py:37`	`rps_inv`	`True`
`main.py:38`	`carbonLimit`	`True`
`main.py:39`	`lulucf_removals_active`	`False`
`main.py:40`	`lulucf_removals_dat`	`'./data/lulucf_removals.csv'`
`main.py:41`	`res_inv`	`True`
`main.py:42`	`hydro_storage`	`True`
`main.py:43`	`save_csv`	`True`
`main.py:44`	`ceep_limit`	`False`
`main.py:45`	`Myopic`	`True`
`main.py:46`	`partial_perfect_foresight`	`True`
`main.py:47`	`simple_curves`	`True`
`main.py:48`	`outside_system_availability`	`True`
`main.py:49`	`outside_system_supply_shares_dat`	`'./data/outside_system_supply_shares.csv'`
`main.py:50`	`outside_system_normalized_curves_dat`	`'./data/outside_system_normalized_curves.csv'`
`main.py:54`	`generation_source_calibration_active`	`True`
`main.py:55`	`hydro_generation_cap_active`	`True`
`main.py:56`	`outside_trade_calibration_active`	`True`
`main.py:57`	`dh_chp_base_year_heat_calibration_active`	`True`
`main.py:58`	`dh_chp_heat_yoy_limit_active`	`True`
`main.py:61`	`model_heating`	`True`
`main.py:62`	`ratio_heat_indv`	`True`
`main.py:63`	`heating_shares_regulate`	`False`

Source	Setting	Value
main.py:64	individual_heat_pump_storage	False
main.py:65	DH_Storage	False
main.py:66	NoResToHeatInv	False
main.py:67	model_Thermal_units	True
main.py:68	model_Thermal_units_DH	True
main.py:69	model_HP_units	True
main.py:70	model_HP_units_DH	True
main.py:71	model_Solar_indv_heating	False
main.py:72	model_Solar_DH	False
main.py:73	model_cooling	False
main.py:74	electric_heating_yoy_limit_active	True
main.py:75	individual_gas_heat_yoy_limit_active	True
main.py:76	fossil_heating_nonincrease_active	True
main.py:77	fossil_boiler_phaseout_active	True
main.py:80	Option_1	False
main.py:81	Option_2	False
main.py:82	Option_3	False
main.py:83	Option_4	False
main.py:84	include_secc_links	False
main.py:87	model_transport_and_industry	True
main.py:88	industry_single_sector	False
main.py:89	model_industry_categories	True
main.py:90	industry_category_electricity_case	'conservative'
main.py:91	industry_complex_electricity_case	'conservative'
main.py:92	industry_storage	True
main.py:94	industry_fuel_share_ramp_active	True
main.py:95	industry_fuel_share_s_curve_active	True
main.py:96	industry_electric_share_yoy_limit_active	True
main.py:104	model_complex_industry	False
main.py:106	model_complex_non_metallic_minerals	False
main.py:107	model_complex_industry	bool(model_complex_industry or model_complex_non_metallic_minerals)
main.py:108	model_complex_non_metallic_minerals	model_complex_industry
main.py:115	complex_industry_dat_efuel	'./data/industry_complex_efuel.xlsx'
main.py:116	complex_industry_dat_no_efuel	'./data/industry_complex.xlsx'
main.py:120	two_stage_auto_mode	False
main.py:124	lightweight_representative_weeks_mode	False

Source	Setting	Value
main.py:125	two_stage_skip_stage1_shadow_exports	True
main.py:126	two_stage_resume_skip_completed_chunks	False
main.py:127	two_stage_resume_reuse_stage1_bundle	False
main.py:130	myopic_resume_from_results	False
main.py:131	validate_completed_heating_policy_history	False
main.py:132	two_stage_timing_logs	True
main.py:133	representative_capacity_mode	False
main.py:134	fixed_capacity_mode	False
main.py:135	dispatch_stage_data_path	'./results/' + scen + '/stage2_dispatch_bundle.pkl'
main.py:136	export_stage2_dispatch_bundle	False
main.py:137	rolling_dispatch_mode	False
main.py:138	rolling_write_solution_quality	False
main.py:139	rolling_write_window_files	True
main.py:140	rolling_keep_results_in_memory	False
main.py:141	data_read_cache_enabled	True
main.py:142	data_read_cache_copy_deep	True
main.py:143	transport_investment_limits_active	False
main.py:144	transport_pathway_corrections_active	True
main.py:145	transport_passenger_car_zero_fossil_2050_sensitivity	True
main.py:148	transport_allow_unconstrained_paid_upgrades_for_test	False
main.py:149	transport_no_base_year_vehicle_investment	False
main.py:152	transport_max_share_limits_active	True
main.py:157	transport_ice_hybrid_stock_share_non_increasing_active	False
main.py:158	transport_electrification_limits_active	True
main.py:159	transport_rail_electrification_link_active	False
main.py:160	rolling_auto_carbon_budget_slack	True
main.py:161	rolling_terminal_fuel_soc	True
main.py:162	rolling_terminal_fuel_soc_hard_balance	False
main.py:163	rolling_terminal_fuel_soc_link_sequential_years	True
main.py:164	rolling_terminal_fuel_soc_ramp_targets	True
main.py:165	rolling_resume_from_window_files	False
main.py:166	rolling_zero_stage1_fuel_soc0	True

Source	Setting	Value
main.py:167	rolling_fuel_soc_stage1_floor	False
main.py:168	stage2_quantity_guidance_active	True
main.py:169	oil_refinery_product_generation_decline_active	False
main.py:170	transport_ice_hybrid_fossil_share_decline_active	False
main.py:171	transport_ice_fuel_share_ramp_active	True
main.py:172	biomass_heating_nonincrease_active	True
main.py:173	biomass_feedstock_limit_active	True
main.py:176	model_efuel	True
main.py:183	complex_industry_dat	complex_industry_dat_efuel if model_efuel else complex_industry_dat_no_efuel
main.py:193	use_HVDC	False
main.py:194	model_terminals	False
main.py:195	h2_terminal_dat	'./data/h2_terminals.xlsx'
main.py:196	model_H2_transmission	False
main.py:197	model_oil_and_gas_products_supply	True
main.py:201	allow_fossil_fuel_exports	True
main.py:202	exogeneous_NH3_demand	False
main.py:204	non_CO2_polutants	False
main.py:205	model_H2_supply_demand_balance	True
main.py:232	model_CC	False
main.py:233	model_CO2_ammine_storeage	False
main.py:234	model_CO2_post_gen_storeage	True
main.py:235	complex_industry_process_capture_required	False
main.py:236	separate_industry_process_carbon_balances	False
main.py:245	shadow_prices	False
main.py:255	solver_stage1_auto_retry_suboptimal	False
main.py:256	solver_stage1_annual_pf_aggregation_retry	True
main.py:257	solver_allow_provisional_solution	True
main.py:258	solver_enable_dual_simplex_fallback	True
main.py:259	solver_finalize_stalled_barrier_with_crossover	False
main.py:260	solver_skip_dual_simplex_after_acceptable_stall	False
main.py:261	write_solver_diagnostics	True
main.py:262	solver_diagnostics_on_problem_only	True

Source	Setting	Value
main.py:263	solver_diagnostics_compute_kappa_exact	False
main.py:264	disable_transport_infrastructure	True
main.py:268	fuel_price_dat	'./data/fuel_prices.csv'
main.py:269	inflows_dat	'./data/hydro_inflows.csv'
main.py:270	heat_demand_dat_indv	'./data/heating_individual.csv'
main.py:271	cool_demand_dat_indv	'./data/cooling_individual.csv'
main.py:272	heat_demand_dat_DH	'./data/heating_district.csv'
main.py:273	cool_demand_dat_DH	'./data/cooling_district.csv'
main.py:274	ev_transpload_dat	'./data/transport_profiles.csv'
main.py:275	transploaddemand_dat	'./data/transport_demand.csv'
main.py:276	importcost_dat	'./data/import_prices.csv'
main.py:277	heat_dem_industry_dat	'./data/industry_heat.csv'
main.py:278	genco_dat	'./data/generators.csv'
main.py:279	policy_dat	'./data/policy_efuel.xlsx' if model_efuel else './data/policy.xlsx'
main.py:284	demand_dat	'./data/electricity_demand.csv'
main.py:285	demand_industry_el	'./data/industry_electricity.csv'
main.py:286	demand_heating_el	'./data/heating_electricity.csv'
main.py:287	solar_thermal_aval	'./data/solar_thermal_availability.csv'
main.py:292	flex_tech_name	'flex_tech'
main.py:307	flex_tech_dat	'./data/' + flex_tech_name + '.xlsx'
main.py:309	heat_dem_industry_dat	'./data/industry_category_profiles.csv' if model_industry_categories else './data/industry_heat.csv'
main.py:321	el_hourly_cost_dat	'./data/electricity_prices.csv'
main.py:348	_data_read_cache	globals().setdefault('_data_read_cache', OrderedDict())
main.py:352	_excel_file_cache	globals().setdefault('_excel_file_cache', OrderedDict())
main.py:356	_data_read_cache_stats	globals().setdefault('_data_read_cache_stats', {'hits': 0, 'misses': 0, 'read_time': 0.0})
policy/model_policy.py:11	generation_annual_build_limit_multipliers	{'HR_Wind': 2.0, 'HR_Wind_1': 2.0, 'HR_PV': 2.0, 'HR_PV_1': 2.0}
policy/model_policy.py:19	dh_chp_heat_max_annual_change	0.25
policy/model_policy.py:20	dh_chp_heat_history_tolerance_mwh	1.0
policy/model_policy.py:21	individual_gas_heat_max_annual_change	0.25
policy/model_policy.py:22	individual_gas_heat_reduction_limit_years	5
policy/model_policy.py:23	fossil_heating_history_tolerance_mwh	1.0

Source	Setting	Value
policy/model_policy.py:24	fossil_boiler_phaseout_years	{'coal': 2030, 'gas': 2035, 'oil': 2035}
policy/model_policy.py:25	dh_chp_base_year_heat_targets_twh	{('TE-TO_Zagreb', 'HR1', 'HR'): 0.9, ('EL-TO_Zagreb', 'HR1', 'HR'): 0.55, ('TE_-TO_Osijek', 'HR1', 'HR'): 0.232}
policy/model_policy.py:31	lulucf_removals_fixed_tco2e_per_year	0.0
policy/model_policy.py:32	electric_heating_max_annual_decrease	0.1
policy/model_policy.py:33	electric_heating_max_annual_increase	0.1
policy/model_policy.py:34	electric_heating_week_hours	24 * 7
policy/model_policy.py:35	electric_heating_individual_units	('Air_to_water_HP', 'Ground_to_water_HP', 'Electric_heater', 'Water_to_water_HP')
policy/model_policy.py:43	industry_fuel_share_max_change_per_year	0.1
policy/model_policy.py:44	industry_fuel_share_ramp_mode	'endogenous_s_curve'
policy/model_policy.py:45	industry_fuel_share_s_curve_initial_fraction	0.02
policy/model_policy.py:46	industry_fuel_share_s_curve_growth_rate	0.5
policy/model_policy.py:50	industry_fuel_share_s_curve_reference_years	5
policy/model_policy.py:54	industry_fuel_share_s_curve_directions	{'Gas': 'decrease', 'Coal': 'decrease', 'Lignite': 'decrease', 'Oil': 'decrease', 'fuel_oil': 'decrease', 'diesel': 'decrease', 'petrol': 'decrease', 'kerosene': 'decrease', 'LPG': 'decrease', 'Biomass': 'decrease', 'lignin': 'increase'}
policy/model_policy.py:61	industry_electric_share_max_annual_decrease	0.1
policy/model_policy.py:62	industry_electric_share_history_tolerance	1e-06
policy/model_policy.py:65	transport_decom_max_share	None
policy/model_policy.py:66	transport_turnover_years	25
policy/model_policy.py:67	transport_turnover_units	('EV_car', 'FCEV_car', 'HYBRID_car', 'ICE_car', 'EV_truck', 'FCEV_truck', 'HYBRID_truck', 'ICE_truck', 'EV_MARINE_SHORT', 'FCEV_MARINE_SHORT', 'ICE_MARINE_SHORT')
policy/model_policy.py:80	transport_turnover_ramp_start_year	2025
policy/model_policy.py:81	transport_turnover_ramp_full_year	2030
policy/model_policy.py:82	transport_turnover_ramp_initial_share	1.5
policy/model_policy.py:83	TRANSPORT_TURNOVER_MAX_SHARE	0.3
policy/model_policy.py:84	transport_existing_fleet_age_spread_years	10
policy/model_policy.py:85	transport_forbidden_investment_units	()

Source	Setting	Value
policy/model_policy.py:86	transport_inv_max_share_by_unit	{'all': 0.2, 'EV_car': 0.42, 'FCEV_car': 0.25, 'ICE_AVIO_SHORT': 0.4, 'ICE_AVIO_LONG': 0.4, 'EL_RAIL': 0.4, 'EL_RAIL_CARGO': 0.4, 'ICE_MARINE_LONG': 0.4, 'ICE_MARINE_SHORT': 0.2}
policy/model_policy.py:97	transport_inv_total_max_by_sector	{}
policy/model_policy.py:98	transport_inv_max_ramp_start_year	2025
policy/model_policy.py:99	transport_inv_max_ramp_full_year	2028
policy/model_policy.py:100	transport_inv_max_ramp_initial_share	0.5
policy/model_policy.py:101	transport_inv_declining_max_share_by_unit	{}
policy/model_policy.py:102	transport_inv_decline_start_year	2025
policy/model_policy.py:103	transport_inv_decline_end_year	2050
policy/model_policy.py:104	transport_inv_decline_final_share	0.1
policy/model_policy.py:105	transport_stock_max_share_year	2050
policy/model_policy.py:111	transport_stock_max_share_by_unit	{}
policy/model_policy.py:112	transport_electrification_target_sectors	('Passenger_cars', 'commercial_vehicles', 'Motor_coaches_buses_and_trolley_buses', 'Rail_transport', 'Rail_metro_and_tram', 'Domestic_shipping', 'Domestic_aviation', 'International_aviation')
policy/model_policy.py:122	transport_non_decreasing_stock_units	('EL_RAIL', 'EL_RAIL_CARGO')
policy/model_policy.py:128	stage2_carbon_budget_mode	'cumulative_weighted'
policy/model_policy.py:129	stage2_carbon_hour_weight	0.25
policy/model_policy.py:130	stage2_carbon_heat_weight	0.75
policy/model_policy.py:131	stage2_carbon_el_weight	0.0
policy/model_policy.py:132	stage2_carbon_auto_slack_cost	1000000.0
policy/model_policy.py:133	stage2_carbon_slack_cost	0.0
policy/model_policy.py:134	rolling_refinery_hourly_cap_multiplier	1.0
policy/model_policy.py:135	rolling_h2_balance_shortage_cost	10000000.0
policy/model_policy.py:136	rolling_industry_cap_shortage_cost	10000000.0
policy/model_policy.py:137	rolling_terminal_fuel_soc_tolerance	0.0
policy/model_policy.py:138	rolling_terminal_fuel_soc_slack_cost	10000000.0
policy/model_policy.py:139	rolling_terminal_fuel_soc_lead_hours	8760
policy/model_policy.py:140	rolling_terminal_fuel_soc_ramp_exponent	1.0
policy/model_policy.py:141	storage_year_continuity_mode	'cyclic'
policy/model_policy.py:142	efuel_dh_heat_feedstock_aliases	('heat', 'dh_heat', 'district_heat', 'district heating')
policy/model_policy.py:143	stage2_quantity_guidance_tolerance	0.15

Source	Setting	Value
policy/model_policy.py:144	stage2_quantity_guidance_abs_tolerance	1e-06
policy/model_policy.py:145	stage2_quantity_guidance_slack_cost	1000000.0
policy/model_policy.py:146	oil_refinery_product_generation_change_limit	0.1
policy/model_policy.py:147	refinery_process_fuel_share	0.08
policy/model_policy.py:148	refinery_process_fuel_co2_key	'Oil'
policy/model_policy.py:149	transport_ice_hybrid_fossil_share_decline_scope	'annual'
policy/model_policy.py:151	transport_ice_fuel_share_change_limit	0.2
policy/model_policy.py:152	biomass_heating_fuel_aliases	('biomass',)
policy/model_policy.py:155	h2_transmission_cost_aliases	{'H2_pipe': 'H2CH'}
policy/model_policy.py:156	transmission_transformer_units	{'400kV_AC': 'transformer_AC', 'HVDC': 'transformer_HVDC'}
policy/model_policy.py:160	h2_pipeline_compressor_unit	'comp_station'
policy/model_policy.py:161	transmission_duplicate_policy	'first'
policy/model_policy.py:163	active_zone_overrides	None
policy/model_policy.py:166	co2_storage_capex_input_unit	'MEUR/t'
policy/model_policy.py:167	co2_amine_tank_capex_input_unit	'MEUR/t'
policy/model_policy.py:168	stat_grid_eff	0.9
policy/model_policy.py:169	ceep_price	1000
policy/model_policy.py:172	noload_el_cost	100000
policy/model_policy.py:173	discount_rate	0.05
policy/model_policy.py:174	discount_base_year	2025
policy/model_policy.py:175	capital_finance_rate	0.05
policy/model_policy.py:181	fuel_storage_capex_aliases	{'N2_stor': 'Gas_stor'}
policy/model_policy.py:182	fuel_storage_fixed_opex_fraction	0.01
policy/model_policy.py:188	oil_gas_initial_soc_fraction	{'Oil': 0.5, 'Gas': 0.5}
policy/model_policy.py:192	industry_electrification_full_applicability	True
policy/model_policy.py:193	industry_heat_pumps_active	True
policy/model_policy.py:194	industry_hp_profiles_dat	'./data/industry_category_profiles.csv'
policy/model_policy.py:200	complex_industry_sectors	('Non_metallic_minerals',)
policy/model_policy.py:203	industry_external_steam_efficiency	0.9
policy/runtime_policy.py:6	partial_foresight_representative_days	4
policy/runtime_policy.py:7	partial_foresight_day_hours	24
policy/runtime_policy.py:8	partial_foresight_representative_day_start_hours	(337, 2521, 4705, 6889)
policy/runtime_policy.py:11	partial_foresight_lookahead_years	10

Source	Setting	Value
policy/runtime_policy.py:12	representative_full_year_hours	8760
policy/runtime_policy.py:13	two_stage_stage1_export_timing	'before_stage2'
policy/runtime_policy.py:14	representative_weeks_count	4
policy/runtime_policy.py:16	representative_week_hours	24
policy/runtime_policy.py:17	representative_week_start_hours	partial_foresight_representative_day_start_hours
policy/runtime_policy.py:18	long_storage_guidance_default_band	0.0
policy/runtime_policy.py:19	rolling_dispatch_granularity	'week'
policy/runtime_policy.py:20	rolling_window_size	48
policy/runtime_policy.py:21	rolling_step_size	24
policy/runtime_policy.py:22	rolling_realized_horizon	24
policy/runtime_policy.py:23	rolling_forecast_avg_dev_percent	5.0
policy/runtime_policy.py:24	stage1_capacity_reserve_margin	0.2
policy/runtime_policy.py:25	rolling_unmet_industry_heat_cost	1000000000.0
policy/runtime_policy.py:26	rolling_unmet_dh_heat_cost	1000000000.0
policy/runtime_policy.py:27	rolling_unmet_indv_heat_cost	1000000000.0
policy/runtime_policy.py:28	data_read_cache_max_entries	256
policy/runtime_policy.py:31	solver_stage1_method	2
policy/runtime_policy.py:32	rolling_solver_method	1
policy/runtime_policy.py:35	solver_stage1_numeric_focus	1
policy/runtime_policy.py:39	solver_diagnostic_method	1
policy/runtime_policy.py:43	solver_diagnostic_time_limit_s	0
policy/runtime_policy.py:44	solver_diagnostic_numeric_focus	3
policy/runtime_policy.py:45	solver_diagnostic_aggregate	0
policy/runtime_policy.py:46	solver_objective_scale	1000.0
policy/runtime_policy.py:47	solver_stage1_scale_flag	2
policy/runtime_policy.py:51	solver_stage1_aggregate	0
policy/runtime_policy.py:54	solver_stage1_annual_pf_retry_aggregate	1
policy/runtime_policy.py:55	annual_accounting_row_scale	1000.0
policy/runtime_policy.py:56	fuel_storage_balance_row_scale	1000.0
policy/runtime_policy.py:57	fuel_storage_capacity_row_scale	1000.0
policy/runtime_policy.py:58	co2_storage_cost_row_scale	1000.0
policy/runtime_policy.py:59	solver_stage1_bar_homogeneous	1
policy/runtime_policy.py:62	solver_stage1_annual_s_curve_bar_homogeneous	0
policy/runtime_policy.py:63	solver_stage1_homogeneous_recovery	True
policy/runtime_policy.py:64	solver_stage1_bar_conv_tol	1e-08

Source	Setting	Value
policy/runtime_policy.py:65	solver_stage1_crossover	0
policy/runtime_policy.py:66	solver_stage1_crossover_basis	-1
policy/runtime_policy.py:67	solver_stage1_time_limit_s	0
policy/runtime_policy.py:68	solver_stage1_auto_extend_attempts	0
policy/runtime_policy.py:69	solver_stage1_auto_extend_s	3600
policy/runtime_policy.py:70	solver_stage1_retry_numeric_focus	3
policy/runtime_policy.py:71	solver_stage1_retry_scale_flag	1
policy/runtime_policy.py:72	solver_stage1_retry_bar_homogeneous	1
policy/runtime_policy.py:73	solver_provisional_max_constr_vio	0.2
policy/runtime_policy.py:76	physical_energy_balance_abs_tolerance	0.001
policy/runtime_policy.py:77	solver_provisional_max_bound_vio	0.0001
policy/runtime_policy.py:78	solver_provisional_max_dual_vio	0.0001
policy/runtime_policy.py:79	solver_provisional_max_dual_residual	3.0
policy/runtime_policy.py:80	solver_provisional_max_barrier_gap	0.03
policy/runtime_policy.py:81	solver_provisional_max_returned_objective_deviation	0.02
policy/runtime_policy.py:85	solver_stage1_bar_iter_limit	1000
policy/runtime_policy.py:87	solver_stage1_threads	32
policy/runtime_policy.py:88	solver_stage1_bar_order	-1
policy/runtime_policy.py:89	solver_auto_iis_max_rows	2000000
policy/runtime_policy.py:90	solver_bar_stall_time_s	0
policy/runtime_policy.py:91	solver_bar_stall_min_runtime_s	2400
policy/runtime_policy.py:92	solver_bar_stall_rel_improve	0.05
policy/runtime_policy.py:93	solver_bar_stall_accept_gap	0.002
policy/runtime_policy.py:94	solver_bar_stall_accept_compl	1e-05
policy/runtime_policy.py:95	solver_bar_stall_accept_primal_inf	0.001
policy/runtime_policy.py:96	solver_bar_stall_accept_dual_inf	0.0001
policy/runtime_policy.py:97	solver_stalled_barrier_finalize_time_s	3600
policy/runtime_policy.py:98	solver_fallback_numeric_focus	3
policy/runtime_policy.py:99	solver_fallback_time_limit_s	0
policy/runtime_policy.py:100	solver_fallback_threads	32
policy/runtime_policy.py:103	electricity_price_zero_threshold_eur_per_mwh	1.0
policy/runtime_policy.py:104	fuel_trade_price_zero_threshold_eur_per_mwh	1.0
policy/runtime_policy.py:106	transport_stock_variable_scale_by_sector	{'Passenger_cars': 1000.0, 'commercial_vehicles': 1000.0, 'Motor_coaches_buses_and_trolley_buses': 10.0, 'Domestic_shipping': 1000.0, 'Domestic_aviation': 100.0}

Source	Setting	Value
policy/runtime_policy.py:115	transport_service_variable_scale	1000.0
policy/runtime_policy.py:116	transport_service_variable_scale_by_sector	{'Passenger_cars': 1.0, 'commercial_vehicles': 10.0, 'Motor_coaches_buses_and_trolley_buses': 10.0, 'Domestic_aviation': 1.0}
policy/runtime_policy.py:122	numeric_abs_min	1e-09
policy/runtime_policy.py:123	numeric_abs_max	100000000.0
policy/runtime_policy.py:124	price_abs_max	10000.0
policy/runtime_policy.py:125	var_cost_abs_max	100000.0
policy/runtime_policy.py:126	transmission_km_abs_max	100000.0
policy/runtime_policy.py:127	fuel_storage_eff_min	0.0001
policy/runtime_policy.py:128	fuel_storage_eff_max	1.0
policy/runtime_policy.py:129	cap_inv_limit_active_max	500000.0
policy/runtime_policy.py:130	transport_coeff_eps	1e-05
policy/runtime_policy.py:131	transport_co2_block_hours	168
policy/runtime_policy.py:132	transport_historical_stock_zero_epsilon	1e-06
policy/runtime_policy.py:134	heating_stock_zero_epsilon_mw	1e-06
policy/runtime_policy.py:135	heating_history_zero_epsilon_mwh	0.001
policy/runtime_policy.py:137	transport_physical_bound_tolerance	1e-06
policy/runtime_policy.py:139	levelized_cost_min_annual_output_mwh	1.0
policy/runtime_policy.py:142	transport_inherited_stock_zero_epsilon	0.1
policy/runtime_policy.py:143	INDUSTRY_SHARE_ROUNDING_TOLERANCE	0.01
policy/runtime_policy.py:144	COMPLEX_INDUSTRY_DEMAND_AUDIT_REL_TOL	0.01
policy/runtime_policy.py:145	COMPLEX_INDUSTRY_DEMAND_AUDIT_ABS_TOL	0.001
policy/runtime_policy.py:147	transport_fixed_stock_abs_tolerance	0.1
policy/runtime_policy.py:150	transport_stock_relative_tolerance	0.001
policy/transport_pathways.py:9	transport_marine_capex_multiplier	{'methanol': 1.09, 'ammonia': 1.1}
policy/transport_pathways.py:10	transport_marine_pilot_energy_share	0.05
policy/transport_pathways.py:13	transport_marine_retrofit_capex_fraction	{'methanol': 0.16, 'ammonia': 0.24}
policy/transport_pathways.py:15	transport_marine_retrofit_remaining_years	10
policy/transport_pathways.py:20	transport_physical_upgrade_capex_fraction	{'bio_diesel': 0.03, 'ethanol': 0.05, 'methanol': 0.1}
policy/transport_pathways.py:23	transport_physical_upgrade_remaining_years_by_sector	{'Passenger_cars': 5, 'commercial_vehicles': 5, 'Motor_coaches_buses_and_trolley_buses': 8, 'Rail_transport': 10, 'Rail_metro_and_tram': 10, 'Domestic_shipping': 10, 'international_shipping': 10}

Source	Setting	Value
`policy/transport_pathways.py:31`	`transport_physical_upgrade_capex_meur_per_vehicle`	`{}`
`policy/transport_pathways.py:32`	`transport_upgraded_max_volume_share`	`{'bio_diesel': 1.0, 'ethanol': 0.85, 'methanol': 1.0}`
`policy/transport_pathways.py:34`	`transport_drop_in_fuels`	`('e_diesel', 'e_petrol', 'e_kerosene', 'SAF')`
`policy/transport_pathways.py:36`	`transport_fame_max_volume_share`	`0.07`
`policy/transport_pathways.py:37`	`transport_ethanol_max_volume_share`	`0.1`
`policy/transport_pathways.py:38`	`transport_methanol_max_volume_share`	`0.03`
`policy/transport_pathways.py:41`	`transport_synthetic_jet_max_volume_share`	`1.0`

Constraint families and locations

See full constraint inventory for flag/function context.

Source	Line	Name / naming expression
`scripts/reading_prev_and_vars_def_dual_industry_categories1.py`	1353	`(unnamed)`
`scripts/reading_prev_and_vars_def_dual_industry_categories1.py`	1404	`(unnamed)`
`scripts/reading_prev_and_vars_def_dual_industry_categories1.py`	1527	`(unnamed)`
`scripts/reading_prev_and_vars_def_dual_industry_categories1.py`	1528	`(unnamed)`
`scripts/scoped_policy.py`	184	`f'{family}[{','.join(map(str, key))}]'`
`scripts/scoped_policy.py`	202	`f'biomass_feedstock_annual_limit[{','.join(map(str, key))}]'`
`scripts/network_checks.py`	101	`f'h2_balance[{z},{s},{year},{p}]'`
`scripts/Build_model_dual_industry_categories_named_debug.py`	19	`(unnamed)`
`scripts/Build_model_dual_industry_categories_named_debug.py`	232	`f'ifsru_{suffix}_{idx}'`
`scripts/Build_model_dual_industry_categories_named_debug.py`	236	`f'ifsrd_{suffix}_{idx}'`
`scripts/Build_model_dual_industry_categories_named_debug.py`	1104	`f'{name}_carry_previous'`
`scripts/Build_model_dual_industry_categories_named_debug.py`	1334	`(unnamed)`
`scripts/Build_model_dual_industry_categories_named_debug.py`	1342	`(unnamed)`
`scripts/Build_model_dual_industry_categories_named_debug.py`	1390	`f'{var_name}_fixed_zero'`

Source	Line	Name / naming expression
`scripts/Build_model_dual_industry_categories_named_debug.py`	1413	`f'{var_name}_fixed_pin_upper'`
`scripts/Build_model_dual_industry_categories_named_debug.py`	1418	`f'{var_name}_fixed_pin_cap'`
`scripts/Build_model_dual_industry_categories_named_debug.py`	1428	`f'{var_name}_fixed_pin'`
`scripts/Build_model_dual_industry_categories_named_debug.py`	1466	`f'{var_name}_guide_lb'`
`scripts/Build_model_dual_industry_categories_named_debug.py`	1474	`f'{var_name}_guide_ub'`
`scripts/Build_model_dual_industry_categories_named_debug.py`	1570	`f'stage2_quantity_guidance_lb_{var_name}'`
`scripts/Build_model_dual_industry_categories_named_debug.py`	1574	`f'stage2_quantity_guidance_ub_{var_name}'`
`scripts/Build_model_dual_industry_categories_named_debug.py`	1696	`f'{_soc_var_name}_rolling_terminal_soc_balance'`
`scripts/Build_model_dual_industry_categories_named_debug.py`	1753	`'h2_electrolysiscap_fixed'`
`scripts/Build_model_dual_industry_categories_named_debug.py`	1765	`'h2_electrolysiscap_unpinned'`
`scripts/Build_model_dual_industry_categories_named_debug.py`	1780	`'h2_electrolysiscap'`
`scripts/Build_model_dual_industry_categories_named_debug.py`	1795	`'h2_steamcapacity_fixed'`
`scripts/Build_model_dual_industry_categories_named_debug.py`	1803	`'h2_steamcapacity'`
`scripts/Build_model_dual_industry_categories_named_debug.py`	1818	`'Power_to_h2eq'`
`scripts/Build_model_dual_industry_categories_named_debug.py`	1828	`'h2_electrolysis'`
`scripts/Build_model_dual_industry_categories_named_debug.py`	1835	`'h2_steam'`
`scripts/Build_model_dual_industry_categories_named_debug.py`	1849	`'steam_h2_generation_nonincrease'`
`scripts/Build_model_dual_industry_categories_named_debug.py`	1895	`'steam_h2_generation_nonincrease_from_previous'`
`scripts/Build_model_dual_industry_categories_named_debug.py`	1913	`'h2_storage_maxcap_fixed'`
`scripts/Build_model_dual_industry_categories_named_debug.py`	1924	`(unnamed)`
`scripts/Build_model_dual_industry_categories_named_debug.py`	1940	`'FC_capacity_fixed'`
`scripts/Build_model_dual_industry_categories_named_debug.py`	1951	`(unnamed)`
`scripts/Build_model_dual_industry_categories_named_debug.py`	1972	`(unnamed)`

Source	Line	Name / naming expression
scripts/Build_model_dual_industry_categories_named_debug.py	1977	(unnamed)
scripts/Build_model_dual_industry_categories_named_debug.py	1982	(unnamed)
scripts/Build_model_dual_industry_categories_named_debug.py	1989	'h2_storage_initial_soc_zero'
scripts/Build_model_dual_industry_categories_named_debug.py	2026	(unnamed)
scripts/Build_model_dual_industry_categories_named_debug.py	2034	(unnamed)
scripts/Build_model_dual_industry_categories_named_debug.py	2035	(unnamed)
scripts/Build_model_dual_industry_categories_named_debug.py	2036	(unnamed)
scripts/Build_model_dual_industry_categories_named_debug.py	2037	(unnamed)
scripts/Build_model_dual_industry_categories_named_debug.py	2039	'h2_electrolysis_base_year_investment_zero'
scripts/Build_model_dual_industry_categories_named_debug.py	2145	(unnamed)
scripts/Build_model_dual_industry_categories_named_debug.py	2155	(unnamed)
scripts/Build_model_dual_industry_categories_named_debug.py	2189	(unnamed)
scripts/Build_model_dual_industry_categories_named_debug.py	2202	(unnamed)
scripts/Build_model_dual_industry_categories_named_debug.py	2227	(unnamed)
scripts/Build_model_dual_industry_categories_named_debug.py	2245	(unnamed)
scripts/Build_model_dual_industry_categories_named_debug.py	2262	(unnamed)
scripts/Build_model_dual_industry_categories_named_debug.py	2295	(unnamed)
scripts/Build_model_dual_industry_categories_named_debug.py	2317	(unnamed)
scripts/Build_model_dual_industry_categories_named_debug.py	2339	(unnamed)
scripts/Build_model_dual_industry_categories_named_debug.py	2360	(unnamed)
scripts/Build_model_dual_industry_categories_named_debug.py	2368	'industries_share_fuel'
scripts/Build_model_dual_industry_categories_named_debug.py	2388	'industries_share_fuel_init2020'
scripts/Build_model_dual_industry_categories_named_debug.py	2408	(unnamed)

Source	Line	Name / naming expression
`scripts/Build_model_dual_industry_categories_named_debug.py`	2424	`(unnamed)`
`scripts/Build_model_dual_industry_categories_named_debug.py`	2442	`(unnamed)`
`scripts/Build_model_dual_industry_categories_named_debug.py`	2468	`'industry_low_carbon_fuel_substitution_cap'`
`scripts/Build_model_dual_industry_categories_named_debug.py`	2580	`(unnamed)`
`scripts/Build_model_dual_industry_categories_named_debug.py`	2590	`(unnamed)`
`scripts/Build_model_dual_industry_categories_named_debug.py`	2615	`(unnamed)`
`scripts/Build_model_dual_industry_categories_named_debug.py`	2625	`(unnamed)`
`scripts/Build_model_dual_industry_categories_named_debug.py`	2647	`(unnamed)`
`scripts/Build_model_dual_industry_categories_named_debug.py`	2657	`(unnamed)`
`scripts/Build_model_dual_industry_categories_named_debug.py`	2668	`(unnamed)`
`scripts/Build_model_dual_industry_categories_named_debug.py`	2681	`(unnamed)`
`scripts/Build_model_dual_industry_categories_named_debug.py`	2704	`(unnamed)`
`scripts/Build_model_dual_industry_categories_named_debug.py`	2725	`(unnamed)`
`scripts/Build_model_dual_industry_categories_named_debug.py`	2735	`(unnamed)`
`scripts/Build_model_dual_industry_categories_named_debug.py`	2746	`(unnamed)`
`scripts/Build_model_dual_industry_categories_named_debug.py`	2758	`(unnamed)`
`scripts/Build_model_dual_industry_categories_named_debug.py`	2769	`(unnamed)`
`scripts/Build_model_dual_industry_categories_named_debug.py`	2781	`(unnamed)`
`scripts/Build_model_dual_industry_categories_named_debug.py`	2793	`(unnamed)`
`scripts/Build_model_dual_industry_categories_named_debug.py`	2809	`(unnamed)`
`scripts/Build_model_dual_industry_categories_named_debug.py`	2828	`(unnamed)`
`scripts/Build_model_dual_industry_categories_named_debug.py`	2838	`(unnamed)`
`scripts/Build_model_dual_industry_categories_named_debug.py`	2855	`(unnamed)`

Source	Line	Name / naming expression
`scripts/Build_model_dual_industry_categories_named_debug.py`	2872	`(unnamed)`
`scripts/Build_model_dual_industry_categories_named_debug.py`	2896	`(unnamed)`
`scripts/Build_model_dual_industry_categories_named_debug.py`	2908	`(unnamed)`
`scripts/Build_model_dual_industry_categories_named_debug.py`	2920	`(unnamed)`
`scripts/Build_model_dual_industry_categories_named_debug.py`	2931	`(unnamed)`
`scripts/Build_model_dual_industry_categories_named_debug.py`	2941	`(unnamed)`
`scripts/Build_model_dual_industry_categories_named_debug.py`	2952	`(unnamed)`
`scripts/Build_model_dual_industry_categories_named_debug.py`	2961	`(unnamed)`
`scripts/Build_model_dual_industry_categories_named_debug.py`	2981	`(unnamed)`
`scripts/Build_model_dual_industry_categories_named_debug.py`	3006	`(unnamed)`
`scripts/Build_model_dual_industry_categories_named_debug.py`	3028	`(unnamed)`
`scripts/Build_model_dual_industry_categories_named_debug.py`	3046	`(unnamed)`
`scripts/Build_model_dual_industry_categories_named_debug.py`	3068	`(unnamed)`
`scripts/Build_model_dual_industry_categories_named_debug.py`	3109	`(unnamed)`
`scripts/Build_model_dual_industry_categories_named_debug.py`	3150	`(unnamed)`
`scripts/Build_model_dual_industry_categories_named_debug.py`	3206	`(unnamed)`
`scripts/Build_model_dual_industry_categories_named_debug.py`	3220	`(unnamed)`
`scripts/Build_model_dual_industry_categories_named_debug.py`	3233	`(unnamed)`
`scripts/Build_model_dual_industry_categories_named_debug.py`	3258	`(unnamed)`
`scripts/Build_model_dual_industry_categories_named_debug.py`	3267	`(unnamed)`
`scripts/Build_model_dual_industry_categories_named_debug.py`	3285	`(unnamed)`
`scripts/Build_model_dual_industry_categories_named_debug.py`	3309	`(unnamed)`
`scripts/Build_model_dual_industry_categories_named_debug.py`	3328	`(unnamed)`

Source	Line	Name / naming expression
scripts/Build_model_dual_industry_categories_named_debug.py	3350	(unnamed)
scripts/Build_model_dual_industry_categories_named_debug.py	3370	(unnamed)
scripts/Build_model_dual_industry_categories_named_debug.py	3396	(unnamed)
scripts/Build_model_dual_industry_categories_named_debug.py	3419	(unnamed)
scripts/Build_model_dual_industry_categories_named_debug.py	3447	(unnamed)
scripts/Build_model_dual_industry_categories_named_debug.py	3465	(unnamed)
scripts/Build_model_dual_industry_categories_named_debug.py	3480	(unnamed)
scripts/Build_model_dual_industry_categories_named_debug.py	3497	(unnamed)
scripts/Build_model_dual_industry_categories_named_debug.py	3512	(unnamed)
scripts/Build_model_dual_industry_categories_named_debug.py	3568	(unnamed)
scripts/Build_model_dual_industry_categories_named_debug.py	3578	(unnamed)
scripts/Build_model_dual_industry_categories_named_debug.py	3611	(unnamed)
scripts/Build_model_dual_industry_categories_named_debug.py	3624	(unnamed)
scripts/Build_model_dual_industry_categories_named_debug.py	3649	(unnamed)
scripts/Build_model_dual_industry_categories_named_debug.py	3667	(unnamed)
scripts/Build_model_dual_industry_categories_named_debug.py	3682	(unnamed)
scripts/Build_model_dual_industry_categories_named_debug.py	3724	(unnamed)
scripts/Build_model_dual_industry_categories_named_debug.py	3746	(unnamed)
scripts/Build_model_dual_industry_categories_named_debug.py	3784	(unnamed)
scripts/Build_model_dual_industry_categories_named_debug.py	3823	(unnamed)
scripts/Build_model_dual_industry_categories_named_debug.py	3835	(unnamed)
scripts/Build_model_dual_industry_categories_named_debug.py	3847	(unnamed)
scripts/Build_model_dual_industry_categories_named_debug.py	3882	'industry_storage_annual_investment_limit'

Source	Line	Name / naming expression
scripts/Build_model_dual_industry_categories_named_debug.py	3899	(unnamed)
scripts/Build_model_dual_industry_categories_named_debug.py	3909	(unnamed)
scripts/Build_model_dual_industry_categories_named_debug.py	3918	(unnamed)
scripts/Build_model_dual_industry_categories_named_debug.py	3927	(unnamed)
scripts/Build_model_dual_industry_categories_named_debug.py	3942	(unnamed)
scripts/Build_model_dual_industry_categories_named_debug.py	3957	(unnamed)
scripts/Build_model_dual_industry_categories_named_debug.py	3978	(unnamed)
scripts/Build_model_dual_industry_categories_named_debug.py	4009	(unnamed)
scripts/Build_model_dual_industry_categories_named_debug.py	4020	'industries_share_fuel_annual_cat'
scripts/Build_model_dual_industry_categories_named_debug.py	4040	'industries_share_fuel_cat'
scripts/Build_model_dual_industry_categories_named_debug.py	4061	'industries_share_fuel_init2020_cat'
scripts/Build_model_dual_industry_categories_named_debug.py	4087	(unnamed)
scripts/Build_model_dual_industry_categories_named_debug.py	4103	(unnamed)
scripts/Build_model_dual_industry_categories_named_debug.py	4121	(unnamed)
scripts/Build_model_dual_industry_categories_named_debug.py	4159	'industry_low_carbon_fuel_substitution_cap_cat'
scripts/Build_model_dual_industry_categories_named_debug.py	4249	'industries_increase_H2_cat'
scripts/Build_model_dual_industry_categories_named_debug.py	4282	'industries_increase_H2_cat'
scripts/Build_model_dual_industry_categories_named_debug.py	4323	(unnamed)
scripts/Build_model_dual_industry_categories_named_debug.py	4333	(unnamed)
scripts/Build_model_dual_industry_categories_named_debug.py	4358	(unnamed)
scripts/Build_model_dual_industry_categories_named_debug.py	4368	(unnamed)
scripts/Build_model_dual_industry_categories_named_debug.py	4390	(unnamed)
scripts/Build_model_dual_industry_categories_named_debug.py	4400	(unnamed)

Source	Line	Name / naming expression
`scripts/Build_model_dual_industry_categories_named_debug.py`	4411	`(unnamed)`
`scripts/Build_model_dual_industry_categories_named_debug.py`	4424	`(unnamed)`
`scripts/Build_model_dual_industry_categories_named_debug.py`	4448	`(unnamed)`
`scripts/Build_model_dual_industry_categories_named_debug.py`	4469	`(unnamed)`
`scripts/Build_model_dual_industry_categories_named_debug.py`	4479	`(unnamed)`
`scripts/Build_model_dual_industry_categories_named_debug.py`	4490	`(unnamed)`
`scripts/Build_model_dual_industry_categories_named_debug.py`	4502	`(unnamed)`
`scripts/Build_model_dual_industry_categories_named_debug.py`	4513	`(unnamed)`
`scripts/Build_model_dual_industry_categories_named_debug.py`	4525	`(unnamed)`
`scripts/Build_model_dual_industry_categories_named_debug.py`	4537	`(unnamed)`
`scripts/Build_model_dual_industry_categories_named_debug.py`	4553	`(unnamed)`
`scripts/Build_model_dual_industry_categories_named_debug.py`	4572	`(unnamed)`
`scripts/Build_model_dual_industry_categories_named_debug.py`	4600	`(unnamed)`
`scripts/Build_model_dual_industry_categories_named_debug.py`	4617	`(unnamed)`
`scripts/Build_model_dual_industry_categories_named_debug.py`	4633	`(unnamed)`
`scripts/Build_model_dual_industry_categories_named_debug.py`	4655	`(unnamed)`
`scripts/Build_model_dual_industry_categories_named_debug.py`	4667	`(unnamed)`
`scripts/Build_model_dual_industry_categories_named_debug.py`	4679	`(unnamed)`
`scripts/Build_model_dual_industry_categories_named_debug.py`	4690	`(unnamed)`
`scripts/Build_model_dual_industry_categories_named_debug.py`	4700	`(unnamed)`
`scripts/Build_model_dual_industry_categories_named_debug.py`	4711	`(unnamed)`
`scripts/Build_model_dual_industry_categories_named_debug.py`	4739	`(unnamed)`
`scripts/Build_model_dual_industry_categories_named_debug.py`	4762	`(unnamed)`

Source	Line	Name / naming expression
`scripts/Build_model_dual_industry_categories_named_debug.py`	4787	`(unnamed)`
`scripts/Build_model_dual_industry_categories_named_debug.py`	4809	`(unnamed)`
`scripts/Build_model_dual_industry_categories_named_debug.py`	4827	`(unnamed)`
`scripts/Build_model_dual_industry_categories_named_debug.py`	4845	`(unnamed)`
`scripts/Build_model_dual_industry_categories_named_debug.py`	4862	`(unnamed)`
`scripts/Build_model_dual_industry_categories_named_debug.py`	4879	`(unnamed)`
`scripts/Build_model_dual_industry_categories_named_debug.py`	4935	`(unnamed)`
`scripts/Build_model_dual_industry_categories_named_debug.py`	4949	`(unnamed)`
`scripts/Build_model_dual_industry_categories_named_debug.py`	4962	`(unnamed)`
`scripts/Build_model_dual_industry_categories_named_debug.py`	4987	`(unnamed)`
`scripts/Build_model_dual_industry_categories_named_debug.py`	4996	`(unnamed)`
`scripts/Build_model_dual_industry_categories_named_debug.py`	5014	`(unnamed)`
`scripts/Build_model_dual_industry_categories_named_debug.py`	5040	`(unnamed)`
`scripts/Build_model_dual_industry_categories_named_debug.py`	5063	`(unnamed)`
`scripts/Build_model_dual_industry_categories_named_debug.py`	5083	`(unnamed)`
`scripts/Build_model_dual_industry_categories_named_debug.py`	5109	`(unnamed)`
`scripts/Build_model_dual_industry_categories_named_debug.py`	5132	`(unnamed)`
`scripts/Build_model_dual_industry_categories_named_debug.py`	5160	`(unnamed)`
`scripts/Build_model_dual_industry_categories_named_debug.py`	5178	`(unnamed)`
`scripts/Build_model_dual_industry_categories_named_debug.py`	5193	`(unnamed)`
`scripts/Build_model_dual_industry_categories_named_debug.py`	5210	`(unnamed)`
`scripts/Build_model_dual_industry_categories_named_debug.py`	5225	`(unnamed)`
`scripts/Build_model_dual_industry_categories_named_debug.py`	5249	`'stat_storage_soc_0'`

Source	Line	Name / naming expression
scripts/Build_model_dual_industry_categories_named_debug.py	5268	'stat_storage_soc'
scripts/Build_model_dual_industry_categories_named_debug.py	5340	'stat_storage_soc_cycle'
scripts/Build_model_dual_industry_categories_named_debug.py	5360	'stat_storage_maxcap_fixed'
scripts/Build_model_dual_industry_categories_named_debug.py	5376	'stat_storage_maxcap'
scripts/Build_model_dual_industry_categories_named_debug.py	5403	'stat_storage_max'
scripts/Build_model_dual_industry_categories_named_debug.py	5414	'stat_storage_inv_max_period'
scripts/Build_model_dual_industry_categories_named_debug.py	5423	'stat_storage_inv_min_period'
scripts/Build_model_dual_industry_categories_named_debug.py	5432	'stat_storage_inv_max'
scripts/Build_model_dual_industry_categories_named_debug.py	5443	'stat_storage_inputmax'
scripts/Build_model_dual_industry_categories_named_debug.py	5461	'stat_storage_outputmax'
scripts/Build_model_dual_industry_categories_named_debug.py	5487	'max_transfer_systexpimp_pos_all'
scripts/Build_model_dual_industry_categories_named_debug.py	5501	'max_transfer_systexpimp_pos_all'
scripts/Build_model_dual_industry_categories_named_debug.py	5515	'max_transfer_systexpimp_pos_all_products'
scripts/Build_model_dual_industry_categories_named_debug.py	5533	'max_transfer_systexpimp_pos'
scripts/Build_model_dual_industry_categories_named_debug.py	5551	'max_transfer_systexpimp_neg'
scripts/Build_model_dual_industry_categories_named_debug.py	5566	'max_transfer_corridor_total'
scripts/Build_model_dual_industry_categories_named_debug.py	5594	'max_transfercap_terminal'
scripts/Build_model_dual_industry_categories_named_debug.py	5637	'max_transfercap_terminal_maxperiod'
scripts/Build_model_dual_industry_categories_named_debug.py	5656	(unnamed)
scripts/Build_model_dual_industry_categories_named_debug.py	5663	(unnamed)
scripts/Build_model_dual_industry_categories_named_debug.py	5670	(unnamed)
scripts/Build_model_dual_industry_categories_named_debug.py	5677	(unnamed)
scripts/Build_model_dual_industry_categories_named_debug.py	5708	'max_transfer_terminal_annual_pos'

Source	Line	Name / naming expression
scripts/Build_model_dual_industry_categories_named_debug.py	5728	'max_transfer_terminal_annual_neg'
scripts/Build_model_dual_industry_categories_named_debug.py	5776	'terminal_ship_route_fuel_export'
scripts/Build_model_dual_industry_categories_named_debug.py	5787	'terminal_ship_route_fuel_import'
scripts/Build_model_dual_industry_categories_named_debug.py	5827	(unnamed)
scripts/Build_model_dual_industry_categories_named_debug.py	5863	(unnamed)
scripts/Build_model_dual_industry_categories_named_debug.py	5872	(unnamed)
scripts/Build_model_dual_industry_categories_named_debug.py	5912	(unnamed)
scripts/Build_model_dual_industry_categories_named_debug.py	5979	(unnamed)
scripts/Build_model_dual_industry_categories_named_debug.py	5996	(unnamed)
scripts/Build_model_dual_industry_categories_named_debug.py	6013	(unnamed)
scripts/Build_model_dual_industry_categories_named_debug.py	6030	(unnamed)
scripts/Build_model_dual_industry_categories_named_debug.py	6050	(unnamed)
scripts/Build_model_dual_industry_categories_named_debug.py	6061	(unnamed)
scripts/Build_model_dual_industry_categories_named_debug.py	6072	(unnamed)
scripts/Build_model_dual_industry_categories_named_debug.py	6084	(unnamed)
scripts/Build_model_dual_industry_categories_named_debug.py	6100	(unnamed)
scripts/Build_model_dual_industry_categories_named_debug.py	6113	(unnamed)
scripts/Build_model_dual_industry_categories_named_debug.py	6124	(unnamed)
scripts/Build_model_dual_industry_categories_named_debug.py	6144	(unnamed)
scripts/Build_model_dual_industry_categories_named_debug.py	6156	(unnamed)
scripts/Build_model_dual_industry_categories_named_debug.py	6168	(unnamed)
scripts/Build_model_dual_industry_categories_named_debug.py	6180	(unnamed)
scripts/Build_model_dual_industry_categories_named_debug.py	6192	(unnamed)

Source	Line	Name / naming expression
`scripts/Build_model_dual_industry_categories_named_debug.py`	6204	`(unnamed)`
`scripts/Build_model_dual_industry_categories_named_debug.py`	6216	`(unnamed)`
`scripts/Build_model_dual_industry_categories_named_debug.py`	6228	`(unnamed)`
`scripts/Build_model_dual_industry_categories_named_debug.py`	6241	`(unnamed)`
`scripts/Build_model_dual_industry_categories_named_debug.py`	6254	`(unnamed)`
`scripts/Build_model_dual_industry_categories_named_debug.py`	6303	`(unnamed)`
`scripts/Build_model_dual_industry_categories_named_debug.py`	6336	`(unnamed)`
`scripts/Build_model_dual_industry_categories_named_debug.py`	6345	`(unnamed)`
`scripts/Build_model_dual_industry_categories_named_debug.py`	6372	`(unnamed)`
`scripts/Build_model_dual_industry_categories_named_debug.py`	6381	`(unnamed)`
`scripts/Build_model_dual_industry_categories_named_debug.py`	6395	`(unnamed)`
`scripts/Build_model_dual_industry_categories_named_debug.py`	6404	`(unnamed)`
`scripts/Build_model_dual_industry_categories_named_debug.py`	6419	`(unnamed)`
`scripts/Build_model_dual_industry_categories_named_debug.py`	6430	`(unnamed)`
`scripts/Build_model_dual_industry_categories_named_debug.py`	6441	`(unnamed)`
`scripts/Build_model_dual_industry_categories_named_debug.py`	6453	`(unnamed)`
`scripts/Build_model_dual_industry_categories_named_debug.py`	6468	`(unnamed)`
`scripts/Build_model_dual_industry_categories_named_debug.py`	6482	`(unnamed)`
`scripts/Build_model_dual_industry_categories_named_debug.py`	6496	`(unnamed)`
`scripts/Build_model_dual_industry_categories_named_debug.py`	6544	`(unnamed)`
`scripts/Build_model_dual_industry_categories_named_debug.py`	6558	`(unnamed)`
`scripts/Build_model_dual_industry_categories_named_debug.py`	6571	`(unnamed)`
`scripts/Build_model_dual_industry_categories_named_debug.py`	6624	`(unnamed)`

Source	Line	Name / naming expression
`scripts/Build_model_dual_industry_categories_named_debug.py`	6643	`(unnamed)`
`scripts/Build_model_dual_industry_categories_named_debug.py`	6680	`(unnamed)`
`scripts/Build_model_dual_industry_categories_named_debug.py`	6687	`(unnamed)`
`scripts/Build_model_dual_industry_categories_named_debug.py`	6711	`(unnamed)`
`scripts/Build_model_dual_industry_categories_named_debug.py`	6737	`(unnamed)`
`scripts/Build_model_dual_industry_categories_named_debug.py`	6758	`(unnamed)`
`scripts/Build_model_dual_industry_categories_named_debug.py`	6773	`(unnamed)`
`scripts/Build_model_dual_industry_categories_named_debug.py`	6795	`(unnamed)`
`scripts/Build_model_dual_industry_categories_named_debug.py`	6804	`(unnamed)`
`scripts/Build_model_dual_industry_categories_named_debug.py`	6813	`(unnamed)`
`scripts/Build_model_dual_industry_categories_named_debug.py`	6825	`(unnamed)`
`scripts/Build_model_dual_industry_categories_named_debug.py`	6831	`(unnamed)`
`scripts/Build_model_dual_industry_categories_named_debug.py`	6841	`(unnamed)`
`scripts/Build_model_dual_industry_categories_named_debug.py`	6856	`(unnamed)`
`scripts/Build_model_dual_industry_categories_named_debug.py`	6890	`(unnamed)`
`scripts/Build_model_dual_industry_categories_named_debug.py`	6896	`(unnamed)`
`scripts/Build_model_dual_industry_categories_named_debug.py`	6904	`(unnamed)`
`scripts/Build_model_dual_industry_categories_named_debug.py`	6913	`(unnamed)`
`scripts/Build_model_dual_industry_categories_named_debug.py`	6964	`(unnamed)`
`scripts/Build_model_dual_industry_categories_named_debug.py`	6973	`(unnamed)`
`scripts/Build_model_dual_industry_categories_named_debug.py`	7078	`(unnamed)`
`scripts/Build_model_dual_industry_categories_named_debug.py`	7161	`'max_output_dispatch'`
`scripts/Build_model_dual_industry_categories_named_debug.py`	7179	`'max_output_nondisp'`

Source	Line	Name / naming expression
scripts/Build_model_dual_industry_categories_named_debug.py	7193	'max_output_nondisp'
scripts/Build_model_dual_industry_categories_named_debug.py	7211	'max_output_hphs_gen'
scripts/Build_model_dual_industry_categories_named_debug.py	7224	'max_output_hphs_pump'
scripts/Build_model_dual_industry_categories_named_debug.py	7241	'max_output_hdam'
scripts/Build_model_dual_industry_categories_named_debug.py	7259	'max_output_hror'
scripts/Build_model_dual_industry_categories_named_debug.py	7273	'max_output_hror'
scripts/Build_model_dual_industry_categories_named_debug.py	7291	'max_capacityyear'
scripts/Build_model_dual_industry_categories_named_debug.py	7312	(unnamed)
scripts/Build_model_dual_industry_categories_named_debug.py	7333	'el_balance'
scripts/Build_model_dual_industry_categories_named_debug.py	7442	'equalityimpexp'
scripts/Build_model_dual_industry_categories_named_debug.py	7461	'equalityimpexp_secc'
scripts/Build_model_dual_industry_categories_named_debug.py	7491	'total_dem_zone_system'
scripts/Build_model_dual_industry_categories_named_debug.py	7568	'total_gen_zone_system'
scripts/Build_model_dual_industry_categories_named_debug.py	7609	'ceep_demand'
scripts/Build_model_dual_industry_categories_named_debug.py	7651	(unnamed)
scripts/Build_model_dual_industry_categories_named_debug.py	7669	(unnamed)
scripts/Build_model_dual_industry_categories_named_debug.py	7685	(unnamed)
scripts/Build_model_dual_industry_categories_named_debug.py	7698	(unnamed)
scripts/Build_model_dual_industry_categories_named_debug.py	7713	'heat_pump_soc_max'
scripts/Build_model_dual_industry_categories_named_debug.py	7725	'heat_pump_ind_capyear_balance_fixed'
scripts/Build_model_dual_industry_categories_named_debug.py	7735	'heat_pump_ind_capyear_balance'
scripts/Build_model_dual_industry_categories_named_debug.py	7748	'heat_pump_indstor_capyear_link'
scripts/Build_model_dual_industry_categories_named_debug.py	7761	'heat_pump_ind_max'

Source	Line	Name / naming expression
scripts/Build_model_dual_industry_categories_named_debug.py	7816	'electric_heat_ind_yoy_min'
scripts/Build_model_dual_industry_categories_named_debug.py	7897	f'{constraint_name}[{zone},{System},{unit},{year},{week_periods[0]}]'
scripts/Build_model_dual_industry_categories_named_debug.py	7928	(unnamed)
scripts/Build_model_dual_industry_categories_named_debug.py	7938	(unnamed)
scripts/Build_model_dual_industry_categories_named_debug.py	7950	(unnamed)
scripts/Build_model_dual_industry_categories_named_debug.py	7960	(unnamed)
scripts/Build_model_dual_industry_categories_named_debug.py	7975	(unnamed)
scripts/Build_model_dual_industry_categories_named_debug.py	7989	(unnamed)
scripts/Build_model_dual_industry_categories_named_debug.py	7999	(unnamed)
scripts/Build_model_dual_industry_categories_named_debug.py	8009	(unnamed)
scripts/Build_model_dual_industry_categories_named_debug.py	8024	(unnamed)
scripts/Build_model_dual_industry_categories_named_debug.py	8033	(unnamed)
scripts/Build_model_dual_industry_categories_named_debug.py	8042	(unnamed)
scripts/Build_model_dual_industry_categories_named_debug.py	8052	(unnamed)
scripts/Build_model_dual_industry_categories_named_debug.py	8061	(unnamed)
scripts/Build_model_dual_industry_categories_named_debug.py	8072	(unnamed)
scripts/Build_model_dual_industry_categories_named_debug.py	8085	(unnamed)
scripts/Build_model_dual_industry_categories_named_debug.py	8112	'heat_pump_soc_balance'
scripts/Build_model_dual_industry_categories_named_debug.py	8125	'heat_pump_soc_balance_0'
scripts/Build_model_dual_industry_categories_named_debug.py	8139	'heat_pump_soc_cycle'
scripts/Build_model_dual_industry_categories_named_debug.py	8152	(unnamed)
scripts/Build_model_dual_industry_categories_named_debug.py	8161	(unnamed)
scripts/Build_model_dual_industry_categories_named_debug.py	8170	'heat_pump_cap_share'

Source	Line	Name / naming expression
scripts/Build_model_dual_industry_categories_named_debug.py	8180	'heat_pump_dh_cap_share'
scripts/Build_model_dual_industry_categories_named_debug.py	8190	(unnamed)
scripts/Build_model_dual_industry_categories_named_debug.py	8199	(unnamed)
scripts/Build_model_dual_industry_categories_named_debug.py	8209	(unnamed)
scripts/Build_model_dual_industry_categories_named_debug.py	8219	(unnamed)
scripts/Build_model_dual_industry_categories_named_debug.py	8229	(unnamed)
scripts/Build_model_dual_industry_categories_named_debug.py	8238	(unnamed)
scripts/Build_model_dual_industry_categories_named_debug.py	8254	(unnamed)
scripts/Build_model_dual_industry_categories_named_debug.py	8264	(unnamed)
scripts/Build_model_dual_industry_categories_named_debug.py	8281	(unnamed)
scripts/Build_model_dual_industry_categories_named_debug.py	8292	(unnamed)
scripts/Build_model_dual_industry_categories_named_debug.py	8306	(unnamed)
scripts/Build_model_dual_industry_categories_named_debug.py	8313	(unnamed)
scripts/Build_model_dual_industry_categories_named_debug.py	8318	(unnamed)
scripts/Build_model_dual_industry_categories_named_debug.py	8325	(unnamed)
scripts/Build_model_dual_industry_categories_named_debug.py	8378	(unnamed)
scripts/Build_model_dual_industry_categories_named_debug.py	8391	(unnamed)
scripts/Build_model_dual_industry_categories_named_debug.py	8416	(unnamed)
scripts/Build_model_dual_industry_categories_named_debug.py	8459	(unnamed)
scripts/Build_model_dual_industry_categories_named_debug.py	8476	(unnamed)
scripts/Build_model_dual_industry_categories_named_debug.py	8492	(unnamed)
scripts/Build_model_dual_industry_categories_named_debug.py	8526	(unnamed)
scripts/Build_model_dual_industry_categories_named_debug.py	8556	(unnamed)

Source	Line	Name / naming expression
scripts/Build_model_dual_industry_categories_named_debug.py	8567	(unnamed)
scripts/Build_model_dual_industry_categories_named_debug.py	8577	(unnamed)
scripts/Build_model_dual_industry_categories_named_debug.py	8603	(unnamed)
scripts/Build_model_dual_industry_categories_named_debug.py	8640	(unnamed)
scripts/Build_model_dual_industry_categories_named_debug.py	8717	(unnamed)
scripts/Build_model_dual_industry_categories_named_debug.py	8734	(unnamed)
scripts/Build_model_dual_industry_categories_named_debug.py	8749	(unnamed)
scripts/Build_model_dual_industry_categories_named_debug.py	8780	(unnamed)
scripts/Build_model_dual_industry_categories_named_debug.py	8794	(unnamed)
scripts/Build_model_dual_industry_categories_named_debug.py	8810	(unnamed)
scripts/Build_model_dual_industry_categories_named_debug.py	8824	(unnamed)
scripts/Build_model_dual_industry_categories_named_debug.py	8839	(unnamed)
scripts/Build_model_dual_industry_categories_named_debug.py	8850	(unnamed)
scripts/Build_model_dual_industry_categories_named_debug.py	8861	(unnamed)
scripts/Build_model_dual_industry_categories_named_debug.py	8901	(unnamed)
scripts/Build_model_dual_industry_categories_named_debug.py	8912	(unnamed)
scripts/Build_model_dual_industry_categories_named_debug.py	8925	(unnamed)
scripts/Build_model_dual_industry_categories_named_debug.py	8938	(unnamed)
scripts/Build_model_dual_industry_categories_named_debug.py	8973	(unnamed)
scripts/Build_model_dual_industry_categories_named_debug.py	8990	(unnamed)
scripts/Build_model_dual_industry_categories_named_debug.py	9010	(unnamed)
scripts/Build_model_dual_industry_categories_named_debug.py	9125	'veh_transp_num_fixed_pin_lb'
scripts/Build_model_dual_industry_categories_named_debug.py	9141	'veh_transp_num_fixed_pin_ub'

Source	Line	Name / naming expression
scripts/Build_model_dual_industry_categories_named_debug.py	9155	'veh_stock_balance'
scripts/Build_model_dual_industry_categories_named_debug.py	9186	(unnamed)
scripts/Build_model_dual_industry_categories_named_debug.py	9211	(unnamed)
scripts/Build_model_dual_industry_categories_named_debug.py	9221	(unnamed)
scripts/Build_model_dual_industry_categories_named_debug.py	9251	(unnamed)
scripts/Build_model_dual_industry_categories_named_debug.py	9267	(unnamed)
scripts/Build_model_dual_industry_categories_named_debug.py	9282	(unnamed)
scripts/Build_model_dual_industry_categories_named_debug.py	9313	(unnamed)
scripts/Build_model_dual_industry_categories_named_debug.py	9344	'veh_sector_total'
scripts/Build_model_dual_industry_categories_named_debug.py	9398	f'transport_ice_hybrid_stock_share_non_increasing_{unit}_{sector}_{zone}_{System}_{year}'
scripts/Build_model_dual_industry_categories_named_debug.py	9411	(unnamed)
scripts/Build_model_dual_industry_categories_named_debug.py	9480	'transport_energy_profile'
scripts/Build_model_dual_industry_categories_named_debug.py	9519	(unnamed)
scripts/Build_model_dual_industry_categories_named_debug.py	9623	(unnamed)
scripts/Build_model_dual_industry_categories_named_debug.py	9636	(unnamed)
scripts/Build_model_dual_industry_categories_named_debug.py	9649	(unnamed)
scripts/Build_model_dual_industry_categories_named_debug.py	9675	f'transport_electrification_prev_floor_{sector}_{zone}_{System}_{year}'
scripts/Build_model_dual_industry_categories_named_debug.py	9711	f'transport_rail_electrification_link_{_base_sector}_{_other_sector}_{zone}_{System}_{year}'
scripts/Build_model_dual_industry_categories_named_debug.py	9733	(unnamed)
scripts/Build_model_dual_industry_categories_named_debug.py	9753	(unnamed)
scripts/Build_model_dual_industry_categories_named_debug.py	9808	(unnamed)
scripts/Build_model_dual_industry_categories_named_debug.py	9856	(unnamed)

Source	Line	Name / naming expression
scripts/Build_model_dual_industry_categories_named_debug.py	9883	(unnamed)
scripts/Build_model_dual_industry_categories_named_debug.py	10060	f'transport_shift_baseline_energy_link_{_transport_baseline_annual_var_no}'
scripts/Build_model_dual_industry_categories_named_debug.py	10109	f'transport_fuel_annual_energy_link_{_transport_fuel_annual_var_no}'
scripts/Build_model_dual_industry_categories_named_debug.py	10148	f'transport_fuel_shift_{_transport_fuel_share_constr_no}_up'
scripts/Build_model_dual_industry_categories_named_debug.py	10155	f'transport_fuel_shift_{_transport_fuel_share_constr_no}_down'
scripts/Build_model_dual_industry_categories_named_debug.py	10217	f'transport_ice_hybrid_fossil_share_decline_{unit}_{sector}_{zone}_{System}_{period}_{year}'
scripts/Build_model_dual_industry_categories_named_debug.py	10245	f'transport_ice_hybrid_fossil_share_decline_annual_{unit}_{sector}_{zone}_{System}_{year}'
scripts/Build_model_dual_industry_categories_named_debug.py	10268	(unnamed)
scripts/Build_model_dual_industry_categories_named_debug.py	10325	(unnamed)
scripts/Build_model_dual_industry_categories_named_debug.py	10334	(unnamed)
scripts/Build_model_dual_industry_categories_named_debug.py	10344	(unnamed)
scripts/Build_model_dual_industry_categories_named_debug.py	10353	(unnamed)
scripts/Build_model_dual_industry_categories_named_debug.py	10402	(unnamed)
scripts/Build_model_dual_industry_categories_named_debug.py	10414	(unnamed)
scripts/Build_model_dual_industry_categories_named_debug.py	10422	(unnamed)
scripts/Build_model_dual_industry_categories_named_debug.py	10450	'efuel_genx'
scripts/Build_model_dual_industry_categories_named_debug.py	10473	(unnamed)
scripts/Build_model_dual_industry_categories_named_debug.py	10528	(unnamed)
scripts/Build_model_dual_industry_categories_named_debug.py	10548	'efuel_inputs1'
scripts/Build_model_dual_industry_categories_named_debug.py	10560	'efuel_inputs2'
scripts/Build_model_dual_industry_categories_named_debug.py	10572	'efuel_inputs3'
scripts/Build_model_dual_industry_categories_named_debug.py	10584	(unnamed)

Source	Line	Name / naming expression
`scripts/Build_model_dual_industry_categories_named_debug.py`	10595	`(unnamed)`
`scripts/Build_model_dual_industry_categories_named_debug.py`	10606	`(unnamed)`
`scripts/Build_model_dual_industry_categories_named_debug.py`	10617	`(unnamed)`
`scripts/Build_model_dual_industry_categories_named_debug.py`	10631	`(unnamed)`
`scripts/Build_model_dual_industry_categories_named_debug.py`	10657	`(unnamed)`
`scripts/Build_model_dual_industry_categories_named_debug.py`	10684	`(unnamed)`
`scripts/Build_model_dual_industry_categories_named_debug.py`	10695	`(unnamed)`
`scripts/Build_model_dual_industry_categories_named_debug.py`	10706	`(unnamed)`
`scripts/Build_model_dual_industry_categories_named_debug.py`	10729	`(unnamed)`
`scripts/Build_model_dual_industry_categories_named_debug.py`	10746	`(unnamed)`
`scripts/Build_model_dual_industry_categories_named_debug.py`	10760	`(unnamed)`
`scripts/Build_model_dual_industry_categories_named_debug.py`	10774	`(unnamed)`
`scripts/Build_model_dual_industry_categories_named_debug.py`	10815	`(unnamed)`
`scripts/Build_model_dual_industry_categories_named_debug.py`	10829	`(unnamed)`
`scripts/Build_model_dual_industry_categories_named_debug.py`	10839	`(unnamed)`
`scripts/Build_model_dual_industry_categories_named_debug.py`	10848	`(unnamed)`
`scripts/Build_model_dual_industry_categories_named_debug.py`	10878	`(unnamed)`
`scripts/Build_model_dual_industry_categories_named_debug.py`	10899	`(unnamed)`
`scripts/Build_model_dual_industry_categories_named_debug.py`	10918	`(unnamed)`
`scripts/Build_model_dual_industry_categories_named_debug.py`	10937	`(unnamed)`
`scripts/Build_model_dual_industry_categories_named_debug.py`	10957	`(unnamed)`
`scripts/Build_model_dual_industry_categories_named_debug.py`	10977	`(unnamed)`
`scripts/Build_model_dual_industry_categories_named_debug.py`	11092	`(unnamed)`

Source	Line	Name / naming expression
scripts/Build_model_dual_industry_categories_named_debug.py	11100	(unnamed)
scripts/Build_model_dual_industry_categories_named_debug.py	11109	(unnamed)
scripts/Build_model_dual_industry_categories_named_debug.py	11129	(unnamed)
scripts/Build_model_dual_industry_categories_named_debug.py	11196	'fuel_storage_capacity_year_fixed'
scripts/Build_model_dual_industry_categories_named_debug.py	11210	'fuel_storage_capacity_year'
scripts/Build_model_dual_industry_categories_named_debug.py	11235	'fuel_storage_inv_period_max'
scripts/Build_model_dual_industry_categories_named_debug.py	11245	'fuel_storage_inv_missing_limit_zero'
scripts/Build_model_dual_industry_categories_named_debug.py	11255	'fuel_storage_total_max'
scripts/Build_model_dual_industry_categories_named_debug.py	11270	'fuel_storage_initial_soc_zero'
scripts/Build_model_dual_industry_categories_named_debug.py	11279	'fuel_storage_capacity'
scripts/Build_model_dual_industry_categories_named_debug.py	11290	'fuel_storage_initial_capacity'
scripts/Build_model_dual_industry_categories_named_debug.py	11368	'fuel_storage_terminal_balance'
scripts/Build_model_dual_industry_categories_named_debug.py	11387	'fuel_storage_terminal_lb'
scripts/Build_model_dual_industry_categories_named_debug.py	11396	'fuel_storage_terminal_ub'
scripts/Build_model_dual_industry_categories_named_debug.py	11415	'fuel_boiler'
scripts/Build_model_dual_industry_categories_named_debug.py	11426	'fuel_boiler_hour'
scripts/Build_model_dual_industry_categories_named_debug.py	11443	'fuel_boiler_DH'
scripts/Build_model_dual_industry_categories_named_debug.py	11455	(unnamed)
scripts/Build_model_dual_industry_categories_named_debug.py	11475	'biomass_heating_nonincrease'
scripts/Build_model_dual_industry_categories_named_debug.py	11500	'biomass_heating_nonincrease_from_previous'
scripts/Build_model_dual_industry_categories_named_debug.py	11524	(unnamed)
scripts/Build_model_dual_industry_categories_named_debug.py	11532	(unnamed)
scripts/Build_model_dual_industry_categories_named_debug.py	11550	(unnamed)

Source	Line	Name / naming expression
`scripts/Build_model_dual_industry_categories_named_debug.py`	11559	`(unnamed)`
`scripts/Build_model_dual_industry_categories_named_debug.py`	11626	`(unnamed)`
`scripts/Build_model_dual_industry_categories_named_debug.py`	11633	`(unnamed)`
`scripts/Build_model_dual_industry_categories_named_debug.py`	11673	`'raff_prod_gen_min'`
`scripts/Build_model_dual_industry_categories_named_debug.py`	11690	`'raff_prod_gen_max'`
`scripts/Build_model_dual_industry_categories_named_debug.py`	11707	`'raff_prod_gen_zero_missing'`
`scripts/Build_model_dual_industry_categories_named_debug.py`	11724	`(unnamed)`
`scripts/Build_model_dual_industry_categories_named_debug.py`	11741	`'raff_prod_gen_sum_max'`
`scripts/Build_model_dual_industry_categories_named_debug.py`	11816	`f'oil_refinery_product_generation_change_up_{product}_{zone}_{system}_{year}'`
`scripts/Build_model_dual_industry_categories_named_debug.py`	11826	`f'oil_refinery_product_generation_change_down_{product}_{zone}_{system}_{year}'`
`scripts/Build_model_dual_industry_categories_named_debug.py`	11871	`(unnamed)`
`scripts/Build_model_dual_industry_categories_named_debug.py`	11878	`(unnamed)`
`scripts/Build_model_dual_industry_categories_named_debug.py`	11886	`(unnamed)`
`scripts/Build_model_dual_industry_categories_named_debug.py`	11891	`(unnamed)`
`scripts/Build_model_dual_industry_categories_named_debug.py`	11898	`(unnamed)`
`scripts/Build_model_dual_industry_categories_named_debug.py`	11906	`(unnamed)`
`scripts/Build_model_dual_industry_categories_named_debug.py`	11961	`'oil_gas_storage_capacity'`
`scripts/Build_model_dual_industry_categories_named_debug.py`	11971	`'oil_gas_initial_soc'`
`scripts/Build_model_dual_industry_categories_named_debug.py`	12002	`(unnamed)`
`scripts/Build_model_dual_industry_categories_named_debug.py`	12031	`(unnamed)`
`scripts/Build_model_dual_industry_categories_named_debug.py`	12048	`(unnamed)`
`scripts/Build_model_dual_industry_categories_named_debug.py`	12072	`(unnamed)`
`scripts/Build_model_dual_industry_categories_named_debug.py`	12095	`(unnamed)`

Source	Line	Name / naming expression
scripts/Build_model_dual_industry_categories_named_debug.py	12109	(unnamed)
scripts/Build_model_dual_industry_categories_named_debug.py	12139	'storage_level_max'
scripts/Build_model_dual_industry_categories_named_debug.py	12165	'storage_level_cap_0'
scripts/Build_model_dual_industry_categories_named_debug.py	12184	'storage_level_balance'
scripts/Build_model_dual_industry_categories_named_debug.py	12203	'hdam_storage_gen_cap'
scripts/Build_model_dual_industry_categories_named_debug.py	12222	'storage_level_cap_0'
scripts/Build_model_dual_industry_categories_named_debug.py	12239	'storage_level_balance'
scripts/Build_model_dual_industry_categories_named_debug.py	12256	'hdam_storage_gen_cap'
scripts/Build_model_dual_industry_categories_named_debug.py	12276	'hphs_storage_level_max'
scripts/Build_model_dual_industry_categories_named_debug.py	12287	'hphs_storage_level_cap_0'
scripts/Build_model_dual_industry_categories_named_debug.py	12306	'hphs_storage_level_period'
scripts/Build_model_dual_industry_categories_named_debug.py	12325	'hphs_storage_gen_cap'
scripts/Build_model_dual_industry_categories_named_debug.py	12342	'hphs_storage_level_cap_0'
scripts/Build_model_dual_industry_categories_named_debug.py	12362	'hphs_storage_level_period'
scripts/Build_model_dual_industry_categories_named_debug.py	12381	'hphs_storage_gen_cap'
scripts/Build_model_dual_industry_categories_named_debug.py	12443	'hydro_soc_cycle'
scripts/Build_model_dual_industry_categories_named_debug.py	12624	(unnamed)
scripts/Build_model_dual_industry_categories_named_debug.py	12645	(unnamed)
scripts/Build_model_dual_industry_categories_named_debug.py	12667	(unnamed)
scripts/Build_model_dual_industry_categories_named_debug.py	12689	(unnamed)
scripts/Build_model_dual_industry_categories_named_debug.py	12710	(unnamed)
scripts/Build_model_dual_industry_categories_named_debug.py	12730	(unnamed)
scripts/Build_model_dual_industry_categories_named_debug.py	12751	(unnamed)

Source	Line	Name / naming expression
scripts/Build_model_dual_industry_categories_named_debug.py	12773	(unnamed)
scripts/Build_model_dual_industry_categories_named_debug.py	12795	(unnamed)
scripts/Build_model_dual_industry_categories_named_debug.py	12818	(unnamed)
scripts/Build_model_dual_industry_categories_named_debug.py	12834	(unnamed)
scripts/Build_model_dual_industry_categories_named_debug.py	12851	(unnamed)
scripts/Build_model_dual_industry_categories_named_debug.py	12871	(unnamed)
scripts/Build_model_dual_industry_categories_named_debug.py	12892	(unnamed)
scripts/Build_model_dual_industry_categories_named_debug.py	12913	(unnamed)
scripts/Build_model_dual_industry_categories_named_debug.py	12940	(unnamed)
scripts/Build_model_dual_industry_categories_named_debug.py	12960	(unnamed)
scripts/Build_model_dual_industry_categories_named_debug.py	12967	(unnamed)
scripts/Build_model_dual_industry_categories_named_debug.py	12974	(unnamed)
scripts/Build_model_dual_industry_categories_named_debug.py	12988	'CO2_emissions_limit'
scripts/Build_model_dual_industry_categories_named_debug.py	12997	'CO2_emissions_limit'
scripts/Build_model_dual_industry_categories_named_debug.py	13026	'CO2_emissions_limit'
scripts/Build_model_dual_industry_categories_named_debug.py	13096	'CO2_emissions_power'
scripts/Build_model_dual_industry_categories_named_debug.py	13105	'CO2_emissions_power_zone'
scripts/Build_model_dual_industry_categories_named_debug.py	13114	(unnamed)
scripts/Build_model_dual_industry_categories_named_debug.py	13126	(unnamed)
scripts/Build_model_dual_industry_categories_named_debug.py	13144	'CO2_emissions_heat'
scripts/Build_model_dual_industry_categories_named_debug.py	13169	'CO2_emissions_heat'
scripts/Build_model_dual_industry_categories_named_debug.py	13327	(unnamed)
scripts/Build_model_dual_industry_categories_named_debug.py	13336	'CO2_emissions_industry'

Source	Line	Name / naming expression
scripts/Build_model_dual_industry_categories_named_debug.py	13352	'CO2_emissions_industry'
scripts/Build_model_dual_industry_categories_named_debug.py	13374	'co2_emissions_transport_total'
scripts/Build_model_dual_industry_categories_named_debug.py	13392	(unnamed)
scripts/Build_model_dual_industry_categories_named_debug.py	13546	(unnamed)
scripts/Build_model_dual_industry_categories_named_debug.py	13558	(unnamed)
scripts/Build_model_dual_industry_categories_named_debug.py	13567	(unnamed)
scripts/Build_model_dual_industry_categories_named_debug.py	13582	(unnamed)
scripts/Build_model_dual_industry_categories_named_debug.py	13611	(unnamed)
scripts/CAPEX_OPEX_Gurobi_settings_dual_industry_categories1.py	289	(unnamed)
scripts/CAPEX_OPEX_Gurobi_settings_dual_industry_categories1.py	316	(unnamed)
scripts/CAPEX_OPEX_Gurobi_settings_dual_industry_categories1.py	344	(unnamed)
scripts/CAPEX_OPEX_Gurobi_settings_dual_industry_categories1.py	375	(unnamed)
scripts/CAPEX_OPEX_Gurobi_settings_dual_industry_categories1.py	403	(unnamed)
scripts/CAPEX_OPEX_Gurobi_settings_dual_industry_categories1.py	457	(unnamed)
scripts/CAPEX_OPEX_Gurobi_settings_dual_industry_categories1.py	515	(unnamed)
scripts/CAPEX_OPEX_Gurobi_settings_dual_industry_categories1.py	572	(unnamed)
scripts/CAPEX_OPEX_Gurobi_settings_dual_industry_categories1.py	607	'h2_terminal_capital_cost'
scripts/CAPEX_OPEX_Gurobi_settings_dual_industry_categories1.py	611	(unnamed)
scripts/CAPEX_OPEX_Gurobi_settings_dual_industry_categories1.py	647	(unnamed)
scripts/CAPEX_OPEX_Gurobi_settings_dual_industry_categories1.py	681	(unnamed)
scripts/CAPEX_OPEX_Gurobi_settings_dual_industry_categories1.py	718	(unnamed)
scripts/CAPEX_OPEX_Gurobi_settings_dual_industry_categories1.py	751	(unnamed)
scripts/CAPEX_OPEX_Gurobi_settings_dual_industry_categories1.py	785	(unnamed)

Source	Line	Name / naming expression
`scripts/CAPEX_OPEX_Gurobi_settings_dual_industry_categories1.py`	813	`(unnamed)`
`scripts/CAPEX_OPEX_Gurobi_settings_dual_industry_categories1.py`	842	`(unnamed)`
`scripts/CAPEX_OPEX_Gurobi_settings_dual_industry_categories1.py`	875	`(unnamed)`
`scripts/CAPEX_OPEX_Gurobi_settings_dual_industry_categories1.py`	916	`(unnamed)`
`scripts/CAPEX_OPEX_Gurobi_settings_dual_industry_categories1.py`	964	`(unnamed)`
`scripts/CAPEX_OPEX_Gurobi_settings_dual_industry_categories1.py`	1008	`(unnamed)`
`scripts/CAPEX_OPEX_Gurobi_settings_dual_industry_categories1.py`	1101	`(unnamed)`
`scripts/CAPEX_OPEX_Gurobi_settings_dual_industry_categories1.py`	1144	`(unnamed)`
`scripts/CAPEX_OPEX_Gurobi_settings_dual_industry_categories1.py`	1187	`(unnamed)`
`scripts/CAPEX_OPEX_Gurobi_settings_dual_industry_categories1.py`	1257	`(unnamed)`
`scripts/CAPEX_OPEX_Gurobi_settings_dual_industry_categories1.py`	1304	`(unnamed)`
`scripts/CAPEX_OPEX_Gurobi_settings_dual_industry_categories1.py`	1340	`(unnamed)`
`scripts/CAPEX_OPEX_Gurobi_settings_dual_industry_categories1.py`	1384	`(unnamed)`
`scripts/CAPEX_OPEX_Gurobi_settings_dual_industry_categories1.py`	1428	`(unnamed)`
`scripts/CAPEX_OPEX_Gurobi_settings_dual_industry_categories1.py`	1458	`(unnamed)`
`scripts/CAPEX_OPEX_Gurobi_settings_dual_industry_categories1.py`	1486	`(unnamed)`
`scripts/CAPEX_OPEX_Gurobi_settings_dual_industry_categories1.py`	1522	`(unnamed)`
`scripts/CAPEX_OPEX_Gurobi_settings_dual_industry_categories1.py`	1560	`(unnamed)`
`scripts/CAPEX_OPEX_Gurobi_settings_dual_industry_categories1.py`	1568	`(unnamed)`
`scripts/CAPEX_OPEX_Gurobi_settings_dual_industry_categories1.py`	1578	`(unnamed)`
`scripts/CAPEX_OPEX_Gurobi_settings_dual_industry_categories1.py`	1616	`(unnamed)`
`scripts/CAPEX_OPEX_Gurobi_settings_dual_industry_categories1.py`	1653	`(unnamed)`
`scripts/CAPEX_OPEX_Gurobi_settings_dual_industry_categories1.py`	1686	`(unnamed)`

Source	Line	Name / naming expression
scripts/CAPEX_OPEX_Gurobi_settings_dual_industry_categories1.py	1720	(unnamed)
scripts/CAPEX_OPEX_Gurobi_settings_dual_industry_categories1.py	1749	(unnamed)
scripts/CAPEX_OPEX_Gurobi_settings_dual_industry_categories1.py	1777	(unnamed)
scripts/CAPEX_OPEX_Gurobi_settings_dual_industry_categories1.py	1813	(unnamed)
scripts/CAPEX_OPEX_Gurobi_settings_dual_industry_categories1.py	1850	(unnamed)
scripts/CAPEX_OPEX_Gurobi_settings_dual_industry_categories1.py	1858	(unnamed)
scripts/CAPEX_OPEX_Gurobi_settings_dual_industry_categories1.py	1868	(unnamed)
scripts/CAPEX_OPEX_Gurobi_settings_dual_industry_categories1.py	1904	(unnamed)
scripts/CAPEX_OPEX_Gurobi_settings_dual_industry_categories1.py	1950	(unnamed)
scripts/CAPEX_OPEX_Gurobi_settings_dual_industry_categories1.py	1978	(unnamed)
scripts/CAPEX_OPEX_Gurobi_settings_dual_industry_categories1.py	2007	(unnamed)
scripts/CAPEX_OPEX_Gurobi_settings_dual_industry_categories1.py	2044	(unnamed)
scripts/CAPEX_OPEX_Gurobi_settings_dual_industry_categories1.py	2088	(unnamed)
scripts/CAPEX_OPEX_Gurobi_settings_dual_industry_categories1.py	2138	(unnamed)
scripts/CAPEX_OPEX_Gurobi_settings_dual_industry_categories1.py	2170	(unnamed)
scripts/CAPEX_OPEX_Gurobi_settings_dual_industry_categories1.py	2201	(unnamed)
scripts/CAPEX_OPEX_Gurobi_settings_dual_industry_categories1.py	2249	(unnamed)
scripts/CAPEX_OPEX_Gurobi_settings_dual_industry_categories1.py	2286	(unnamed)
scripts/CAPEX_OPEX_Gurobi_settings_dual_industry_categories1.py	2320	(unnamed)
scripts/CAPEX_OPEX_Gurobi_settings_dual_industry_categories1.py	2373	(unnamed)
scripts/CAPEX_OPEX_Gurobi_settings_dual_industry_categories1.py	2432	(unnamed)
scripts/CAPEX_OPEX_Gurobi_settings_dual_industry_categories1.py	2491	(unnamed)
scripts/CAPEX_OPEX_Gurobi_settings_dual_industry_categories1.py	2545	(unnamed)

Source	Line	Name / naming expression
scripts/CAPEX_OPEX_Gurobi_settings_dual_industry_categories1.py	2687	(unnamed)
scripts/CAPEX_OPEX_Gurobi_settings_dual_industry_categories1.py	2688	(unnamed)
scripts/CAPEX_OPEX_Gurobi_settings_dual_industry_categories1.py	2693	(unnamed)
scripts/CAPEX_OPEX_Gurobi_settings_dual_industry_categories1.py	2694	(unnamed)
scripts/CAPEX_OPEX_Gurobi_settings_dual_industry_categories1.py	2695	'HP_fOPEX_years'
scripts/CAPEX_OPEX_Gurobi_settings_dual_industry_categories1.py	2696	(unnamed)
scripts/CAPEX_OPEX_Gurobi_settings_dual_industry_categories1.py	2697	(unnamed)
scripts/CAPEX_OPEX_Gurobi_settings_dual_industry_categories1.py	2698	(unnamed)
scripts/CAPEX_OPEX_Gurobi_settings_dual_industry_categories1.py	2700	(unnamed)
scripts/CAPEX_OPEX_Gurobi_settings_dual_industry_categories1.py	2704	(unnamed)
scripts/CAPEX_OPEX_Gurobi_settings_dual_industry_categories1.py	2708	(unnamed)
scripts/CAPEX_OPEX_Gurobi_settings_dual_industry_categories1.py	2709	(unnamed)
scripts/CAPEX_OPEX_Gurobi_settings_dual_industry_categories1.py	2718	(unnamed)
scripts/CAPEX_OPEX_Gurobi_settings_dual_industry_categories1.py	2719	(unnamed)
scripts/CAPEX_OPEX_Gurobi_settings_dual_industry_categories1.py	2722	(unnamed)
scripts/CAPEX_OPEX_Gurobi_settings_dual_industry_categories1.py	2739	(unnamed)
scripts/CAPEX_OPEX_Gurobi_settings_dual_industry_categories1.py	2757	(unnamed)
scripts/CAPEX_OPEX_Gurobi_settings_dual_industry_categories1.py	2776	(unnamed)
scripts/CAPEX_OPEX_Gurobi_settings_dual_industry_categories1.py	2794	(unnamed)
scripts/CAPEX_OPEX_Gurobi_settings_dual_industry_categories1.py	2812	(unnamed)
scripts/CAPEX_OPEX_Gurobi_settings_dual_industry_categories1.py	2831	(unnamed)
scripts/CAPEX_OPEX_Gurobi_settings_dual_industry_categories1.py	2862	(unnamed)
scripts/CAPEX_OPEX_Gurobi_settings_dual_industry_categories1.py	2883	(unnamed)

Source	Line	Name / naming expression
scripts/CAPEX_OPEX_Gurobi_settings_dual_industry_categories1.py	2899	(unnamed)
scripts/CAPEX_OPEX_Gurobi_settings_dual_industry_categories1.py	2919	(unnamed)
scripts/CAPEX_OPEX_Gurobi_settings_dual_industry_categories1.py	2920	(unnamed)
scripts/CAPEX_OPEX_Gurobi_settings_dual_industry_categories1.py	2922	'h2_terminal_fixed_cost'
scripts/CAPEX_OPEX_Gurobi_settings_dual_industry_categories1.py	2926	(unnamed)
scripts/CAPEX_OPEX_Gurobi_settings_dual_industry_categories1.py	2946	(unnamed)
scripts/CAPEX_OPEX_Gurobi_settings_dual_industry_categories1.py	2957	(unnamed)
scripts/CAPEX_OPEX_Gurobi_settings_dual_industry_categories1.py	2971	(unnamed)
scripts/CAPEX_OPEX_Gurobi_settings_dual_industry_categories1.py	3230	(unnamed)
scripts/CAPEX_OPEX_Gurobi_settings_dual_industry_categories1.py	3338	(unnamed)
scripts/CAPEX_OPEX_Gurobi_settings_dual_industry_categories1.py	3344	(unnamed)
scripts/CAPEX_OPEX_Gurobi_settings_dual_industry_categories1.py	3351	(unnamed)
scripts/CAPEX_OPEX_Gurobi_settings_dual_industry_categories1.py	3357	(unnamed)
scripts/CAPEX_OPEX_Gurobi_settings_dual_industry_categories1.py	3364	(unnamed)
scripts/CAPEX_OPEX_Gurobi_settings_dual_industry_categories1.py	3369	(unnamed)
scripts/CAPEX_OPEX_Gurobi_settings_dual_industry_categories1.py	3386	(unnamed)
scripts/CAPEX_OPEX_Gurobi_settings_dual_industry_categories1.py	3402	(unnamed)
scripts/CAPEX_OPEX_Gurobi_settings_dual_industry_categories1.py	3428	(unnamed)
scripts/CAPEX_OPEX_Gurobi_settings_dual_industry_categories1.py	3440	(unnamed)
scripts/CAPEX_OPEX_Gurobi_settings_dual_industry_categories1.py	3453	(unnamed)
scripts/CAPEX_OPEX_Gurobi_settings_dual_industry_categories1.py	3498	'noloadel_penalty_cost_scaled'
scripts/complex_industry_accounting.py	135	f'industry_source_CO2_annual_{len(ledger)}'
scripts/complex_industry_investment_limits.py	87	f'complex_{variable_name}_limit_{sha1(repr(key).encode()).hexdigest()[:12]}'

Source	Line	Name / naming expression
scripts/industry_heat_pumps.py	183	f'industry_hp_capacity_{index}_{period}_{year}'
scripts/industry_heat_pumps.py	206	f'industry_hp_eligible_{index}_{threshold}_{wi}_{year}'
scripts/industry_heat_pumps.py	210	f'industry_hp_source_{index}_{period}_{year}'
scripts/industry_annual_s_curve.py	134	f'industry_electric_annual_link_{suffix}_{service_idx}_{year}'
scripts/industry_annual_s_curve.py	150	f'industry_electric_annual_decrease_{suffix}_{idx}'
scripts/industry_annual_s_curve.py	164	f'industry_electric_annual_growth_{suffix}_{idx}'
scripts/industry_annual_s_curve.py	167	f'industry_electric_annual_taper_{suffix}_{idx}'
scripts/industry_annual_s_curve.py	171	f'industry_electric_annual_growth_{suffix}_{idx}'
scripts/industry_annual_s_curve.py	287	f'ifs_annual_link_{suffix}_{index}_{year}'
scripts/industry_annual_s_curve.py	323	f'ifs_adoption_floor_{suffix}_{idx}_{year}'
scripts/industry_annual_s_curve.py	328	f'ifs_phaseout_ceiling_{suffix}_{idx}_{year}'
scripts/industry_annual_s_curve.py	351	f'ifsra_{suffix}_{idx}'
scripts/industry_annual_s_curve.py	355	f'ifsrt_{suffix}_{idx}'
scripts/industry_annual_s_curve.py	357	f'ifsrn_{suffix}_{idx}'
scripts/industry_annual_s_curve.py	359	f'ifsrd_{suffix}_{idx}'
scripts/industry_annual_s_curve.py	363	f'ifsrz_{suffix}_{idx}'
scripts/industry_annual_s_curve.py	365	f'ifsri_{suffix}_{idx}'
scripts/transport_pathways.py	330	f'retrofit_source_exit_{parent}_{zone}_{system}_{year}'
scripts/transport_pathways.py	340	f'retrofit_existing_stock_{parent}_{zone}_{system}_{year}'
scripts/transport_pathways.py	482	f'compat_annual_{label}_{unit}_{zone}_{system}_{year}'
scripts/transport_pathways.py	530	f'transport_ice_fuel_annual_max_share_{fuel}_{unit}_{zone}_{system}_{year}'
scripts/transport_pathways.py	561	f'synthetic_jet_build_pool_{zone}_{system}_{year}'
scripts/transport_pathways.py	568	f'synthetic_jet_stock_pool_{zone}_{system}_{year}'
scripts/transport_emissions.py	108	f'transport_co2_block[{unit},{sector},{zone},{system},{_block_index},{year}]'
scripts/transport_emissions.py	117	'transport_co2_by_unit[' + ','.join(map(str, annual_key)) + ']'
scripts/h2_terminals.py	183	f'h2_terminal_capacity[{y},{z},{s}]'

Source	Line	Name / naming expression
`scripts/h2_terminals.py`	184	`f'h2_terminal_max_capacity[{y},{z},{s}]'`
`scripts/h2_terminals.py`	185	`f'h2_terminal_build_limit[{y},{z},{s}]'`
`scripts/h2_terminals.py`	191	`f'h2_terminal_import_off[{y},{p},{z},{s}]'`
`scripts/h2_terminals.py`	193	`f'h2_terminal_export_off[{y},{p},{z},{s}]'`
`scripts/h2_terminals.py`	195	`f'h2_terminal_shared_throughput[{y},{p},{z},{s}]'`
`scripts/h2_terminals.py`	203	`f'h2_terminal_annual_throughput[{y},{z},{s}]'`
`scripts/h2_terminals.py`	218	`f'h2_terminal_ship_arrival[{y},{p},{r}]'`
`scripts/h2_terminals.py`	225	`f'h2_terminal_ship_departure[{y},{p},{r}]'`
`scripts/h2_terminals.py`	229	`f'h2_terminal_ship_average[{y},{p},{r}]'`
`scripts/h2_terminals.py`	244	`f'h2_terminal_fleet_capacity[{y},{p},{fleet}]'`

Variable definitions

See variable inventory for bounds, types and flag context.

Source	Line	Name / naming expression	Type
`scripts/reading_prev_and_vars_def_dual_industry_categories1.py`	1019	`(unnamed)`	`Gurobi default continuous`
`scripts/reading_prev_and_vars_def_dual_industry_categories1.py`	1036	`(unnamed)`	`Gurobi default continuous`
`scripts/reading_prev_and_vars_def_dual_industry_categories1.py`	1047	`'gen_output'`	`GRB.CONTINUOUS`
`scripts/reading_prev_and_vars_def_dual_industry_categories1.py`	1048	`'hphs_pumpback'`	`GRB.CONTINUOUS`
`scripts/reading_prev_and_vars_def_dual_industry_categories1.py`	1049	`'ceep_period'`	`GRB.CONTINUOUS`
`scripts/reading_prev_and_vars_def_dual_industry_categories1.py`	1050	`'capacity_Investment'`	`GRB.CONTINUOUS`
`scripts/reading_prev_and_vars_def_dual_industry_categories1.py`	1051	`'noloadel'`	`GRB.CONTINUOUS`

Source	Line	Name / naming expression	Type
`scripts/reading_prev_and_vars_def_dual_industry_categories1.py`	1053	`'import_pay_additional_cost_ETS'`	`GRB.CONTINUOUS`
`scripts/reading_prev_and_vars_def_dual_industry_categories1.py`	1055	`'import_pay_additional_cost_ETS_secc'`	`GRB.CONTINUOUS`
`scripts/reading_prev_and_vars_def_dual_industry_categories1.py`	1057	`'generationcapyear'`	`GRB.CONTINUOUS`
`scripts/reading_prev_and_vars_def_dual_industry_categories1.py`	1059	`'CBAM_cost'`	`GRB.CONTINUOUS`
`scripts/reading_prev_and_vars_def_dual_industry_categories1.py`	1061	`'CBAM_cost_secc'`	`GRB.CONTINUOUS`
`scripts/reading_prev_and_vars_def_dual_industry_categories1.py`	1063	`'CBAM_cost_out'`	`GRB.CONTINUOUS`
`scripts/reading_prev_and_vars_def_dual_industry_categories1.py`	1067	`'transfer_cap'`	`GRB.CONTINUOUS`
`scripts/reading_prev_and_vars_def_dual_industry_categories1.py`	1068	`'transfer_cap_hour_prim'`	`GRB.CONTINUOUS`
`scripts/reading_prev_and_vars_def_dual_industry_categories1.py`	1069	`'transfer_cap_new'`	`GRB.CONTINUOUS`
`scripts/reading_prev_and_vars_def_dual_industry_categories1.py`	1071	`'transfer_cap_outside'`	`GRB.CONTINUOUS`
`scripts/reading_prev_and_vars_def_dual_industry_categories1.py`	1073	`'transfer_cap_outside_new'`	`GRB.CONTINUOUS`
`scripts/reading_prev_and_vars_def_dual_industry_categories1.py`	1075	`'import_outside'`	`GRB.CONTINUOUS`
`scripts/reading_prev_and_vars_def_dual_industry_categories1.py`	1076	`'export_outside'`	`GRB.CONTINUOUS`
`scripts/reading_prev_and_vars_def_dual_industry_categories1.py`	1078	`'import_zone_system'`	`GRB.CONTINUOUS`
`scripts/reading_prev_and_vars_def_dual_industry_categories1.py`	1079	`'export_zone_system'`	`GRB.CONTINUOUS`
`scripts/reading_prev_and_vars_def_dual_industry_categories1.py`	1083	`'transfer_cap_hour_secc'`	`GRB.CONTINUOUS`
`scripts/reading_prev_and_vars_def_dual_industry_categories1.py`	1086	`'ETS_cost'`	`GRB.CONTINUOUS`

Source	Line	Name / naming expression	Type
scripts/reading_prev_and_vars_def_dual_industry_categories1.py	1087	'national_carbon_cost'	GRB.CONTINUOUS
scripts/reading_prev_and_vars_def_dual_industry_categories1.py	1089	'CO2_total_PP'	GRB.CONTINUOUS
scripts/reading_prev_and_vars_def_dual_industry_categories1.py	1090	'CO2_total_PP_zone'	GRB.CONTINUOUS
scripts/reading_prev_and_vars_def_dual_industry_categories1.py	1092	'CO2_total_elect'	GRB.CONTINUOUS
scripts/reading_prev_and_vars_def_dual_industry_categories1.py	1093	'CO2_total_heat'	GRB.CONTINUOUS
scripts/reading_prev_and_vars_def_dual_industry_categories1.py	1094	'CO2_total_industry'	GRB.CONTINUOUS
scripts/reading_prev_and_vars_def_dual_industry_categories1.py	1095	'CO2_total_refinery'	GRB.CONTINUOUS
scripts/reading_prev_and_vars_def_dual_industry_categories1.py	1096	'CO2_total_refinery_zone'	GRB.CONTINUOUS
scripts/reading_prev_and_vars_def_dual_industry_categories1.py	1097	'CO2_total_transport'	GRB.CONTINUOUS
scripts/reading_prev_and_vars_def_dual_industry_categories1.py	1098	'CO2_total_steamgas'	GRB.CONTINUOUS
scripts/reading_prev_and_vars_def_dual_industry_categories1.py	1101	'Storage_level_Hydro'	GRB.CONTINUOUS
scripts/reading_prev_and_vars_def_dual_industry_categories1.py	1102	'storage_level0'	GRB.CONTINUOUS
scripts/reading_prev_and_vars_def_dual_industry_categories1.py	1104	'sto_out_flow_Hydro'	GRB.CONTINUOUS
scripts/reading_prev_and_vars_def_dual_industry_categories1.py	1109	'power_to_h2'	GRB.CONTINUOUS
scripts/reading_prev_and_vars_def_dual_industry_categories1.py	1110	'h2_sto_soc'	GRB.CONTINUOUS
scripts/reading_prev_and_vars_def_dual_industry_categories1.py	1111	'h2_sto_soc0'	GRB.CONTINUOUS
scripts/reading_prev_and_vars_def_dual_industry_categories1.py	1112	'h2_storage_out'	GRB.CONTINUOUS

Source	Line	Name / naming expression	Type
scripts/reading_prev_and_vars_def_dual_industry_categories1.py	1113	'h2_storage_cap'	GRB.CONTINUOUS
scripts/reading_prev_and_vars_def_dual_industry_categories1.py	1114	'h2_electrolysis_cap'	GRB.CONTINUOUS
scripts/reading_prev_and_vars_def_dual_industry_categories1.py	1115	'h2_electrolysis_gen'	GRB.CONTINUOUS
scripts/reading_prev_and_vars_def_dual_industry_categories1.py	1116	'H2_sto_cap'	GRB.CONTINUOUS
scripts/reading_prev_and_vars_def_dual_industry_categories1.py	1117	'cap_H2'	GRB.CONTINUOUS
scripts/reading_prev_and_vars_def_dual_industry_categories1.py	1118	'FC_cap'	GRB.CONTINUOUS
scripts/reading_prev_and_vars_def_dual_industry_categories1.py	1119	'generation_H2_steam'	GRB.CONTINUOUS
scripts/reading_prev_and_vars_def_dual_industry_categories1.py	1120	'new_H2_steam'	GRB.CONTINUOUS
scripts/reading_prev_and_vars_def_dual_industry_categories1.py	1121	'H2_steamcap'	GRB.CONTINUOUS
scripts/reading_prev_and_vars_def_dual_industry_categories1.py	1124	'FC_capacity_Investment'	GRB.CONTINUOUS
scripts/reading_prev_and_vars_def_dual_industry_categories1.py	1125	'gen_output_FuelCell'	GRB.CONTINUOUS
scripts/reading_prev_and_vars_def_dual_industry_categories1.py	1131	'fuel_ind_consumpt'	GRB.CONTINUOUS
scripts/reading_prev_and_vars_def_dual_industry_categories1.py	1132	'boiler_industry_generation'	GRB.CONTINUOUS
scripts/reading_prev_and_vars_def_dual_industry_categories1.py	1133	'boiler_industry_generation_cap_year'	GRB.CONTINUOUS
scripts/reading_prev_and_vars_def_dual_industry_categories1.py	1136	'industry_share_fuel_annual'	GRB.CONTINUOUS
scripts/reading_prev_and_vars_def_dual_industry_categories1.py	1151	'industry_share_fuel'	GRB.CONTINUOUS
scripts/reading_prev_and_vars_def_dual_industry_categories1.py	1155	'boiler_industry_cap_inv'	GRB.CONTINUOUS

Source	Line	Name / naming expression	Type
scripts/reading_prev_and_vars_def_dual_industry_categories1.py	1156	'elInd'	GRB.CONTINUOUS
scripts/reading_prev_and_vars_def_dual_industry_categories1.py	1159	'CC_industry_cap_inv'	GRB.CONTINUOUS
scripts/reading_prev_and_vars_def_dual_industry_categories1.py	1160	'CC_industry_gen'	GRB.CONTINUOUS
scripts/reading_prev_and_vars_def_dual_industry_categories1.py	1161	'CC_industry_cap_year'	GRB.CONTINUOUS
scripts/reading_prev_and_vars_def_dual_industry_categories1.py	1162	'CO2_CC_el_dem'	GRB.CONTINUOUS
scripts/reading_prev_and_vars_def_dual_industry_categories1.py	1163	'CO2_CC_el_dem_tot'	GRB.CONTINUOUS
scripts/reading_prev_and_vars_def_dual_industry_categories1.py	1167	'CC_tank_cap_inv'	GRB.CONTINUOUS
scripts/reading_prev_and_vars_def_dual_industry_categories1.py	1168	'CC_tank_cap_year'	GRB.CONTINUOUS
scripts/reading_prev_and_vars_def_dual_industry_categories1.py	1170	'CO2_amine_stor_SOC'	GRB.CONTINUOUS
scripts/reading_prev_and_vars_def_dual_industry_categories1.py	1171	'CO2_amine_stor_SOC0'	GRB.CONTINUOUS
scripts/reading_prev_and_vars_def_dual_industry_categories1.py	1173	'CC_reboiler_cap_inv'	GRB.CONTINUOUS
scripts/reading_prev_and_vars_def_dual_industry_categories1.py	1174	'CO2_stripper_react'	GRB.CONTINUOUS
scripts/reading_prev_and_vars_def_dual_industry_categories1.py	1175	'CC_reboiler_cap_year'	GRB.CONTINUOUS
scripts/reading_prev_and_vars_def_dual_industry_categories1.py	1176	'CO2_stripper_react_heat_dem'	GRB.CONTINUOUS
scripts/reading_prev_and_vars_def_dual_industry_categories1.py	1177	'CO2_stripper_react_el_dem'	GRB.CONTINUOUS
scripts/reading_prev_and_vars_def_dual_industry_categories1.py	1178	'CO2_gen_tot'	GRB.CONTINUOUS
scripts/reading_prev_and_vars_def_dual_industry_categories1.py	1181	'Carbon_DAC'	GRB.CONTINUOUS

Source	Line	Name / naming expression	Type
`scripts/reading_prev_and_vars_def_dual_industry_categories1.py`	1182	`'Carbon_DAC_new_cap'`	`GRB.CONTINUOUS`
`scripts/reading_prev_and_vars_def_dual_industry_categories1.py`	1183	`'Carbon_DAC_cap'`	`GRB.CONTINUOUS`
`scripts/reading_prev_and_vars_def_dual_industry_categories1.py`	1187	`'CO2_storage_cap_year'`	`GRB.CONTINUOUS`
`scripts/reading_prev_and_vars_def_dual_industry_categories1.py`	1188	`'CO2_storage_cap_inv'`	`GRB.CONTINUOUS`
`scripts/reading_prev_and_vars_def_dual_industry_categories1.py`	1189	`'CO2_storage_SOC'`	`GRB.CONTINUOUS`
`scripts/reading_prev_and_vars_def_dual_industry_categories1.py`	1190	`'CO2_storage_SOC0'`	`GRB.CONTINUOUS`
`scripts/reading_prev_and_vars_def_dual_industry_categories1.py`	1191	`'CO2_storage_in'`	`GRB.CONTINUOUS`
`scripts/reading_prev_and_vars_def_dual_industry_categories1.py`	1192	`'CO2_storage_out'`	`GRB.CONTINUOUS`
`scripts/reading_prev_and_vars_def_dual_industry_categories1.py`	1197	`'efuel_gen_new_cap_gen'`	`GRB.CONTINUOUS`
`scripts/reading_prev_and_vars_def_dual_industry_categories1.py`	1201	`'veh_energy_prim'`	`GRB.CONTINUOUS`
`scripts/reading_prev_and_vars_def_dual_industry_categories1.py`	1203	`'ICE_energy_transp_prim'`	`GRB.CONTINUOUS`
`scripts/reading_prev_and_vars_def_dual_industry_categories1.py`	1206	`'ICE_energy_transp_primsecc_sum_fuel'`	`GRB.CONTINUOUS`
`scripts/reading_prev_and_vars_def_dual_industry_categories1.py`	1210	`'fuel_industry_hours'`	`GRB.CONTINUOUS`
`scripts/reading_prev_and_vars_def_dual_industry_categories1.py`	1211	`'fuel_industry_hour_CO2'`	`GRB.CONTINUOUS`
`scripts/reading_prev_and_vars_def_dual_industry_categories1.py`	1212	`'fuel_dem'`	`GRB.CONTINUOUS`
`scripts/reading_prev_and_vars_def_dual_industry_categories1.py`	1213	`'fuel_demyear'`	`GRB.CONTINUOUS`
`scripts/reading_prev_and_vars_def_dual_industry_categories1.py`	1214	`'fuel_dem_prod'`	`GRB.CONTINUOUS`

Source	Line	Name / naming expression	Type
`scripts/reading_prev_and_vars_def_dual_industry_categories1.py`	1215	`'fuel_demand'`	`GRB.CONTINUOUS`
`scripts/reading_prev_and_vars_def_dual_industry_categories1.py`	1216	`'fuel_dem_prod_cap'`	`GRB.CONTINUOUS`
`scripts/reading_prev_and_vars_def_dual_industry_categories1.py`	1217	`'fuel_dem_cap'`	`GRB.CONTINUOUS`
`scripts/reading_prev_and_vars_def_dual_industry_categories1.py`	1219	`'efuelinput1'`	`GRB.CONTINUOUS`
`scripts/reading_prev_and_vars_def_dual_industry_categories1.py`	1220	`'efuelinput2'`	`GRB.CONTINUOUS`
`scripts/reading_prev_and_vars_def_dual_industry_categories1.py`	1221	`'efuelinput3'`	`GRB.CONTINUOUS`
`scripts/reading_prev_and_vars_def_dual_industry_categories1.py`	1223	`'feedstock_tot'`	`GRB.CONTINUOUS`
`scripts/reading_prev_and_vars_def_dual_industry_categories1.py`	1224	`'feedstock_tot1'`	`GRB.CONTINUOUS`
`scripts/reading_prev_and_vars_def_dual_industry_categories1.py`	1225	`'feedstock_tot2'`	`GRB.CONTINUOUS`
`scripts/reading_prev_and_vars_def_dual_industry_categories1.py`	1226	`'feedstock_tot3'`	`GRB.CONTINUOUS`
`scripts/reading_prev_and_vars_def_dual_industry_categories1.py`	1230	`'fuel_boil'`	`GRB.CONTINUOUS`
`scripts/reading_prev_and_vars_def_dual_industry_categories1.py`	1231	`'fuel_boil_hour'`	`GRB.CONTINUOUS`
`scripts/reading_prev_and_vars_def_dual_industry_categories1.py`	1233	`'fuel_boil_DH'`	`GRB.CONTINUOUS`
`scripts/reading_prev_and_vars_def_dual_industry_categories1.py`	1234	`'fuel_boil_DH_hour'`	`GRB.CONTINUOUS`
`scripts/reading_prev_and_vars_def_dual_industry_categories1.py`	1236	`'fuel_gen_power'`	`GRB.CONTINUOUS`
`scripts/reading_prev_and_vars_def_dual_industry_categories1.py`	1238	`'PE_power_tot'`	`GRB.CONTINUOUS`
`scripts/reading_prev_and_vars_def_dual_industry_categories1.py`	1240	`'PE_power_CHP_tot'`	`GRB.CONTINUOUS`

Source	Line	Name / naming expression	Type
scripts/reading_prev_and_vars_def_dual_industry_categories1.py	1242	'fuel_SOC'	GRB.CONTINUOUS
scripts/reading_prev_and_vars_def_dual_industry_categories1.py	1243	'fuel_SOC_0'	GRB.CONTINUOUS
scripts/reading_prev_and_vars_def_dual_industry_categories1.py	1249	'fuel_storage_cap_year'	GRB.CONTINUOUS
scripts/reading_prev_and_vars_def_dual_industry_categories1.py	1250	'fuel_storage_cap_inv'	GRB.CONTINUOUS
scripts/reading_prev_and_vars_def_dual_industry_categories1.py	1254	'oil_products_gen'	GRB.CONTINUOUS
scripts/reading_prev_and_vars_def_dual_industry_categories1.py	1255	'oil_gas_storage'	GRB.CONTINUOUS
scripts/reading_prev_and_vars_def_dual_industry_categories1.py	1256	'raf_oil_dem'	GRB.CONTINUOUS
scripts/reading_prev_and_vars_def_dual_industry_categories1.py	1259	'oil_gas_SOC'	GRB.CONTINUOUS
scripts/reading_prev_and_vars_def_dual_industry_categories1.py	1260	'oil_gas_SOC0'	GRB.CONTINUOUS
scripts/reading_prev_and_vars_def_dual_industry_categories1.py	1262	'oil_gas_extraction'	GRB.CONTINUOUS
scripts/reading_prev_and_vars_def_dual_industry_categories1.py	1263	'oil_gas_reserves'	GRB.CONTINUOUS
scripts/reading_prev_and_vars_def_dual_industry_categories1.py	1266	'generation_H2_steam_gas'	GRB.CONTINUOUS
scripts/reading_prev_and_vars_def_dual_industry_categories1.py	1270	'transfer_cap'	GRB.CONTINUOUS
scripts/reading_prev_and_vars_def_dual_industry_categories1.py	1271	'transfer_cap_hour_prim'	GRB.CONTINUOUS
scripts/reading_prev_and_vars_def_dual_industry_categories1.py	1272	'transfer_cap_new'	GRB.CONTINUOUS
scripts/reading_prev_and_vars_def_dual_industry_categories1.py	1274	'transfer_cap_outside'	GRB.CONTINUOUS
scripts/reading_prev_and_vars_def_dual_industry_categories1.py	1276	'transfer_cap_outside_new'	GRB.CONTINUOUS

Source	Line	Name / naming expression	Type
`scripts/reading_prev_and_vars_def_dual_industry_categories1.py`	1278	`'import_outside'`	`GRB.CONTINUOUS`
`scripts/reading_prev_and_vars_def_dual_industry_categories1.py`	1279	`'export_outside'`	`GRB.CONTINUOUS`
`scripts/reading_prev_and_vars_def_dual_industry_categories1.py`	1281	`'import_zone_system'`	`GRB.CONTINUOUS`
`scripts/reading_prev_and_vars_def_dual_industry_categories1.py`	1282	`'export_zone_system'`	`GRB.CONTINUOUS`
`scripts/reading_prev_and_vars_def_dual_industry_categories1.py`	1285	`'import_zone_system_terminal'`	`GRB.CONTINUOUS`
`scripts/reading_prev_and_vars_def_dual_industry_categories1.py`	1286	`'export_zone_system_terminal'`	`GRB.CONTINUOUS`
`scripts/reading_prev_and_vars_def_dual_industry_categories1.py`	1287	`'import_zone_system_terminal_outside'`	`GRB.CONTINUOUS`
`scripts/reading_prev_and_vars_def_dual_industry_categories1.py`	1288	`'export_zone_system_terminal_outside'`	`GRB.CONTINUOUS`
`scripts/reading_prev_and_vars_def_dual_industry_categories1.py`	1290	`'transfer_cap_terminal'`	`GRB.CONTINUOUS`
`scripts/reading_prev_and_vars_def_dual_industry_categories1.py`	1291	`'transfer_cap_new_terminal'`	`GRB.CONTINUOUS`
`scripts/reading_prev_and_vars_def_dual_industry_categories1.py`	1294	`'emiss_generation_transport'`	`GRB.CONTINUOUS`
`scripts/reading_prev_and_vars_def_dual_industry_categories1.py`	1295	`'emiss_generation_industry'`	`GRB.CONTINUOUS`
`scripts/reading_prev_and_vars_def_dual_industry_categories1.py`	1296	`'emiss_generation_power_generation'`	`GRB.CONTINUOUS`
`scripts/reading_prev_and_vars_def_dual_industry_categories1.py`	1297	`'emiss_generation_CHP'`	`GRB.CONTINUOUS`
`scripts/reading_prev_and_vars_def_dual_industry_categories1.py`	1298	`'emiss_generation_individual_heating'`	`GRB.CONTINUOUS`
`scripts/reading_prev_and_vars_def_dual_industry_categories1.py`	1299	`'emiss_generation_DH'`	`GRB.CONTINUOUS`
`scripts/reading_prev_and_vars_def_dual_industry_categories1.py`	1300	`'transport_emiss_cleaning'`	`GRB.CONTINUOUS`

Source	Line	Name / naming expression	Type
`scripts/reading_prev_and_vars_def_dual_industry_categories1.py`	1304	`'ind_sto_soc'`	`GRB.CONTINUOUS`
`scripts/reading_prev_and_vars_def_dual_industry_categories1.py`	1305	`'ind_sto_soc0'`	`GRB.CONTINUOUS`
`scripts/reading_prev_and_vars_def_dual_industry_categories1.py`	1306	`'ind_sto_capacity'`	`GRB.CONTINUOUS`
`scripts/reading_prev_and_vars_def_dual_industry_categories1.py`	1307	`'ind_sto_capacity_new'`	`GRB.CONTINUOUS`
`scripts/reading_prev_and_vars_def_dual_industry_categories1.py`	1315	`'import_zone_system_H2'`	`GRB.CONTINUOUS`
`scripts/reading_prev_and_vars_def_dual_industry_categories1.py`	1316	`'export_zone_system_H2'`	`GRB.CONTINUOUS`
`scripts/reading_prev_and_vars_def_dual_industry_categories1.py`	1318	`'h2_comp_cap'`	`GRB.CONTINUOUS`
`scripts/reading_prev_and_vars_def_dual_industry_categories1.py`	1319	`'h2_comp_cap_new'`	`GRB.CONTINUOUS`
`scripts/reading_prev_and_vars_def_dual_industry_categories1.py`	1321	`'transfer_cap_H2'`	`GRB.CONTINUOUS`
`scripts/reading_prev_and_vars_def_dual_industry_categories1.py`	1322	`'transfer_cap_new_H2'`	`GRB.CONTINUOUS`
`scripts/reading_prev_and_vars_def_dual_industry_categories1.py`	1324	`'transfer_cap_outside_H2'`	`GRB.CONTINUOUS`
`scripts/reading_prev_and_vars_def_dual_industry_categories1.py`	1325	`'transfer_cap_outside_new_H2'`	`GRB.CONTINUOUS`
`scripts/reading_prev_and_vars_def_dual_industry_categories1.py`	1327	`'import_outside_H2'`	`GRB.CONTINUOUS`
`scripts/reading_prev_and_vars_def_dual_industry_categories1.py`	1328	`'export_outside_H2'`	`GRB.CONTINUOUS`
`scripts/reading_prev_and_vars_def_dual_industry_categories1.py`	1333	`'transformer_cap'`	`GRB.CONTINUOUS`
`scripts/reading_prev_and_vars_def_dual_industry_categories1.py`	1334	`'substation_cap_new'`	`GRB.CONTINUOUS`
`scripts/reading_prev_and_vars_def_dual_industry_categories1.py`	1342	`'stat_sto_in_out'`	`Gurobi default continuous`

Source	Line	Name / naming expression	Type
`scripts/reading_prev_and_vars_def_dual_industry_categories1.py`	1350	`'stat_charge'`	`Gurobi default continuous`
`scripts/reading_prev_and_vars_def_dual_industry_categories1.py`	1351	`'stat_discharge'`	`Gurobi default continuous`
`scripts/reading_prev_and_vars_def_dual_industry_categories1.py`	1363	`'stat_sto_soc'`	`GRB.CONTINUOUS`
`scripts/reading_prev_and_vars_def_dual_industry_categories1.py`	1364	`'stat_sto_soc0'`	`GRB.CONTINUOUS`
`scripts/reading_prev_and_vars_def_dual_industry_categories1.py`	1365	`'stat_sto_cap'`	`GRB.CONTINUOUS`
`scripts/reading_prev_and_vars_def_dual_industry_categories1.py`	1368	`'stat_sto_cap_inv'`	`GRB.CONTINUOUS`
`scripts/reading_prev_and_vars_def_dual_industry_categories1.py`	1370	`'demand_el_zone_system'`	`GRB.CONTINUOUS`
`scripts/reading_prev_and_vars_def_dual_industry_categories1.py`	1371	`'gen_el_zone_system'`	`GRB.CONTINUOUS`
`scripts/reading_prev_and_vars_def_dual_industry_categories1.py`	1377	`'import_zone_system_secc'`	`GRB.CONTINUOUS`
`scripts/reading_prev_and_vars_def_dual_industry_categories1.py`	1378	`'export_zone_system_secc'`	`GRB.CONTINUOUS`
`scripts/reading_prev_and_vars_def_dual_industry_categories1.py`	1384	`'veh_sto_soc'`	`GRB.CONTINUOUS`
`scripts/reading_prev_and_vars_def_dual_industry_categories1.py`	1385	`'veh_sto_soc0'`	`GRB.CONTINUOUS`
`scripts/reading_prev_and_vars_def_dual_industry_categories1.py`	1387	`'veh_sto_max'`	`GRB.CONTINUOUS`
`scripts/reading_prev_and_vars_def_dual_industry_categories1.py`	1388	`'veh_Grid_P_max'`	`GRB.CONTINUOUS`
`scripts/reading_prev_and_vars_def_dual_industry_categories1.py`	1392	`'veh_sto_in_out'`	`Gurobi default continuous`
`scripts/reading_prev_and_vars_def_dual_industry_categories1.py`	1401	`'veh_charge'`	`Gurobi default continuous`
`scripts/reading_prev_and_vars_def_dual_industry_categories1.py`	1402	`'veh_discharge'`	`Gurobi default continuous`

Source	Line	Name / naming expression	Type
scripts/reading_prev_and_vars_def_dual_industry_categories1.py	1422	'veh_transp_num'	GRB.CONTINUOUS
scripts/reading_prev_and_vars_def_dual_industry_categories1.py	1423	'veh_inv'	GRB.CONTINUOUS
scripts/reading_prev_and_vars_def_dual_industry_categories1.py	1424	'veh_decom'	GRB.CONTINUOUS
scripts/reading_prev_and_vars_def_dual_industry_categories1.py	1426	'EV_infr_cap'	GRB.CONTINUOUS
scripts/reading_prev_and_vars_def_dual_industry_categories1.py	1427	'H2_infr_cap'	GRB.CONTINUOUS
scripts/reading_prev_and_vars_def_dual_industry_categories1.py	1429	'EV_infr_new'	GRB.CONTINUOUS
scripts/reading_prev_and_vars_def_dual_industry_categories1.py	1430	'H2_infr_new'	GRB.CONTINUOUS
scripts/reading_prev_and_vars_def_dual_industry_categories1.py	1433	'veh_energy_prim'	GRB.CONTINUOUS
scripts/reading_prev_and_vars_def_dual_industry_categories1.py	1435	'CO2_transp'	GRB.CONTINUOUS
scripts/reading_prev_and_vars_def_dual_industry_categories1.py	1450	'heat_pump_dh'	GRB.CONTINUOUS
scripts/reading_prev_and_vars_def_dual_industry_categories1.py	1451	'heat_pump_ind'	GRB.CONTINUOUS
scripts/reading_prev_and_vars_def_dual_industry_categories1.py	1454	'heat_pump_out'	GRB.CONTINUOUS
scripts/reading_prev_and_vars_def_dual_industry_categories1.py	1455	'heat_pump_indstor_capyear'	GRB.CONTINUOUS
scripts/reading_prev_and_vars_def_dual_industry_categories1.py	1456	'heatpump_sto_soc'	GRB.CONTINUOUS
scripts/reading_prev_and_vars_def_dual_industry_categories1.py	1457	'heat_pump_sto_soc0'	GRB.CONTINUOUS
scripts/reading_prev_and_vars_def_dual_industry_categories1.py	1464	'Heat_pump_df_cap_inv'	GRB.CONTINUOUS
scripts/reading_prev_and_vars_def_dual_industry_categories1.py	1465	'Heat_pump_ind_cap_inv'	GRB.CONTINUOUS

Source	Line	Name / naming expression	Type
`scripts/reading_prev_and_vars_def_dual_industry_categories1.py`	1467	`'heat_pump_ind_capyear'`	`GRB.CONTINUOUS`
`scripts/reading_prev_and_vars_def_dual_industry_categories1.py`	1468	`'heat_pump_dh_capyear'`	`GRB.CONTINUOUS`
`scripts/reading_prev_and_vars_def_dual_industry_categories1.py`	1474	`'heat_pump_ind_share'`	`GRB.CONTINUOUS`
`scripts/reading_prev_and_vars_def_dual_industry_categories1.py`	1475	`'heat_pump_ind_maxhour'`	`GRB.CONTINUOUS`
`scripts/reading_prev_and_vars_def_dual_industry_categories1.py`	1476	`'heat_pump_dh_share'`	`GRB.CONTINUOUS`
`scripts/reading_prev_and_vars_def_dual_industry_categories1.py`	1477	`'heat_pump_dh_maxhour'`	`GRB.CONTINUOUS`
`scripts/reading_prev_and_vars_def_dual_industry_categories1.py`	1480	`'cool_pump_dh'`	`GRB.CONTINUOUS`
`scripts/reading_prev_and_vars_def_dual_industry_categories1.py`	1481	`'cool_pump_ind'`	`GRB.CONTINUOUS`
`scripts/reading_prev_and_vars_def_dual_industry_categories1.py`	1485	`'trad_boilers_Heat_dh'`	`GRB.CONTINUOUS`
`scripts/reading_prev_and_vars_def_dual_industry_categories1.py`	1486	`'trad_boilers_Heat_ind'`	`GRB.CONTINUOUS`
`scripts/reading_prev_and_vars_def_dual_industry_categories1.py`	1487	`'capacity_Investment_ind_boiler'`	`GRB.CONTINUOUS`
`scripts/reading_prev_and_vars_def_dual_industry_categories1.py`	1488	`'trad_boiler_dh_cap_inv'`	`GRB.CONTINUOUS`
`scripts/reading_prev_and_vars_def_dual_industry_categories1.py`	1490	`'trad_boiler_ind_capyear'`	`GRB.CONTINUOUS`
`scripts/reading_prev_and_vars_def_dual_industry_categories1.py`	1491	`'boiler_dh_capyear'`	`GRB.CONTINUOUS`
`scripts/reading_prev_and_vars_def_dual_industry_categories1.py`	1495	`'boiler_ind_share'`	`GRB.CONTINUOUS`
`scripts/reading_prev_and_vars_def_dual_industry_categories1.py`	1496	`'boiler_ind_maxhour'`	`GRB.CONTINUOUS`
`scripts/reading_prev_and_vars_def_dual_industry_categories1.py`	1497	`'boiler_dh_share'`	`GRB.CONTINUOUS`

Source	Line	Name / naming expression	Type
`scripts/reading_prev_and_vars_def_dual_industry_categories1.py`	1498	`'boiler_dh_maxhour'`	`GRB.CONTINUOUS`
`scripts/reading_prev_and_vars_def_dual_industry_categories1.py`	1502	`'Res_to_Heat'`	`GRB.CONTINUOUS`
`scripts/reading_prev_and_vars_def_dual_industry_categories1.py`	1505	`'unmet_dh_heat'`	`GRB.CONTINUOUS`
`scripts/reading_prev_and_vars_def_dual_industry_categories1.py`	1506	`'unmet_indv_heat'`	`GRB.CONTINUOUS`
`scripts/reading_prev_and_vars_def_dual_industry_categories1.py`	1518	`'unmet_industry_heat'`	`GRB.CONTINUOUS`
`scripts/reading_prev_and_vars_def_dual_industry_categories1.py`	1533	`'solar_gen_heat'`	`GRB.CONTINUOUS`
`scripts/reading_prev_and_vars_def_dual_industry_categories1.py`	1534	`'solar_gen_heat_new'`	`GRB.CONTINUOUS`
`scripts/reading_prev_and_vars_def_dual_industry_categories1.py`	1535	`'solar_gen_heat_year'`	`GRB.CONTINUOUS`
`scripts/reading_prev_and_vars_def_dual_industry_categories1.py`	1536	`'solar_gen_heat_year_cap'`	`GRB.CONTINUOUS`
`scripts/reading_prev_and_vars_def_dual_industry_categories1.py`	1539	`'solar_stor_heat_new'`	`GRB.CONTINUOUS`
`scripts/reading_prev_and_vars_def_dual_industry_categories1.py`	1540	`'solar_stor_soc'`	`GRB.CONTINUOUS`
`scripts/reading_prev_and_vars_def_dual_industry_categories1.py`	1541	`'solar_stor_soc0'`	`GRB.CONTINUOUS`
`scripts/reading_prev_and_vars_def_dual_industry_categories1.py`	1542	`'solar_stor_heat_year'`	`GRB.CONTINUOUS`
`scripts/reading_prev_and_vars_def_dual_industry_categories1.py`	1544	`'solar_stor_output'`	`GRB.CONTINUOUS`
`scripts/reading_prev_and_vars_def_dual_industry_categories1.py`	1546	`'solar_gen_indv_heat'`	`GRB.CONTINUOUS`
`scripts/reading_prev_and_vars_def_dual_industry_categories1.py`	1547	`'solar_gen_heat_indv_new'`	`GRB.CONTINUOUS`
`scripts/reading_prev_and_vars_def_dual_industry_categories1.py`	1548	`'solar_gen_indv_heat_cap'`	`GRB.CONTINUOUS`

Source	Line	Name / naming expression	Type
`scripts/reading_prev_and_vars_def_dual_industry_categories1.py`	1552	`'solar_ind_share'`	`GRB.CONTINUOUS`
`scripts/reading_prev_and_vars_def_dual_industry_categories1.py`	1558	`'heat_from_elect_storage_input'`	`GRB.CONTINUOUS`
`scripts/reading_prev_and_vars_def_dual_industry_categories1.py`	1559	`'heat_storage_level'`	`GRB.CONTINUOUS`
`scripts/reading_prev_and_vars_def_dual_industry_categories1.py`	1560	`'heat_sto_level0'`	`GRB.CONTINUOUS`
`scripts/reading_prev_and_vars_def_dual_industry_categories1.py`	1563	`'STOCapacity_heat_year'`	`GRB.CONTINUOUS`
`scripts/reading_prev_and_vars_def_dual_industry_categories1.py`	1564	`'STOCapacity_heat_new'`	`GRB.CONTINUOUS`
`scripts/reading_prev_and_vars_def_dual_industry_categories1.py`	1569	`'cool_storage_level'`	`GRB.CONTINUOUS`
`scripts/reading_prev_and_vars_def_dual_industry_categories1.py`	1570	`'cool_sto_level0'`	`GRB.CONTINUOUS`
`scripts/reading_prev_and_vars_def_dual_industry_categories1.py`	1573	`'STOCapacity_cool_year'`	`GRB.CONTINUOUS`
`scripts/reading_prev_and_vars_def_dual_industry_categories1.py`	1574	`'STOCapacity_cool_new'`	`GRB.CONTINUOUS`
`scripts/Build_model_dual_industry_categories_named_debug.py`	1560	`'stage2_quantity_guidance_slack_under'`	`GRB.CONTINUOUS`
`scripts/Build_model_dual_industry_categories_named_debug.py`	1565	`'stage2_quantity_guidance_slack_over'`	`GRB.CONTINUOUS`
`scripts/Build_model_dual_industry_categories_named_debug.py`	10048	`f'transport_shift_baseline_energy_gwh_{_transport_baseline_annual_var_no}'`	`GRB.CONTINUOUS`
`scripts/Build_model_dual_industry_categories_named_debug.py`	10096	`f'transport_fuel_annual_energy_gwh_{_transport_fuel_annual_var_no}'`	`GRB.CONTINUOUS`
`scripts/Build_model_dual_industry_categories_named_debug.py`	11382	`'rolling_terminal_fuel_soc_slack'`	`GRB.CONTINUOUS`
`scripts/Build_model_dual_industry_categories_named_debug.py`	13551	`'rolling_carbon_budget_slack'`	`GRB.CONTINUOUS`
`scripts/Build_model_dual_industry_categories_named_debug.py`	13581	`'fuel_dem_zone'`	`GRB.CONTINUOUS`

Source	Line	Name / naming expression	Type
`scripts/CAPEX_OPEX_Gurobi_settings_dual_industry_categories1.py`	202	`'inv_cost_year'`	`GRB.CONTINUOUS`
`scripts/CAPEX_OPEX_Gurobi_settings_dual_industry_categories1.py`	203	`'inv_cost_disp_no_CHP_year'`	`GRB.CONTINUOUS`
`scripts/CAPEX_OPEX_Gurobi_settings_dual_industry_categories1.py`	204	`'inv_cost_CHP_year'`	`GRB.CONTINUOUS`
`scripts/CAPEX_OPEX_Gurobi_settings_dual_industry_categories1.py`	205	`'statCapInvCost_year'`	`GRB.CONTINUOUS`
`scripts/CAPEX_OPEX_Gurobi_settings_dual_industry_categories1.py`	207	`'CAPEX_year'`	`GRB.CONTINUOUS`
`scripts/CAPEX_OPEX_Gurobi_settings_dual_industry_categories1.py`	209	`'el_line_infr_year'`	`GRB.CONTINUOUS`
`scripts/CAPEX_OPEX_Gurobi_settings_dual_industry_categories1.py`	210	`'el_line_infr_yearout'`	`GRB.CONTINUOUS`
`scripts/CAPEX_OPEX_Gurobi_settings_dual_industry_categories1.py`	211	`'product_line_infr_year'`	`GRB.CONTINUOUS`
`scripts/CAPEX_OPEX_Gurobi_settings_dual_industry_categories1.py`	212	`'product_line_infr_yearout'`	`GRB.CONTINUOUS`
`scripts/CAPEX_OPEX_Gurobi_settings_dual_industry_categories1.py`	213	`'terminal_line_infr_year'`	`GRB.CONTINUOUS`
`scripts/CAPEX_OPEX_Gurobi_settings_dual_industry_categories1.py`	215	`'el_substation_costyear'`	`GRB.CONTINUOUS`
`scripts/CAPEX_OPEX_Gurobi_settings_dual_industry_categories1.py`	216	`'h2_pump_costyear'`	`GRB.CONTINUOUS`
`scripts/CAPEX_OPEX_Gurobi_settings_dual_industry_categories1.py`	218	`'H2_line_infr_year'`	`GRB.CONTINUOUS`
`scripts/CAPEX_OPEX_Gurobi_settings_dual_industry_categories1.py`	219	`'H2_line_infr_yearout'`	`GRB.CONTINUOUS`
`scripts/CAPEX_OPEX_Gurobi_settings_dual_industry_categories1.py`	221	`'inv_cost_efuel_year'`	`GRB.CONTINUOUS`
`scripts/CAPEX_OPEX_Gurobi_settings_dual_industry_categories1.py`	222	`'inv_cost_DAC_year'`	`GRB.CONTINUOUS`
`scripts/CAPEX_OPEX_Gurobi_settings_dual_industry_categories1.py`	224	`'inv_cost_solar_heat_year'`	`GRB.CONTINUOUS`

Source	Line	Name / naming expression	Type
`scripts/CAPEX_OPEX_Gurobi_settings_dual_industry_categories1.py`	225	`'inv_cost_solar_stor_year'`	`GRB.CONTINUOUS`
`scripts/CAPEX_OPEX_Gurobi_settings_dual_industry_categories1.py`	226	`'inv_cost_solar_indv_heat_year'`	`GRB.CONTINUOUS`
`scripts/CAPEX_OPEX_Gurobi_settings_dual_industry_categories1.py`	227	`'ind_boiler_inv_year'`	`GRB.CONTINUOUS`
`scripts/CAPEX_OPEX_Gurobi_settings_dual_industry_categories1.py`	228	`'inv_cost_FuelCell_year'`	`GRB.CONTINUOUS`
`scripts/CAPEX_OPEX_Gurobi_settings_dual_industry_categories1.py`	229	`'heat_pump_cap_cost_year'`	`GRB.CONTINUOUS`
`scripts/CAPEX_OPEX_Gurobi_settings_dual_industry_categories1.py`	230	`'inv_cost_boiler_year'`	`GRB.CONTINUOUS`
`scripts/CAPEX_OPEX_Gurobi_settings_dual_industry_categories1.py`	231	`'h2_storage_cost_year'`	`GRB.CONTINUOUS`
`scripts/CAPEX_OPEX_Gurobi_settings_dual_industry_categories1.py`	232	`'h2_CAPEX_cost_year'`	`GRB.CONTINUOUS`
`scripts/CAPEX_OPEX_Gurobi_settings_dual_industry_categories1.py`	233	`'transport_cost_year'`	`GRB.CONTINUOUS`
`scripts/CAPEX_OPEX_Gurobi_settings_dual_industry_categories1.py`	234	`'transport_ev_infr_year'`	`GRB.CONTINUOUS`
`scripts/CAPEX_OPEX_Gurobi_settings_dual_industry_categories1.py`	235	`'transport_fcev_infr_year'`	`GRB.CONTINUOUS`
`scripts/CAPEX_OPEX_Gurobi_settings_dual_industry_categories1.py`	237	`'steam_gas_inv_year_year'`	`GRB.CONTINUOUS`
`scripts/CAPEX_OPEX_Gurobi_settings_dual_industry_categories1.py`	238	`'el_substation_inv_year_year'`	`GRB.CONTINUOUS`
`scripts/CAPEX_OPEX_Gurobi_settings_dual_industry_categories1.py`	240	`'inv_cost_heatstor_dh_year'`	`GRB.CONTINUOUS`
`scripts/CAPEX_OPEX_Gurobi_settings_dual_industry_categories1.py`	241	`'inv_cost_coolstor_dh_year'`	`GRB.CONTINUOUS`
`scripts/CAPEX_OPEX_Gurobi_settings_dual_industry_categories1.py`	242	`'ind_sto_cost_year'`	`GRB.CONTINUOUS`
`scripts/CAPEX_OPEX_Gurobi_settings_dual_industry_categories1.py`	245	`'ind_CC_inv_year'`	`GRB.CONTINUOUS`

Source	Line	Name / naming expression	Type
scripts/CAPEX_OPEX_Gurobi_settings_dual_industry_categories1.py	246	'ind_CC_tank_inv_year'	GRB.CONTINUOUS
scripts/CAPEX_OPEX_Gurobi_settings_dual_industry_categories1.py	247	'ind_CC_reboiler_inv_year'	GRB.CONTINUOUS
scripts/CAPEX_OPEX_Gurobi_settings_dual_industry_categories1.py	250	'CO2_storage_inv_year'	GRB.CONTINUOUS
scripts/CAPEX_OPEX_Gurobi_settings_dual_industry_categories1.py	2599	'generators_fOPEX_year'	GRB.CONTINUOUS
scripts/CAPEX_OPEX_Gurobi_settings_dual_industry_categories1.py	2600	'statCapInvCost_fix_OPEX_year'	GRB.CONTINUOUS
scripts/CAPEX_OPEX_Gurobi_settings_dual_industry_categories1.py	2606	'solar_gen_DH_fOPEX_year'	GRB.CONTINUOUS
scripts/CAPEX_OPEX_Gurobi_settings_dual_industry_categories1.py	2607	'solar_gen_indv_fOPEX_year'	GRB.CONTINUOUS
scripts/CAPEX_OPEX_Gurobi_settings_dual_industry_categories1.py	2608	'HP_fOPEX_year'	GRB.CONTINUOUS
scripts/CAPEX_OPEX_Gurobi_settings_dual_industry_categories1.py	2609	'HP_indv_fOPEX_year'	GRB.CONTINUOUS
scripts/CAPEX_OPEX_Gurobi_settings_dual_industry_categories1.py	2610	'boiler_indv_fOPEX_year'	GRB.CONTINUOUS
scripts/CAPEX_OPEX_Gurobi_settings_dual_industry_categories1.py	2611	'boiler_DH_fOPEX_year'	GRB.CONTINUOUS
scripts/CAPEX_OPEX_Gurobi_settings_dual_industry_categories1.py	2612	'heatstor_DH_fOPEX_year'	GRB.CONTINUOUS
scripts/CAPEX_OPEX_Gurobi_settings_dual_industry_categories1.py	2619	'coolstor_DH_fOPEX_year'	GRB.CONTINUOUS
scripts/CAPEX_OPEX_Gurobi_settings_dual_industry_categories1.py	2624	'industry_boil_fOPEX_year'	GRB.CONTINUOUS
scripts/CAPEX_OPEX_Gurobi_settings_dual_industry_categories1.py	2625	'FC_fOPEX_year'	GRB.CONTINUOUS
scripts/CAPEX_OPEX_Gurobi_settings_dual_industry_categories1.py	2626	'veh_transp_fOPEX_year'	GRB.CONTINUOUS
scripts/CAPEX_OPEX_Gurobi_settings_dual_industry_categories1.py	2628	'CO2_stor_fOPEX_year'	GRB.CONTINUOUS

Source	Line	Name / naming expression	Type
scripts/CAPEX_OPEX_Gurobi_settings_dual_industry_categories1.py	2632	'H2_stor_fOPEX_year'	GRB.CONTINUOUS
scripts/CAPEX_OPEX_Gurobi_settings_dual_industry_categories1.py	2633	'H2_gen_fOPEX_year'	GRB.CONTINUOUS
scripts/CAPEX_OPEX_Gurobi_settings_dual_industry_categories1.py	2637	'industry_CC_fOPEX_year'	GRB.CONTINUOUS
scripts/CAPEX_OPEX_Gurobi_settings_dual_industry_categories1.py	2638	'CC_tank_cap_fOPEX_year'	GRB.CONTINUOUS
scripts/CAPEX_OPEX_Gurobi_settings_dual_industry_categories1.py	2639	'reboiler_fOPEX_year'	GRB.CONTINUOUS
scripts/CAPEX_OPEX_Gurobi_settings_dual_industry_categories1.py	2643	'efuel_fOPEX_year'	GRB.CONTINUOUS
scripts/CAPEX_OPEX_Gurobi_settings_dual_industry_categories1.py	2646	'el_line_infr_fix_OPEX_year'	GRB.CONTINUOUS
scripts/CAPEX_OPEX_Gurobi_settings_dual_industry_categories1.py	2647	'el_line_infr_out_fix_OPEX_year'	GRB.CONTINUOUS
scripts/CAPEX_OPEX_Gurobi_settings_dual_industry_categories1.py	2648	'terminal_line_infr_fix_OPEX_year'	GRB.CONTINUOUS
scripts/CAPEX_OPEX_Gurobi_settings_dual_industry_categories1.py	2652	'H2_line_infr_fix_OPEX_year'	GRB.CONTINUOUS
scripts/CAPEX_OPEX_Gurobi_settings_dual_industry_categories1.py	2653	'H2_line_infr_out_fix_OPEX_year'	GRB.CONTINUOUS
scripts/CAPEX_OPEX_Gurobi_settings_dual_industry_categories1.py	2657	'fix_OPEX_year'	GRB.CONTINUOUS
scripts/CAPEX_OPEX_Gurobi_settings_dual_industry_categories1.py	3015	'per_mw1_year'	GRB.CONTINUOUS
scripts/CAPEX_OPEX_Gurobi_settings_dual_industry_categories1.py	3020	'ramp_up_cost_var_year'	GRB.CONTINUOUS
scripts/CAPEX_OPEX_Gurobi_settings_dual_industry_categories1.py	3021	'ramp_down_cost_var_year'	GRB.CONTINUOUS
scripts/CAPEX_OPEX_Gurobi_settings_dual_industry_categories1.py	3022	'statToPowert_cost_year'	GRB.CONTINUOUS
scripts/CAPEX_OPEX_Gurobi_settings_dual_industry_categories1.py	3023	'V2G_cost_year'	GRB.CONTINUOUS

Source	Line	Name / naming expression	Type
`scripts/CAPEX_OPEX_Gurobi_settings_dual_industry_categories1.py`	3027	`'costCO2per_year'`	`GRB.CONTINUOUS`
`scripts/CAPEX_OPEX_Gurobi_settings_dual_industry_categories1.py`	3029	`'imp_zone_var_cost_year'`	`GRB.CONTINUOUS`
`scripts/CAPEX_OPEX_Gurobi_settings_dual_industry_categories1.py`	3030	`'exp_zone_var_cost_year'`	`GRB.CONTINUOUS`
`scripts/CAPEX_OPEX_Gurobi_settings_dual_industry_categories1.py`	3033	`'imp_zone_secc_var_cost_year'`	`GRB.CONTINUOUS`
`scripts/CAPEX_OPEX_Gurobi_settings_dual_industry_categories1.py`	3034	`'exp_zone_secc_var_cost_year'`	`GRB.CONTINUOUS`
`scripts/CAPEX_OPEX_Gurobi_settings_dual_industry_categories1.py`	3038	`'export_cost_year_H2'`	`GRB.CONTINUOUS`
`scripts/CAPEX_OPEX_Gurobi_settings_dual_industry_categories1.py`	3039	`'import_cost_year_H2'`	`GRB.CONTINUOUS`
`scripts/CAPEX_OPEX_Gurobi_settings_dual_industry_categories1.py`	3207	`'ceep_cost_year'`	`GRB.CONTINUOUS`
`scripts/CAPEX_OPEX_Gurobi_settings_dual_industry_categories1.py`	3208	`'noload_el_penalty'`	`GRB.CONTINUOUS`
`scripts/complex_industry_accounting.py`	133	`f'industry_source_CO2_{len(ledger)}'`	`Gurobi default continuous`
`scripts/industry_annual_s_curve.py`	125	`f'industry_electric_annual_{suffix}_{service_idx}_{year}'`	`Gurobi default continuous`
`scripts/industry_annual_s_curve.py`	285	`f'ifs_annual_{suffix}_{index}_{year}'`	`Gurobi default continuous`
`scripts/transport_emissions.py`	80	`'transport_co2_block'`	`Gurobi default continuous`
`scripts/h2_terminals.py`	112	`name`	`Gurobi default continuous`
`scripts/h2_terminals.py`	114	`name`	`Gurobi default continuous`
`scripts/h2_terminals.py`	116	`name`	`Gurobi default continuous`